

Recent advances on carbazole-based photoinitiators of polymerization

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ABSTRACT

Photoinitiators of polymerization that can be activated under visible light and low intensity are actively researched as these compounds constitute the next generation of photoinitiators that will replace the controverted UV-photoinitiators. Over the years, a family of compounds has been identified as being highly promising in light of its remarkable photoinitiating ability, photochemical stability, low cost and easiness to functionalize, namely carbazoles. In this review, an overview of the different carbazole-based photoinitiators reported to date is presented. Interestingly, carbazoles have mostly been used as electron donors in push-pull structures, as photosensitizers for iodonium salts, as ligands for ferrocenium salts or as chromophores for phenacyl onium salts. These different applications clearly demonstrate the versatility of this structure used as the elemental building block for the design of UV, near-UV or visible light photoinitiators.

1. Introduction

During the past decades, photopolymerization carried out under low light intensity and visible light has been the focus of numerous research efforts due to the wide range of applications in which this polymerization technique is involved, ranging from coatings, adhesives, high-tech applications [1], 3D printing [2] and more recently the polymerization of thick samples comprising a high content of fillers [3,4]. From a technical point of point, photopolymerization exhibits several advantages compared to the traditional thermal polymerization done in solution. Notably, a temporal and a spatial control of the polymerization process can be obtained [5–7]. The polymerization process can also be extremely fast, the polymerization being ended after a few seconds (See Fig. 1) [8,9]. Parallel to this, photopolymerization can be carried out in solvent-free conditions, avoiding the release of volatile organic compounds (VOC) [10]. At present, most of the polymerization processes done in industries are based on UV-photoinitiators giving rise to safety concerns for the manipulators. Notably, UV-light can be responsible from eye damages under accidental exposure [11,12]. This unwanted exposure can nonetheless occur in industry but also during dental treatments in dental clinics so that alternatives to UV light photoinitiators are actively researched [13].

Besides, UV photoinitiators remains appealing candidates for photopolymerization due to their lack of color, these compounds being white powders. As a result of this, colorless polymers can be obtained, their incorporation into photocurable resins unaffected the color of the resins and thus that of the final polymers. Conversely, visible light photoinitiators are colored compounds and the final polymers are most

of the time highly colored even if low amounts of photoinitiators are classically used. In this context, a great deal of efforts has been devoted to develop photoinitiators capable to bleach i.e. to become colorless after a prolonged light irradiation or during the polymerization time. At present, examples of photoinitiators capable to bleach remain scarce [14]. In this field, metal complexes are excellent candidates and acylstannanes 1, 2 [15,16]. or acylgermanes (ivocerin 3, tetrakis(2-methylbenzoyl)germane 4) [17,18] can be cited as type I photoinitiators exhibiting the most remarkable photobleaching properties (See Fig. 2). But, examples of type II photoinitiators also exist as exemplified with a substituted (dithiophene)diketopyrrolopyrrole 5 [19], a zinc complex bearing oxadiazoles as pendant groups 6 [20], a cyclohexan-1-one derivative 7 [21] or a carboxylated camphorquinone 8 [22].

Parallel to safety concerns, light penetration inside the photocurable resin is another parameter governing the polymerization efficiency [23]. As evidenced in the Fig. 3 for a polystyrene latex with an average diameter of 112 nm, the light penetration ranges from 600 μm at 320 nm (i.e. in the UV region) to 5 cm at 800 nm (i.e. in the near-infrared region), completely modifying the scope of applications of photopolymerization. Indeed, if historically, photopolymerization which was based on UV photoinitiators was limited to coatings due to the poor light penetration inside the polymerizable resins, the access to thick samples containing fillers is now possible, revolutionizing the possibilities of photopolymerization (See Fig. 3).

At present, numerous families of organic dyes have been developed as type II photoinitiators activable under low light intensity and visible light. Among them, chalcones 9 [24], chromones 10 [25–27], perylenes 11 [28–30], diketopyrrolopyrroles 12 [31–33], acridine-1,8-diones 13



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最新 基于咔唑的聚合光引发剂的研究进展

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唑鎇盐

摘 要

能够在可见光和低强度下激活的光聚合引发剂正被积极研究, 因为这些化合物构成了下一代光引发剂, 将取代有争议的紫外光引发剂。多年来, 一系列化合物因其卓越的光引发能力、光化学稳定性、低成本和易功能化而被认为极具前景, 即咔唑。本综述概述了迄今为止报道的不同咔唑基光引发剂。有趣的是, 咔唑主要被用作推拉结构中的电子给体、碘鎇盐的光敏剂、二茂铁鎇盐的配体或苯甲酰鎇盐的发色团。这些不同的应用清楚地表明了该结构作为设计紫外、近紫外或可见光光引发剂的基本构建模块的多功能性。

1. 引言

在过去的几十年中, 在低光强度和可见光下进行的光聚合已成为众多研究工作的焦点, 原因是这种聚合技术在从涂料、粘合剂、高科技应用到3D打印以及最近涉及高填料含量厚样品聚合的广泛应用中扮演重要角色 [1], 33D打印 [2] 以及最近涉及高填料含量厚样品聚合的应用填料的厚样品聚合[3,4]。从技术角度来看, 与传统的溶液热聚合相比, 光聚合具有若干优势。特别是, 可以实现聚合过程的时空控制 [5—7]。聚合过程也可以非常快, 只需几秒钟即可完成(见图1) [8,9]。与此并行的是, 光聚合可以在无溶剂条件下进行, 避免挥发性有机化合物(VOC)的释放 [10]。目前, 工业上进行的大多数聚合过程基于紫外线光引发剂, 这给操作人员带来了安全隐患。值得注意的是, 意外暴露于紫外光下可能导致眼睛损伤[11,12]。这种意外暴露不仅可能发生在工业中, 也可能发生在牙科诊所的治疗过程中, 因此人们正在积极研究紫外线光引发剂的替代品 [13]。

此外, 紫外光引发剂由于其无色特性(这些化合物为白色粉末), 仍然是光聚合反应中具有吸引力的候选物。因此, 可以获得无色聚合物, 将其掺入光固化树脂中不会影响树脂的颜色, 从而也不会影响最终聚合物的颜色。相反, 可见光引发剂是有色化合物, 最终聚合物通常高度着色, 即使通常使用低含量的光引发剂也是如此。

在这种背景下, 大量研究致力于开发能够漂白的光引发剂, 即经过长时间光照或在聚合时间内变为无色。目前, 能够漂白的光引发剂实例仍然很少[14]。在该领域, 金属配合物是最佳的候选物, 而酰基锡烷 1, 2 [15,16]或酰基锆烷 (ivocerin 3、四(2-甲基苯甲酰基)锆烷4) [17,18] 可以作为具有最显著光漂白性能的I型光引发剂实例(参见图2)。但是, 也存在II型光引发剂的实例, 例如以取代的(二噻吩)二酮吡咯并吡咯 5 [19]、带有噁二唑侧基的锌配合物 6 [20]、环己酮衍生物 7 [21] 或羧化樟脑醌 8 [22]为例。

与安全性问题并行, 光固化树脂内部的光穿透性是影响聚合效率的另一个参数。[23]如图3中平均直径为112 nm的聚苯乙烯乳胶所示, 光穿透范围从320 nm(即紫外区域)的600 μm到800 nm(即近红外区域)的5 cm, 这彻底改变了光聚合的应用范围。实际上, 如果历史上基于紫外光引发剂的光聚合由于在可聚合树脂中光穿透性差而被限制于涂层, 那么现在可以获取含有填料的厚样品, 这彻底革新了光聚合的可能性(见图。。3)。

目前, 已开发出众多有机染料家族作为可在低光强和可见光下活化的II型光引发剂。其中, 查尔酮 9 [24], 色酮 10 [25–27], 花11 [28–30], 二酮吡咯并吡咯 12 [31–33], 吡啶-1,8-二酮13

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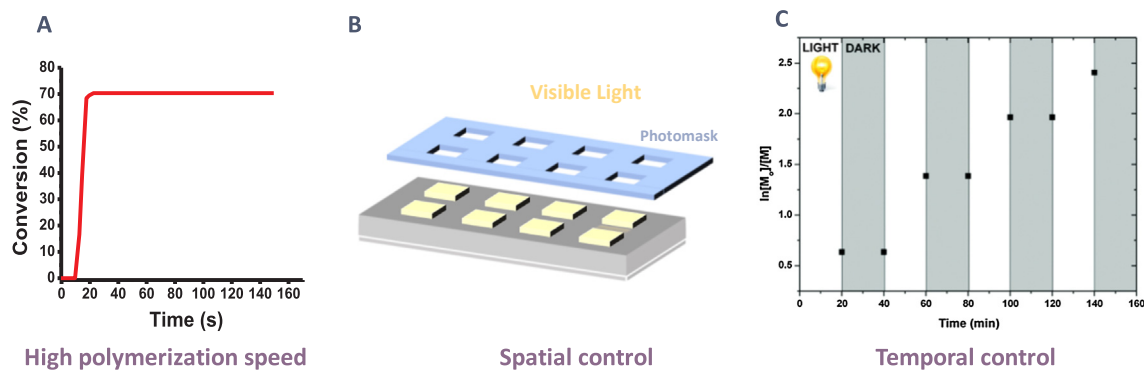


Fig. 1. The different advantages of photopolymerization.

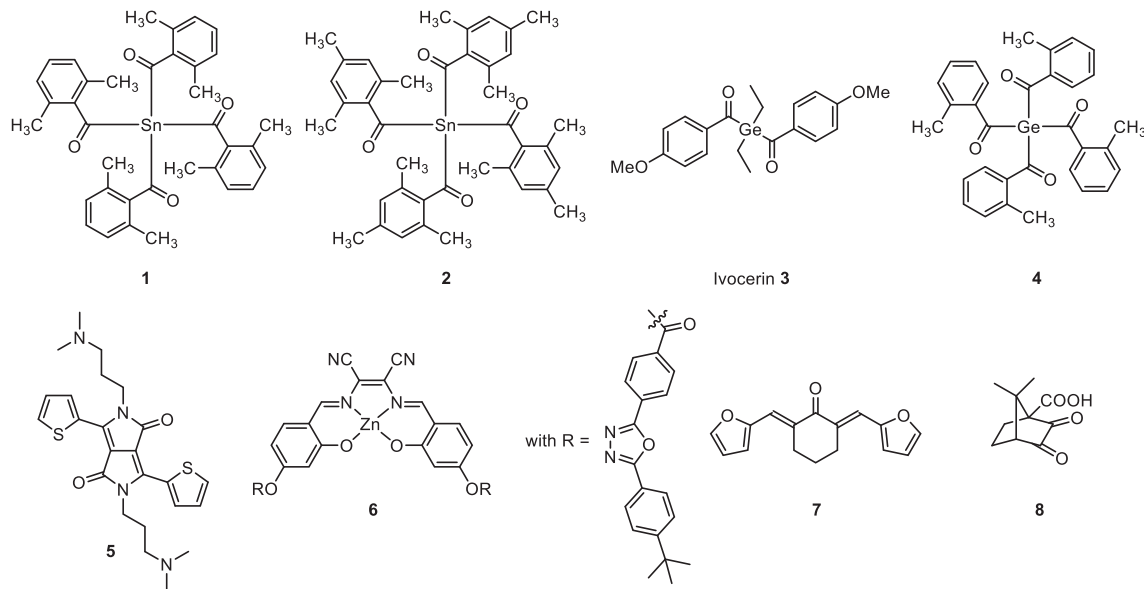


Fig. 2. Examples of type I and type II photoinitiators reported in the literature as being capable to bleach under light irradiation.

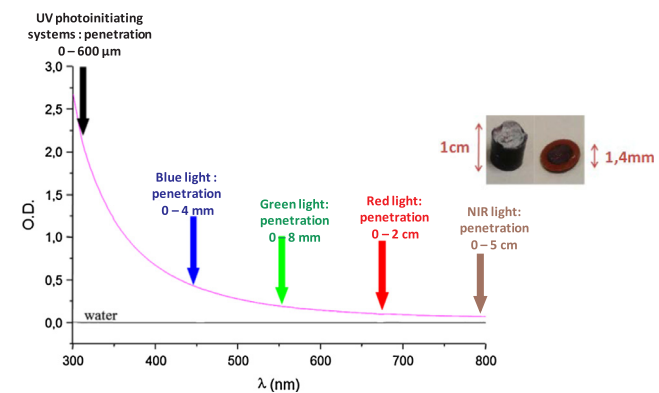


Fig. 3. Light penetration inside a polystyrene latex with an average diameter of 112 nm. Inset: examples of polymerized resins containing fillers. Reproduced with permission from Bonardi et al. [23]. Copyright 2018 American Chemical Society.

[34–36], naphthalimides 14 [37–47], iodonium salts 15 [48,49], squaraines 16, bodipys 17 and porphyrins 18 [50,51], pyrenes 19 [52–55], benzophenones 20 [56–59], push-pull dyes 21 [60–69], coumarines 22 [70], 2,3-diphenylquinoxaline derivatives 23 [71], pyridinium derivatives 24 [72], or dihydroxy-anthraquinones 25 [73] can be cited the most relevant examples (See Fig. 4).

Since a decade and due to the fact that the traditional families of UV

photoinitiators are not adapted for the design of visible light photoinitiators, even after extensive works of chemical engineering, new families of compounds and especially dyes that have never been explored in the context of photopolymerization are now classically tested by polymerists. Considering that visible light photoinitiators should strongly absorb in the visible range, be chemically stable, and that these compounds should also be capable to be reduced or oxidized in an acceptable electrochemical window, all organic dyes examined in Organic Electronics (organic solar cells, organic light-emitting diodes, organic field-effect transistors) are potential candidates for photoinitiation [74]. In this field, carbazole is a representative example. This symmetric structure which is extremely cheap has been extensively studied in Organic Electronics due to its remarkable electron-donating ability, its low oxidation potential and its facile chemical modification so that a wide range of hole-transport materials [75,76], molecular glasses [77–80], or light-emitting materials [81–85] have been designed with carbazole. Considering that push-pull dyes can be easily synthesized with this electron donor, carbazole was also an excellent candidate for the design of light-harvesting materials for solar cells, and numerous dyes exhibiting high molar extinction coefficients and broad absorption spectra were obtained with this planar electron donor [86,87]. Carbazole is a polyaromatic structure comprising two benzene rings fused on both sides of a nitrogen-containing five-membered ring (See Fig. 5) [88]. The central position of the nitrogen atom allows to write resonance forms with an electronic delocalization extending over the two aromatic rings. By the presence of the amine, the nitrogen atom

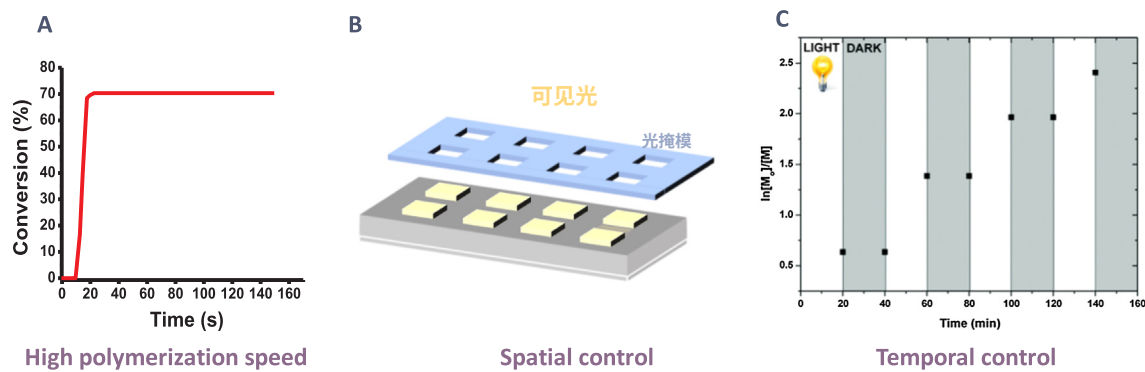


图1. 光聚合的不同优势。

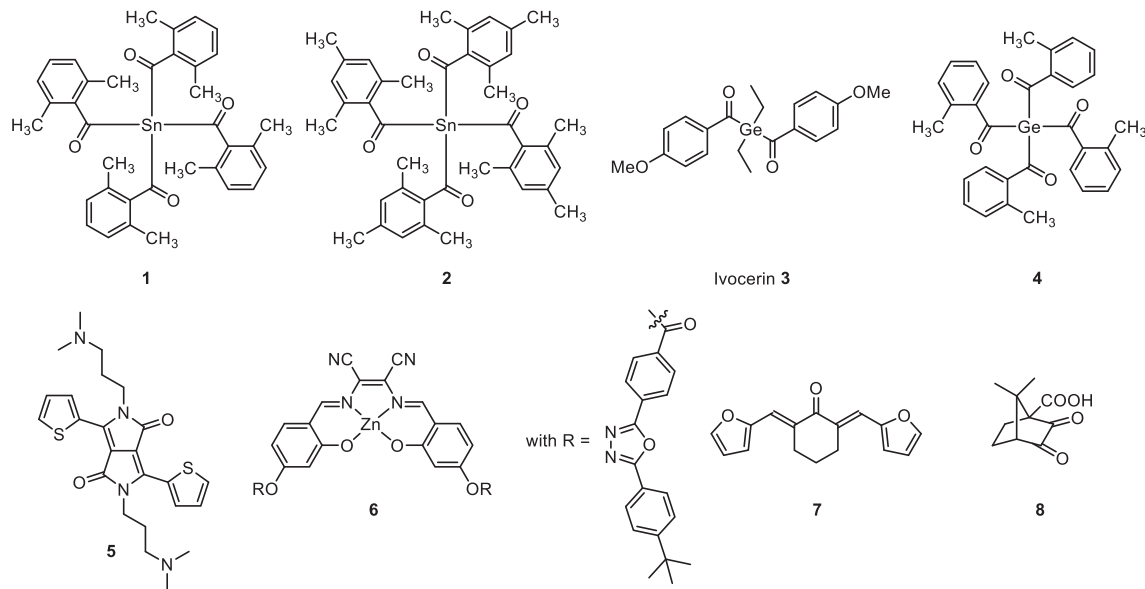


图2. 文献中报道的在光照射下漂白的I型和II型光引发剂的示例能够

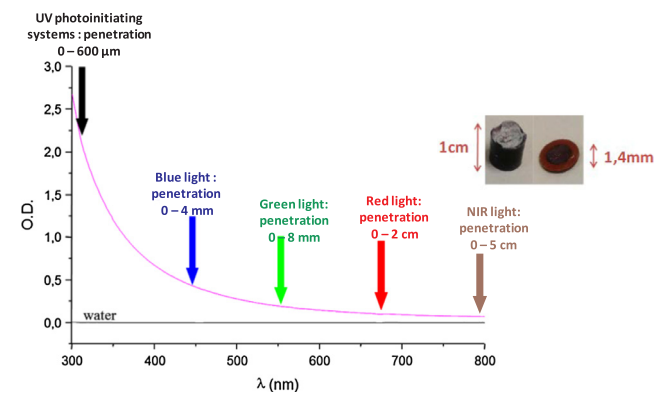


图3. 光在平均直径为112 nm的聚苯乙烯乳胶内的穿透。插图：含有填料的聚合树脂实例。经Bonardi等人许可转载。 [23]。版权所有2018美国化学学会。

[34–36], 萘酰亚胺 14 [37–47], 碘鎓盐 15 [48,49], 方酸16, 氟硼二吡咯17 和卟啉类 18 [50,51], 芪19[52–55], 二苯甲酮 20 [56–59], 推拉型染料 21 [60–69], 香豆素 22 [70], 2,3-二苯基喹啉衍生物 23 [71], 吡啶衍生物 24 [72], 或二羟基蒽醌 25 [73] 可作为最相关的例子 (参见图4)。

近十年来, 由于传统UV光源家族

光引发剂不适用于设计可见光光引发剂, 即使在大量化学工程工作之后, 新的化合物类别, 尤其是从未在光聚合背景下研究过的染料, 现在也常规地被高分子学家测试。考虑到可见光光引发剂应在可见光范围内有强吸收、化学稳定, 并且这些化合物还能够在可接受的电化学窗口中被还原或氧化, 所有在有机电子学 (有机太阳能电池、有机发光二极管、有机场效应晶体管) 中研究的有机染料都是光引发的潜在候选者 [74]。在这个领域中, 咔唑是一个典型例子。这种对称结构极其廉价, 由于其卓越的给电子能力、低氧化电位和易于化学修饰, 已在有机电子学中广泛研究, 因此设计了一系列空穴传输材料 [75,76], 分子玻璃 [77–80], 或发光材料 [81–85] 使用咔唑。考虑到使用这种给电子体容易合成推拉染料, 咔唑也是设计太阳能电池捕光材料的优秀候选者, 并且许多具有高摩尔消光系数和宽吸收光谱的染料已用这种平面给电子体制得 [86,87]。咔唑是一种多芳环结构, 包含两个苯环融合在含氮五元环的两侧 (见图5) [88]。氮原子的中心位置允许写出电子离域扩展到两个芳环的共振形式。由于胺的存在, 氮原子

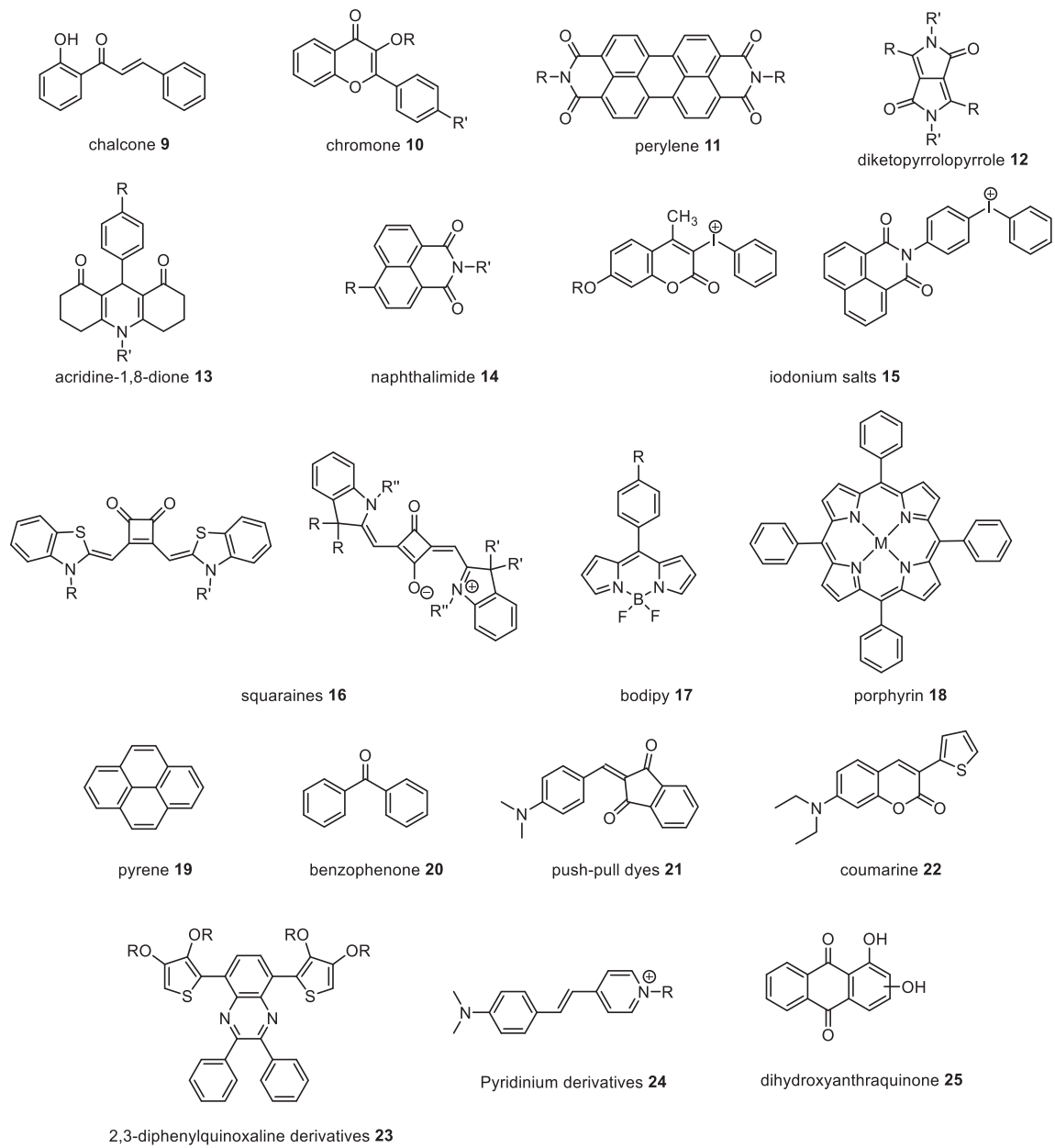


Fig. 4. Different families of compounds (**9–25**) examined as visible light photoinitiators of polymerization activable under low light intensity.

can be functionalized with alkyl chains to improve the solubility of photoinitiators. By quaternization of the amine, carbazolium salts can also be prepared [89,90]. The 3- and 6-positions of carbazole can also be easily functionalized with various groups such as aldehyde, halogens and nitro groups so that an extension of conjugation or the introduction of various groups can be carried out. Parallel to the modification of the carbazole core, carbazole is also an electron-rich polyaromatic group that can be used for the design of ferrocenium salts, these iron complexes being prepared by ligand exchange on ferrocene [91].

Concerning photopolymerization, *N*-vinylcarbazole (NVK) is classically used as an additive improving the final monomer conversion due to the possibility for the photogenerated radicals to add onto the vinylic group of NVK [92,93]. In fact, NVK is capable to convert hardly oxidizable radicals formed by the photoinitiating systems to easily oxidizable ones so that it often constitutes the key element enabling to get high final monomer conversions. To get this impressive result, a careful examination of the polymerization mechanism should be carried out. Typically, to get a significant enhancement of the final monomer

conversion with NVK, this latter should be added to two-component systems comprising a photoinitiator (PI) and an iodonium salt (Ph_2I^+) (See Scheme 1) [94,95]. When the photosensitizer is excited to its triplet state by light irradiation, it can thus reduce Ph_2I^+ to $\text{Ph}_2\text{I}^\cdot$ which subsequently decompose to $\text{Ph}^\cdot$ (and $\text{Ph-I}^\cdot$), constituting an initiating species for the free radical polymerization (FRP) of acrylates. However, by reacting with the phenyl radical $\text{Ph}^\cdot$, NVK can also give a highly reactive species $\text{Ph-NVK}^\cdot$, initiating more efficiently the FRP of acrylates than $\text{Ph}^\cdot$. Parallel to photoinitiation, $\text{Ph-NVK}^\cdot$ can also react with the oxidized form of the photoinitiator PI^+ , generating the Ph-NVK^+ species capable to initiate the free radical promoted cationic polymerization (FRPCP) of epoxides. $\text{Ph-NVK}^\cdot$ can also be oxidized by Ph_2I^+ which is used in excess so that Ph-NVK^+ species can be obtained by this second route. Therefore, due to the possibility of a dual cure process, the three-component system $\text{PI}/\text{Ph}_2\text{I}^+/\text{NVK}$ has been extensively studied in the literature as it opens the way towards the generation of interpenetrated networks (IPNs) where the simultaneous polymerization of acrylates and epoxides provides crosslinked polymers of unique

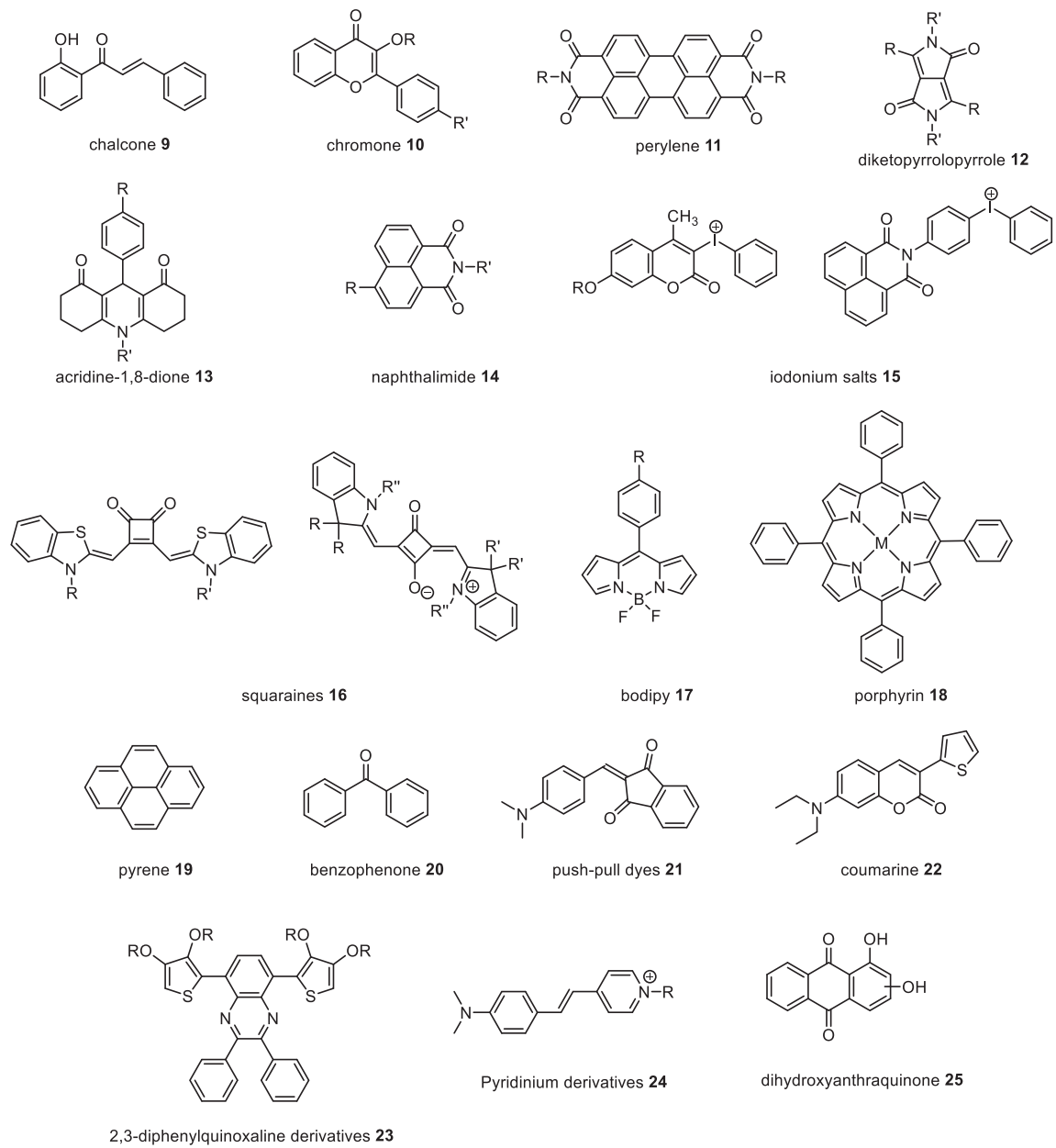


图4. 在低光强度下可作为可见光光引发剂的不同化合物家族（9–25）。

可以通过烷基链功能化以提高光引发剂的溶解度。通过胺的季铵化，也可以制备咪唑鎓盐 [89,90]。咪唑的3位和6位也可以轻松用各种基团（如醛基、卤素和硝基）功能化，从而实现共轭延伸或引入各种基团。除了修饰咪唑核心外，咪唑也是一种富电子多芳环基团，可用于设计二茂铁鎓盐，这些铁配合物通过二茂铁上的配体交换制备 [91]。

关于光聚合，*N*-乙烯基咪唑（NVK）通常被用作添加剂，由于光生自由基能够加成到NVK的乙烯基上，从而改善最终单体的转化率 [92,93]。事实上，NVK能够将光引发体系形成的难氧化自由基转化为易氧化自由基，因此它常常成为实现高最终单体转化率的关键元素。为了获得这一显著结果，应对聚合机理进行仔细研究。通常，要显著提高最终单体

通过与NVK的转化，后者应添加到包含光引发剂(PI)和碘鎓盐(Ph_2I^+)的双组分体系中（见方案1） [94,95]。当光敏剂通过光照射激发到其三线态时，它可以还原 Ph_2I^+ 为 $\text{Ph}_2\text{I}^\cdot$ ，后者随后分解为 $\text{Ph}^\cdot$ （和 $\text{Ph-I}^\cdot$ ），构成丙烯酸酯自由基聚合（FRP）的引发种。然而，通过与苯基自由基 $\text{Ph}^\cdot$ 反应，NVK也可以生成高活性物种 $\text{Ph-NVK}^\cdot$ ，比 $\text{Ph}^\cdot$ 更有效地引发丙烯酸酯的FRP。与光引发平行， $\text{Ph-NVK}^\cdot$ 还可以与氧化形式的光引发剂 PI^+ 反应，生成能够引发环氧化物自由基促进阳离子聚合（FRPCP）的 Ph-NVK^+ 物种。 $\text{Ph-NVK}^\cdot$ 也可被过量的 Ph_2I^+ 氧化，因此通过第二种途径可得到 Ph-NVK^+ 物种。因此，由于双重固化过程的可能性，三组分体系 $\text{PI}/\text{Ph}_2\text{I}^+/\text{NVK}$ 在文献中得到了广泛研究，因为它为实现互穿网络（IPN）开辟了道路，其中丙烯酸酯和环氧化合物的同时聚合提供了具有独特性能 of 的交联聚合物。

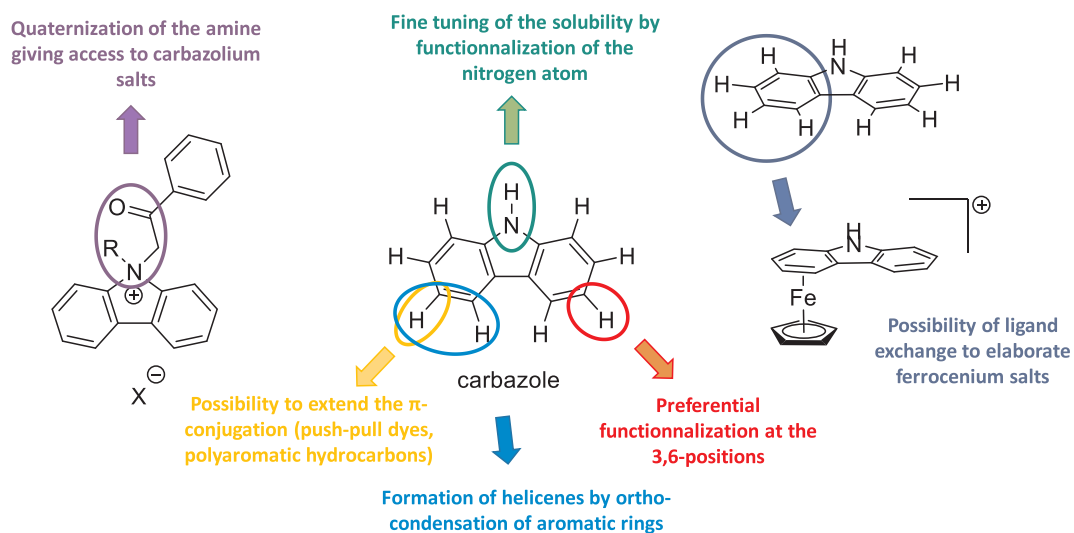


Fig. 5. The different possible chemical modifications on the carbazole core.

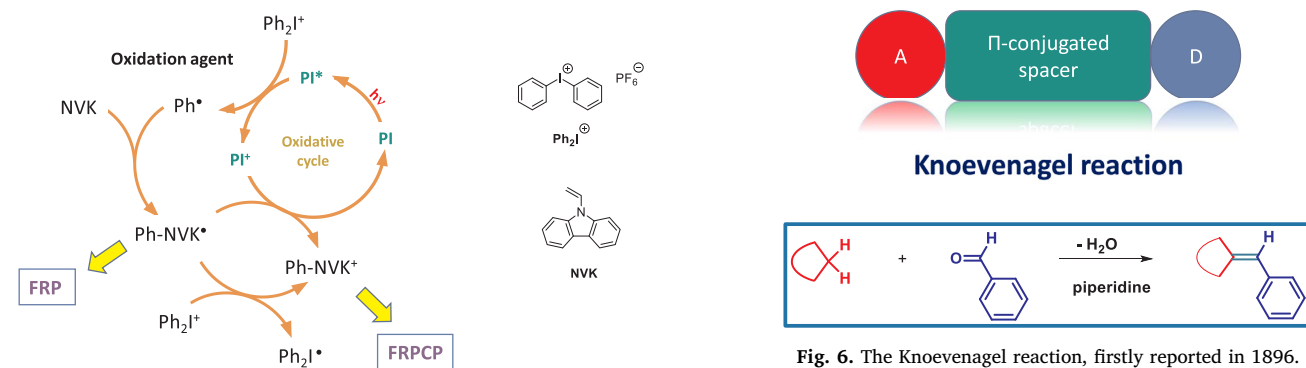


Fig. 6. The Knoevenagel reaction, firstly reported in 1896.

Scheme 1. Photoredox catalytic cycle involving a three-component system.

photophysical and mechanical properties. Due to the reduction of PI^+ to PI and thus its regeneration, the photoinitiator can thus be introduced into catalytic amount in the photoinitiating system. The catalytic cycle reported in the Scheme 1 is therefore a typical photoredox catalytic cycle. Parallel to the development of photoinitiators based on carbazole, a new application has recently been reported for a carbazole derivative, namely, its use as a fluorescent probe to monitor the cationic photopolymerization process by fluorescence measurements. It therefore evidences the versatility of carbazoles in polymerization [96].

In this review, an overview of the different photoinitiators comprising the carbazole moiety is presented. Over the years, different strategies have been developed to take advantage of the carbazole scaffold and push-pull dyes, nitrogen-containing polyaromatic hydrocarbons, helicenes, ferrocenium or carbazolium, and more recently photoinitiators exhibiting a thermally activated delayed fluorescence (TADF) property have been synthesized as tested in photopolymerization. To evidence the interest of these different families of photoinitiators, performances of these photoinitiators in FRP and FRPCP is presented.

2. Carbazoles as photoinitiators of polymerization

2.1. Push-pull dyes

As mentioned in the introduction section, carbazole has been extensively used for the design of push-pull dyes due to its remarkable electron-donating ability. Push-pull dyes are composed of an electron-donating and an electron accepting moiety connected by mean of a

conjugated spacer (See Fig. 6).

Among all possible chemical reactions to create a double bond, the Knoevenagel reaction is a reaction of choice, opposing an aldehyde to an activated methylene group. The first report mentioning this coupling mode was published in 1896 by Knoevenagel, consisting in the condensation of benzaldehyde with ethyl acetoacetate in the presence of piperidine [97]. The first product synthesized by mean of this reaction could be obtained in 95% yield. From an industrial point of view, this reaction is of crucial interest as this reaction is easy to perform, can be done in Green solvents such as ethanol and even in solvent-free conditions, by mechanosynthesis [98]. Irrespective of the synthetic approach, the side product of this reaction is safe and limited to a water molecule. Considering that both the electron donor and the electron acceptor can be commercially available, numerous push-pull dyes have been designed and examined as photoinitiators of polymerization.

As far as carbazoles are concerned, the first report mentioning the design of carbazole-based push-pull dyes for photopolymerization applications was published in 2013 by Lalevée and coworkers [99]. **Cbz_1** and **Cbz_2** were prepared in one step, starting from 9-octyl-9H-carbazole-3-carbaldehyde **26**, barbituric acid **27** and thiobarbituric acid **28** respectively. The two dyes **Cbz_1** and **Cbz_2** could be obtained by refluxing the two precursors in ethanol, in 78 and 85% yields respectively (See Scheme 2). Interestingly, **Cbz_1** and **Cbz_2** exhibited a broad intramolecular charge transfer (ITC) band ranging between 350 and 500 nm for **Cbz_1**, 375 and 550 nm for **Cbz_2** respectively so that photopolymerization experiments could be carried out upon irradiation with laser diodes emitting at 457 and 473 nm or a LED bulb emitting at 462 nm (See Fig. 7).

High molar extinction coefficients were determined at the absorption maxima of the two complexes, $45\,000\text{ M}^{-1}\text{ cm}^{-1}$ at 431 nm for

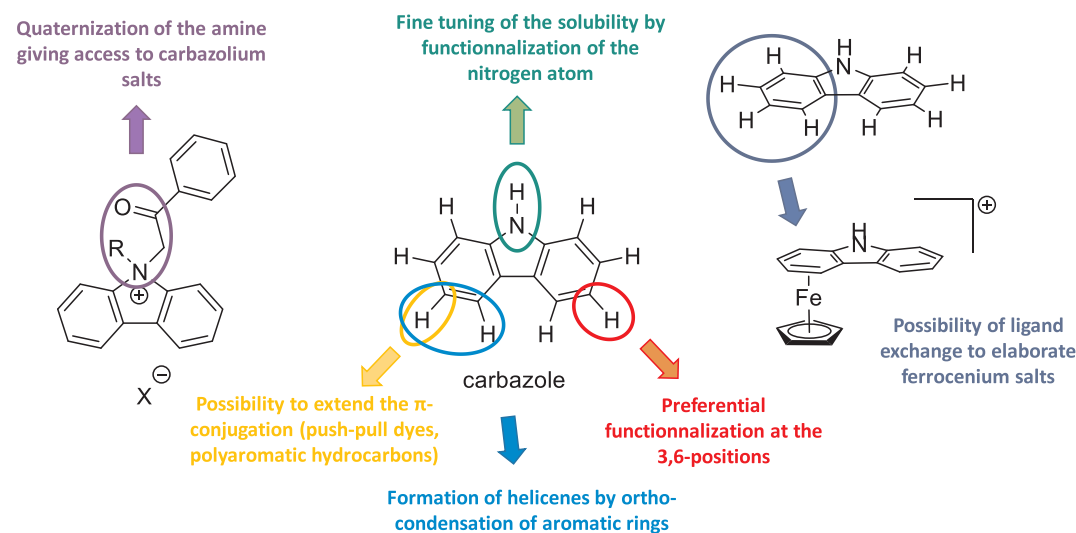


图5. 咔唑核上可能的不同化学修饰。

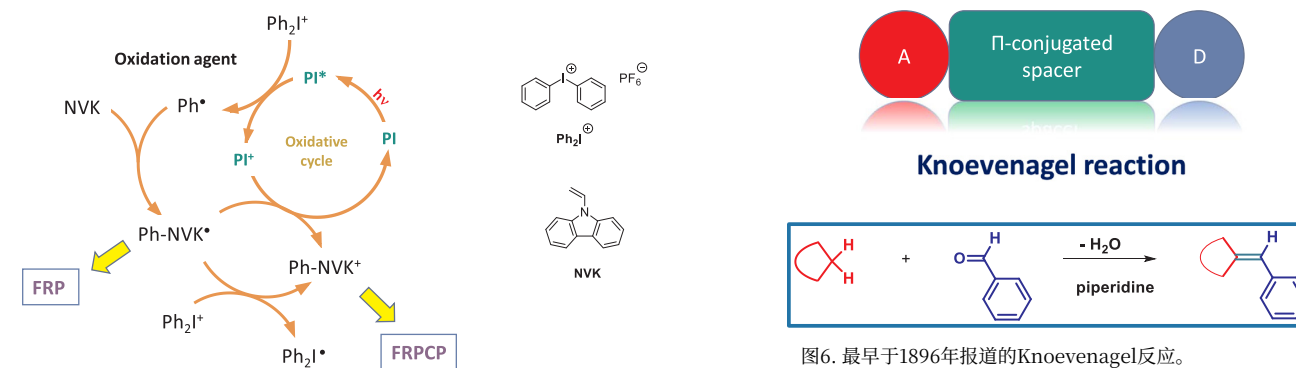


图6. 最早于1896年报道的Knoevenagel反应。

方案1. 涉及三组分体系的光氧化还原催化循环。

光物理和机械性能。^{*}由于 PI 被还原为 PI 及其再生, 光引发剂可以以催化量引入光引发体系中。在方案1中报道的催化循环因此是一个典型的光氧化还原催化循环。与基于咔唑的光引发剂的发展并行, 最近报道了一种咔唑衍生物的新应用, 即其作为荧光探针, 通过荧光测量监测阳离子光聚合过程。因此, 它证明了咔唑在聚合反应中的多功能性 [96]。

本综述概述了包含咔唑基团的不同光引发剂。多年来, 已开发出多种策略来利用咔唑骨架和推拉染料、含氮多环芳烃、螺烯、二茂铁或咔唑鎓, 以及最近合成的具有热激活延迟荧光 (TADF) 特性的光引发剂, 并对其在光聚合中的性能进行了测试。为了证明这些不同光引发剂家族的价值, 本文介绍了它们在自由基光聚合 (FRP) 和自由基促进阳离子光聚合 (FRPCP) 中的性能。

2. 咔唑作为聚合光引发剂

2.1. 推拉染料

如引言所述, 咔唑因其显著的给电子能力而被广泛用于推拉型染料的设计。推拉型染料由给电子部分和受电子部分通过

共轭间隔基 (参见图6)。

在所有可能生成双键的化学反应中, Knoevenagel反应是一种优选反应, 它使醛与活化亚甲基发生反应。首次报道这种偶联模式的文献发表于1896年, 由Knoevenagel提出, 涉及在哌啶存在下苯甲醛与乙酰乙酸乙酯的缩合 [97]。通过该反应合成的首个产物产率可达95%。从工业角度来看, 该反应至关重要, 因为它易于操作, 可在绿色溶剂如乙醇中进行, 甚至可通过机械合成在无溶剂条件下进行 [98]。无论采用何种合成方法, 该反应的副产物安全且仅限于水分子。考虑到给电子体和受电子体均可商业获得, 众多推拉型染料已被设计并被研究作为聚合光引发剂。

就咔唑而言, 首次报道将咔唑基推拉染料用于光聚合应用的研究发表于2013年, 由Lalevée及其同事完成 [99]。Cbz_1和Cbz_2通过一步法制备, 分别以9-辛基-9H-咔唑-3-甲醛26、巴比妥酸27和硫代巴比妥酸28为原料。两种染料Cbz_1和Cbz_2可通过将两种前体在乙醇中回流获得, 产率分别为78%和85% (见方案2)。有趣的是, Cbz_1和Cbz_2显示出宽泛的分子内电荷转移 (ITC) 带, Cbz_1的吸收范围在350至500 nm之间, Cbz_2的吸收范围在375至550 nm之间, 因此可以使用发射波长为457 nm和473 nm的激光二极管或发射波长为462 nm的LED灯泡进行光聚合实验 (见图7)。

在两种配合物的吸收最大值处测定出高摩尔消光系数, 分别为431 nm处的 $45\,000\text{ M}^{-1}\text{ cm}^{-1}$ (对于

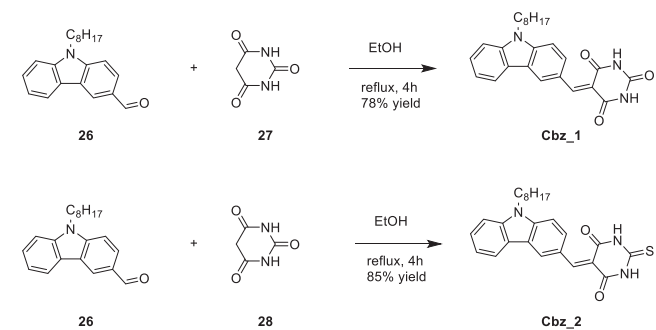
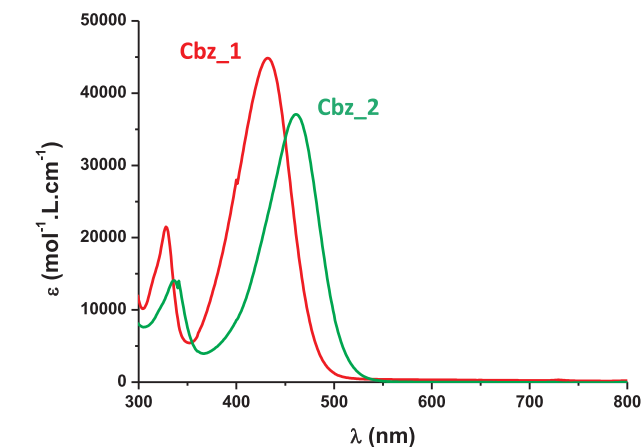
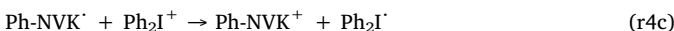
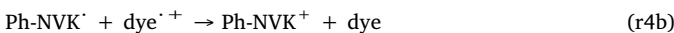
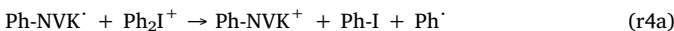
Scheme 2. Synthetic route to **Cbz_1** and **Cbz_2**.

Fig. 7. UV–visible absorption spectra of **Cbz_1** and **Cbz_2** in acetonitrile. Reprinted with permission from Tehfe et al. [99] Copyright © 2013 John Wiley & Sons, Inc.

Cbz_1 and $37\,000\text{ M}^{-1}\text{ cm}^{-1}$ at 462 nm for **Cbz_2** respectively. However, despite the similarity of structures, completely different behaviors were observed upon photolysis of **Cbz_1** and **Cbz_2**. Upon exposure to a laser diode at 473 nm, a fast photobleaching of the **Cbz_2**/diphenyliodonium hexafluorophosphate (Ph_2I^+) mixture was observed, indicating the dyes to be oxidized by the iodonium salt and to result in its decomposition (see equations **r(1)** and **r(2)**). In the presence of NVK, $\text{Ph}^\cdot$ is rapidly converted to $\text{Ph-NVK}^\cdot$ (**r3**) by addition on the double bond. This radical can be easily oxidized by Ph_2I^+ , providing Ph-NVK^+ (**r4a**) but $\text{Ph-NVK}^\cdot$ can also reduce the oxidized form of the dye (**r4b**), regenerating the dye. Overall, by the formation of $\text{Ph-NVK}^\cdot$ and Ph-NVK^+ , both the free radical polymerization (FRP) of acrylates (**r6**) or the free radical promoted cationic polymerization (FRPCP) of

epoxides (**r5**) can be initiated.

$\text{dye} \rightarrow {}^1\text{dye}(\text{h}\nu)$



On the opposite, a completely different behavior was observed with **Cbz_1**, the photolysis of the **Cbz_1**/ Ph_2I^+ mixture being extremely slow (See **Fig. 8**). This counter-performance was assigned to a back electron-transfer between **Cbz_1** $^{\cdot+}$ and $\text{Ph}_2\text{I}^\cdot$ in **r1**.

While examining the polymerization profiles of (3,4-epoxycyclohexane)methyl 3,4-epoxycyclohexylcarboxylate (EPOX) upon irradiation at 473 nm, the three-component system **Cbz_1**/NVK/ Ph_2I^+ (0.3%/3%/2% w/w) proved to be ineffective to initiate a polymerization whereas a final monomer conversion of 55% could be obtained within 800 s for the three-component system **Cbz_2**/NVK/ Ph_2I^+ (0.3%/3%/2% w/w). A similar monomer conversion could be obtained with the LED bulb at 462 nm with **Cbz_2**, the absorption of **Cbz_2** being similar at 462 and 473 nm.

While examining the FRP of trimethylolpropane triacrylate (TMPTA) (See **Fig. 9**), no polymerization was observed with **Cbz_1** whereas a final monomer conversion of 45% could be obtained within 300 s with the three-component system **Cbz_2**/NVK/ Ph_2I^+ (0.3%/3%/2% w/w).

Finally, an interesting finding was determined while examining the solvatochromism of **Cbz_1** and **Cbz_2** in solvents of different polarity. Surprisingly, if a linear correlation could be established with different empirical scales of solvent polarity such as the Bakhshiev's [100], the Lippert–Mataga [101], the Kowski–Chamma–Viallet's [102], the McRae's [103] or the Suppan's [104] scales with **Cbz_2**, no linear correlation could be established for **Cbz_1**. This unexpected behavior for **Cbz_1** was assigned to the formation of hydrogen bonds with the solvents, altering the solvatochromic behavior of this compound (See **Scheme 3**). Besides, a direct correlation between solvatochromism and photoinitiating ability can be established and this point is an important finding.

As evidenced by the authors in this work but also in other studies

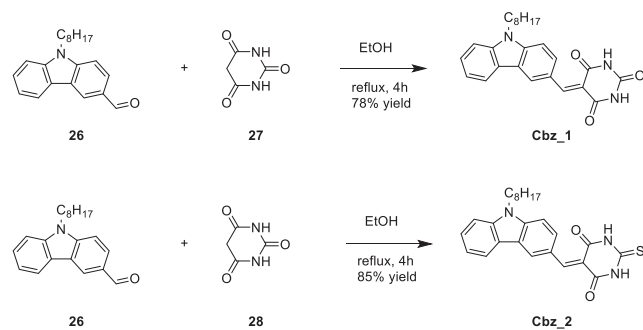
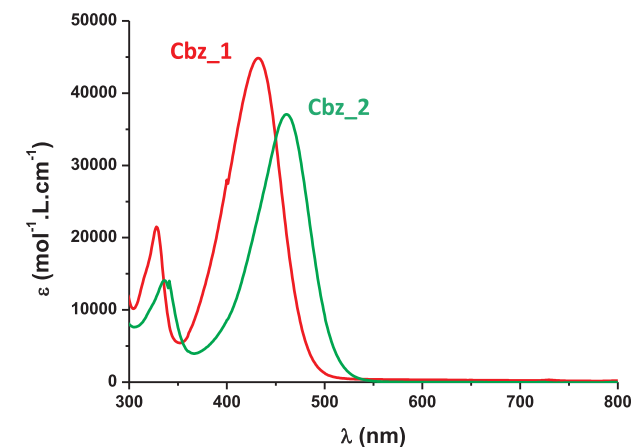
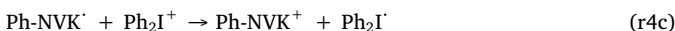
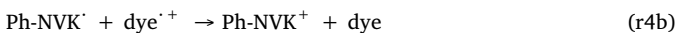
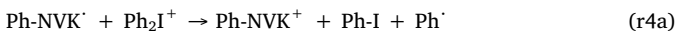
方案2. **Cbz_1** 和 **Cbz_2** 的合成路线。

图7. **Cbz_1** 和 **Cbz_2** 在乙腈中的紫外–可见吸收光谱。经Tehfe等人许可转载。[99]版权所有 © 2013 John Wiley & Sons, Inc.

Cbz_1) 和 462 nm 处的 $37\,000\text{ M}^{-1}\text{ cm}^{-1}$ (对于 **Cbz_2**)。然而, 尽管结构相似, 在光解 **Cbz_1** 和 **Cbz_2** 时观察到完全不同的行为。当暴露于 473 nm 激光二极管时, 观察到 **Cbz_2**/二苯基碘鎓六氟磷酸盐 (Ph_2I^+) 混合物快速光漂白, 表明染料被碘鎓盐氧化并导致其分解 (见方程 **r(1)** 和 **r(2)**)。在 NVK 存在下, $\text{Ph}^\cdot$ 通过加成到双键上迅速转化为 $\text{Ph-NVK}^\cdot$ (**r3**)。该自由基可被 Ph_2I^+ 轻易氧化, 生成 Ph-NVK^+ (**r4a**), 但 $\text{Ph-NVK}^\cdot$ 也能还原染料的氧化形式 (**r4b**), 再生染料。总的来说, 通过形成 $\text{Ph-NVK}^\cdot$ 和 Ph-NVK^+ , 既可以进行丙烯酸酯的自由基聚合 (FRP) (**r6**), 也可以进行

环氧化物 (**r5**) 可以被引发。

$\text{dye} \rightarrow {}^1\text{dye}(\text{h}\nu)$



相反, 使用 **Cbz** 观察到完全不同的行为 **Cbz_1**, 光解反应 **Cbz_1**/ Ph_2I^+ 混合物极其缓慢 (参见 **图8**)。这种不良性能归因于 **Cbz_1** $^{\cdot+}$ 和 $\text{Ph}_2\text{I}^\cdot$ 在 **r1** 中的反向电子转移。

在检查 (3,4-环氧环己基) 甲基 3,4-环氧环己基甲酸酯 (EPOX) 在 473 nm 辐照下的聚合曲线时, 三组分体系 **Cbz_1**/NVK/ Ph_2I^+ (0.3%/3%/2% w/w) 被证明无法引发聚合, 而三组分体系 **Cbz_2**/NVK/ Ph_2I^+ (0.3%/3%/2% w/w) 在 800 秒内可实现 55% 的单体转化率。使用 462 nm 的 LED 灯泡时, **Cbz_2** 可获得相似的单体转化率, 因为 **Cbz_2** 在 462 nm 和 473 nm 处的吸收相似。

在检查三羟甲基丙烷三丙烯酸酯 (TMPTA) 的 FRP 时 (参见 **图9**), 使用 **Cbz_1** 未观察到聚合反应, 而使用三组分体系 **fb** 可在 300 秒内获得 45% 的最终单体转化率 **Cbz_2**/NVK/ Ph_2I^+ (0.3%/3%/2% w/w)。

最后, 在检查 **Cbz_1** 和 **Cbz_2** 在不同极性溶剂中的溶剂化显色时, 发现了一个有趣的现象。令人惊讶的是, 虽然对 **Cbz_2** 使用不同的溶剂极性经验标度 (如 Bakhshiev 的 [100], 、Lippert–Mataga 的 [101], 、Kawski–Chamma–Viallet 的 [102], 、McRae 的 [103] 或 Suppan 的 [104] 标度) 可以建立线性相关, 但对 **Cbz_1** 却无法建立任何线性相关。**Cbz_1** 的这种意外行为归因于与溶剂形成氢键, 改变了该化合物的溶剂化显色行为 (参见 **方案3**)。此外, 可以直接建立溶剂化显色与光引发能力之间的相关性, 这是一个重要的发现。

正如作者在本研究及其他研究中所证明的,



Fig. 8. Photolysis of a **Cbz_2**/Iod (**A**) or **Cbz_1**/Iod (**B**) mixture in acetonitrile upon irradiation with a laser diode at 473 nm. The different UV–visible spectra were recorded at the same irradiation times under air. Reprinted with permission from Tehfe et al. [99] Copyright © 2013 John Wiley & Sons, Inc.

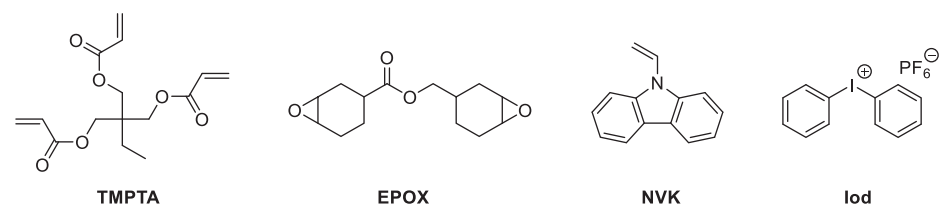


Fig. 9. Chemical structures of the different monomers and additives used with **Cbz_1** and **Cbz_2**.

making use of push-pull dyes as photoinitiators, only the dyes exhibiting a solvatochromism that can be rationalized with classical solvatochromic scales can initiate a polymerization process. Finally, authors proved that by simply examining the solvatochromism of the newly synthesized push-pull dyes, the photoinitiating abilities of these chromophores could be efficiently anticipated.

Considering the strong structure/reactivity relationship exists for this class of dyes, another carbazole-based dye **Cbz_3** was examined with a series of three other dyes **29–30** comprising the same electron acceptor (See Fig. 10) [69].

Synthesis of **Cbz_3** proved to be more difficult than that of **Cbz_1** and **Cbz_2** since addition of a base (piperidine) was required to generate the anion of indane-1,3-dione **31**. A reflux of 24 h in ethanol was also required to get the product in 86% yield (see Scheme 4). On the opposite, synthesis of **29** and **30** starting from the aldehydes **32** and **33** was easier, the different dyes being obtained with reaction yields ranging from 82 to 94% after only four hours of reflux.

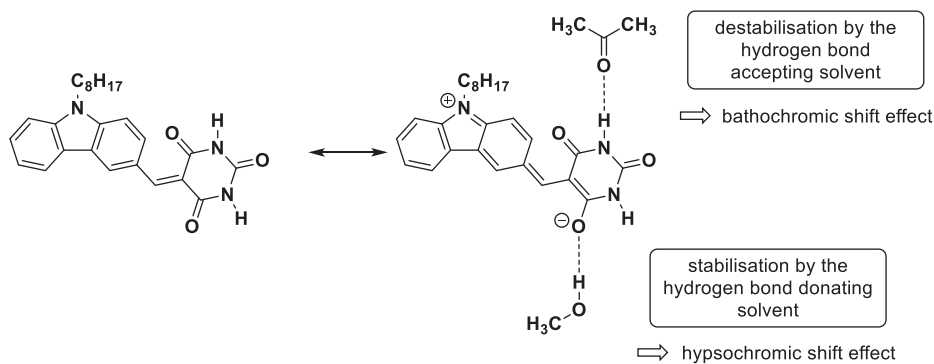
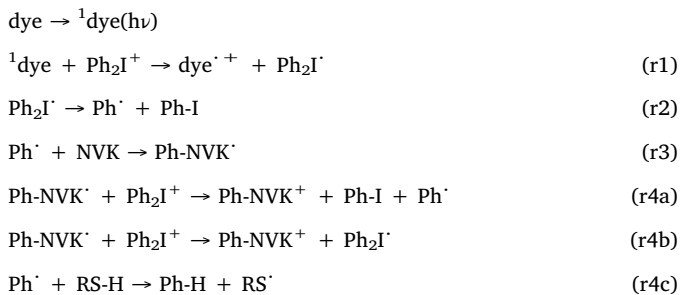
Compared to **Cbz_1** and **Cbz_2**, **Cbz_3** exhibited an absorption maximum located at 448 nm which is an intermediate situation between that of **Cbz_1** (431 nm) and **Cbz_2** (462 nm). Similarly, the molar extinction coefficient of **Cbz_3** ($42\,000\text{ M}^{-1}\cdot\text{cm}^{-1}$) is an intermediate situation between that of **Cbz_1** ($45\,000\text{ M}^{-1}\cdot\text{cm}^{-1}$) and **Cbz_2** ($37\,000\text{ M}^{-1}\cdot\text{cm}^{-1}$). Conversely, due to the weak electron donor, the absorption of **29** is UV-centered (351 nm, $29\,900\text{ M}^{-1}\cdot\text{cm}^{-1}$) whereas the absorption of **30** which possesses a stronger electron donor than **29** is centered in the visible range (467 nm, $7700\text{ M}^{-1}\cdot\text{cm}^{-1}$) but exhibit a low molar extinction coefficient. Once again, the crucial role of the electron donor on both the absorption maximum and the molar extinction coefficient was clearly evidenced. Polymerization of EPOX using the three-component system **Cbz_3**/NVK/ Ph_2I^+ (0.3%/3%/2% w/w) furnished a final monomer conversion of 59% after 800 s of irradiation with a laser diode at 457 nm ($100\text{ mW}/\text{cm}^2$) under air. This result is similar to that obtained with **Cbz_2** in the same conditions. While using a LED emitting at 385 nm, a final monomer conversion of 61% could be determined after 1000 s of irradiation under air, resulting from the high molar extinction coefficient of **Cbz_3** at this wavelength ($\epsilon > 10\,000\text{ M}^{-1}\cdot\text{cm}^{-1}$). Overall, the polymerization can be efficiently initiated in the 350–450 nm range, even with light sources of low light intensity as in the case of the LED bulb emitting at 385 nm ($9\text{ mW}/\text{cm}^2$). Comparison of the polymerization efficiencies of **29** and **30** in the same conditions revealed these two dyes to be unable to initiate the

polymerization of EPOX. Finally, polymerization of TMPTA using the same three-component system **Cbz_3**/NVK/ Ph_2I^+ (0.3%/3%/2% w/w) furnished a final monomer conversion of 58% while irradiating at 457 nm. Considering that $\text{Ph-NVK}^\cdot$ and Ph-NVK^+ species are formed during polymerization, the concomitant cationic/radical photopolymerization of an EPOX/TMPTA blend (50%/50% w/w) under air and in laminate was examined (See Fig. 11).

As anticipated, under air, the FRP of TMPTA was inhibited by the presence of oxygen (See Fig. 11a). Besides, the final monomer conversion of 50% after 5 min highlights the remarkable photoinitiating ability of the **Cbz_3**-based system, this result being obtained with a LED at 462 nm of low light intensity ($10\text{ mW}/\text{cm}^2$). In laminates i.e. in oxygen-free conditions, the monomer conversion could be increased up to 70%, producing tack-free polymers (See Fig. 11b).

Classically, thiol-ene photopolymerizations is carried out with UV light [105–108]. In the present case, the polymerization profiles obtained for the copolymerization of trithiol/DVE-3 upon irradiation with a laser diode at 457 nm ($100\text{ mW}/\text{cm}^2$) or a LED bulb at 462 nm ($10\text{ mW}/\text{cm}^2$) were similar, even though the light intensity of the LED was 10 times lower than that of the laser diode (See Fig. 12). It therefore demonstrates the high reactivity of the two-component **Cbz_3**/ Ph_2I^+ system. Samples with thickness ranging between 1 and 5 mm could also be polymerized using this approach.

In line of the mechanism proposed for the FRP of acrylates or the FRCPCP of epoxides (See before), the following mechanism was proposed to support the thiol-ene photopolymerization of the DVE-3/trithiol blend. Notably, the phenyl radical $\text{Ph}^\cdot$ can induce a hydrogen abstraction from the thiol monomer, producing $\text{RS}^\cdot$ than can subsequently add onto the vinyl groups of DVE-3, initiating the polymerization process.



Scheme 3. The different parameters affecting the solvatochromic behavior of **Cbz_1**.

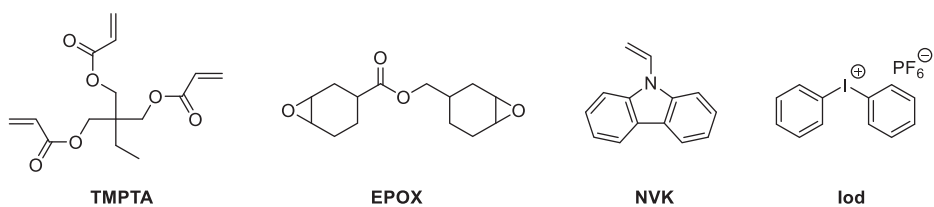


图9. 与Cbz 1和Cbz 2共同使用的不同单体及添加剂的化学结构。 - -

使用推拉染料作为光引发剂时，只有那些溶剂化显色现象可以用经典溶剂化显色标度合理化的染料才能引发聚合过程。最后，作者证明，只需检查新合成的推拉染料的溶剂化显色性，就能有效预测这些发色团的光引发能力。

考虑到这类染料存在强大的结构/反应活性关系，另一种基于咪唑的染料**Cbz_3**与另外三种染料**29–30**（具有相同的电子受体）一起进行了考察（参见图10） [69]。

合成**Cbz_3**比合成**Cbz_1**和**Cbz_2**更困难，因为需要添加碱（哌啶）来生成茚-1,3-二酮**31**的阴离子。还需要在乙醇中回流24小时，才能以86%的产率获得产物（参见方案4）。相反，从醛**32**和**33**开始合成**29**和**30**更容易，不同的染料仅需回流四小时，反应产率即可达到82%至94%。

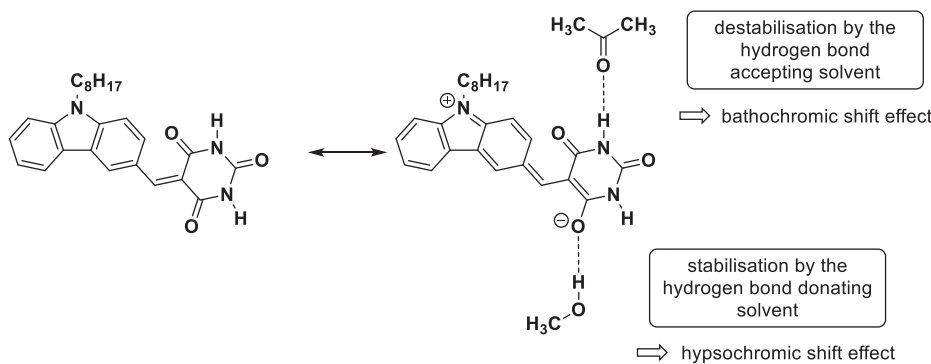
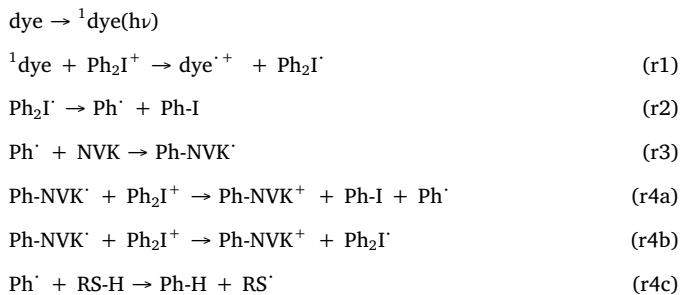
与**Cbz_1**和**Cbz_2**相比，**Cbz_3**的吸收最大值位于448 nm，介于**Cbz_1** (431 nm)和**Cbz_2** (462 nm)之间。同样，**Cbz_3**的摩尔消光系数($42\,000\text{ M}^{-1}\cdot\text{cm}^{-1}$)也介于**Cbz_1**($45\,000\text{ M}^{-1}\cdot\text{cm}^{-1}$)和**Cbz_2** ($37\,000\text{ M}^{-1}\cdot\text{cm}^{-1}$)之间。相反，由于供电子能力弱，化合物**29**的吸收集中在紫外区(351 nm, $29\,900\text{ M}^{-1}\cdot\text{cm}^{-1}$)，而化合物**30**由于具有比**29**更强的供电子基团，其吸收位于可见光区(467 nm, $7700\text{ M}^{-1}\cdot\text{cm}^{-1}$)，但摩尔消光系数较低。再次证明，供电子基团对吸收最大值和摩尔消光系数均起关键作用。使用三组分体系**Cbz_3**/NVK/ Ph_2I^+ (0.3%/3%/2% w/w)在空气下以457 nm激光二极管($100\text{ mW}/\text{cm}^2$)照射，EPOX聚合800秒后单体最终转化率为59%。该结果与相同条件下使用**Cbz_2**得到的结果相似。当使用385 nm LED灯时，在空气下照射1000秒后单体最终转化率为61%，这是由于**Cbz_3**在该波长下具有高摩尔消光系数($\epsilon > 10\,000\text{ M}^{-1}\cdot\text{cm}^{-1}$)。总体而言，即使在低光强光源（如385 nm LED灯泡， $9\text{ mW}/\text{cm}^2$ ）下，该聚合反应也能在350–450 nm范围内有效引发。在相同条件下比较**29**和**30**的聚合效率，发现这两种染料无法引发

EPOX的聚合。最后，使用相同三组分体系 **Cbz_3**/NVK/ Ph_2I^+ (0.3%/3%/2% w/w)在457 nm照射下，TMPTA聚合的最终单体转化率为58%。考虑到聚合过程中会形成 $\text{Ph-NVK}^\cdot$ 和 Ph-NVK^+ 物种，在空气和层压条件下研究了EPOX/TMPTA共混物(50%/50% w/w)的同步阳离子/自由基光聚合（参见图11）。

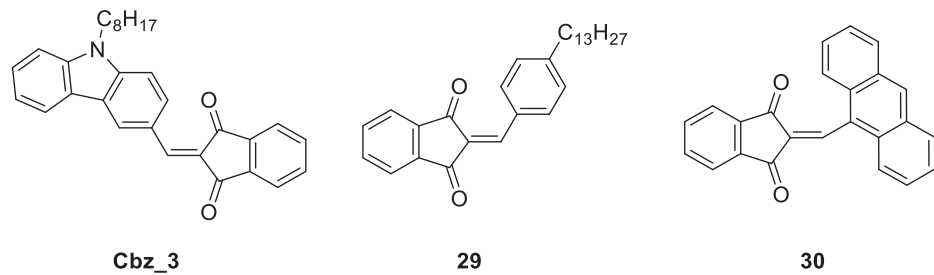
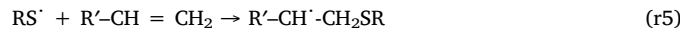
正如预期的那样，在空气条件下，TMPTA的FRP受到氧气的抑制（参见图11a）。此外，5分钟后50%的单体最终转化率突显了**Cbz_3**体系卓越的光引发能力，这一结果是在低光强度（ $10\text{ mW}/\text{cm}^2$ ）的462 nm LED下获得的。在层压条件下（即无氧环境下），单体转化率可提高到70%，得到无粘性聚合物（参见图11b）。

经典上，硫醇-烯光聚合是用紫外光进行的 [105–108]。在本例中，使用457 nm激光二极管（ $100\text{ mW}/\text{cm}^2$ ）或462 nm LED灯泡（ $10\text{ mW}/\text{cm}^2$ ）照射三硫醇/DVE-3共聚合得到的聚合曲线相似，即使LED的光强度比激光二极管低10倍（见图12）。因此，这证明了双组分**Cbz_3**/ Ph_2I^+ 体系的高反应性。使用该方法，厚度在1至5 nm之间的样品也可聚合。

根据前面提出的丙烯酸酯FRP或环氧化物FRCPCP机理，提出了以下机理解释DVE-3/三硫醇共混物的硫醇-烯光聚合。值得注意的是，苯基自由基 $\text{Ph}^\cdot$ 可以从硫醇单体中夺取氢原子，生成 $\text{RS}^\cdot$ ，随后 $\text{RS}^\cdot$ 加成到DVE-3的乙烯基上，引发聚合过程。

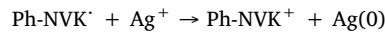


方案3. 影响Cbz_1溶剂化显色行为的不同参数。

Fig. 10. Chemical structures of **Cbz_3** and references compounds **29** and **30**.

While getting a deeper insight into the polymerization mechanism, examination of the photolysis of the **Cbz_3**/ Ph_2I^+ mixture in acetonitrile revealed the photobleaching to be relatively fast with this dye whereas no photobleaching was observed after 10 min for the **29**/ Ph_2I^+ and **30**/ Ph_2I^+ mixtures (See Fig. 13). These results are consistent with a back electron-transfer in r1 and the poor efficiency observed in photopolymerization for these two dyes.

Finally, the possibility to incorporate Ag nanoparticles into EPOX and TMPTA-based polymers was examined as it can deeply modify the mechanical properties of the resulting polymers. To generate these nanoparticles, a silver salt i.e. AgSbF_6 was used (See Fig. 14). Introduction of Ag nanoparticles was notably obtained by oxidation of $\text{Ph-NVK}^{\cdot}$ with this silver salt, reducing Ag^+ to Ag (0) according to the following equation:



In 2017, carbazole was revisited in the context of a new approach where carbazole was not used as an electron donor but as a conjugated spacer connecting electron-donating and accepting groups introduced on both sides of its skeleton. In this innovative approach, nitro, dialkylamino and aldehyde groups were introduced as peripheral groups of **Cbz_4-Cbz_7** (See Fig. 15) [109]. As specificity, the four carbazole derivatives **Cbz_4-Cbz_7** could be used as photoredox catalysts so that low photoinitiator contents could be used. To evidence the validity of this approach, a comparison with two benchmark photoinitiators, namely *bis*(2,4,6-trimethylbenzoyl)phenylphosphine oxide (BAPO) and 9*H*-carbazole-9-ethanol (CARET) was established.

As first interesting feature, UV-visible absorption spectra of **Cbz_4**-

Cbz_7 in dichloromethane revealed **Cbz_7** to exhibit the most extended absorption, this latter ranging from 350 to 500 nm. This broad absorption band can be assigned to the simultaneous presence of both the electron accepting nitro groups and the electron-donating alkylamino groups, inducing an efficient electronic delocalization. For the other dyes (**Cbz_4-Cbz_6**), an absorption extending between 350 and 440 nm was only found (See Fig. 16). Finally, comparison of the absorption spectra of **Cbz_4-Cbz_7** with that of CARET and BAPO revealed the absorption of CARET to be inexistent between 360 and 550 nm whereas a low absorption was detected for BAPO between 350 and 450 nm.

Interestingly, the two-component systems **Cbz_4-Cbz_7**/ $(t\text{-Bu})\text{Ph}_2\text{I}^+$ (0.5%/1% w/w) showed a high reactivity for the polymerization of EPOX upon irradiation with a LED at 405 nm (110 mW/cm²) since final monomer conversions of 76, 50, 58 and 70% could be obtained with **Cbz_4-Cbz_7** respectively. Interestingly, no direct correlation with the molar extinction coefficients and the monomer conversions was found, the best performances being obtained with the dye exhibiting the lowest absorption at 405 nm. In fact, authors tentatively assigned the higher photoinitiating ability of **Cbz_4** compared to **Cbz_6** to the bulkiness of the octyl chain in **Cbz_6**, the two dyes only differing by the length of the alkyl chain on the central amine. During polymerization and the formation of the crosslinked polymer, the radical cation on **Cbz_4** remains more available than that of **Cbz_6** due to the smaller size of the alkyl chain, enabling to reach in turn higher final monomer conversions. Besides, in this work, the usual three-component systems **Cbz_x** ($x = 4\text{--}7$)/ $(t\text{-Bu})\text{Ph}_2\text{I}^+$ /NVK was not examined for the polymerization of EPOX, due to the exceptional efficacy of the two-component systems.

Compared to the final monomer conversions obtained with

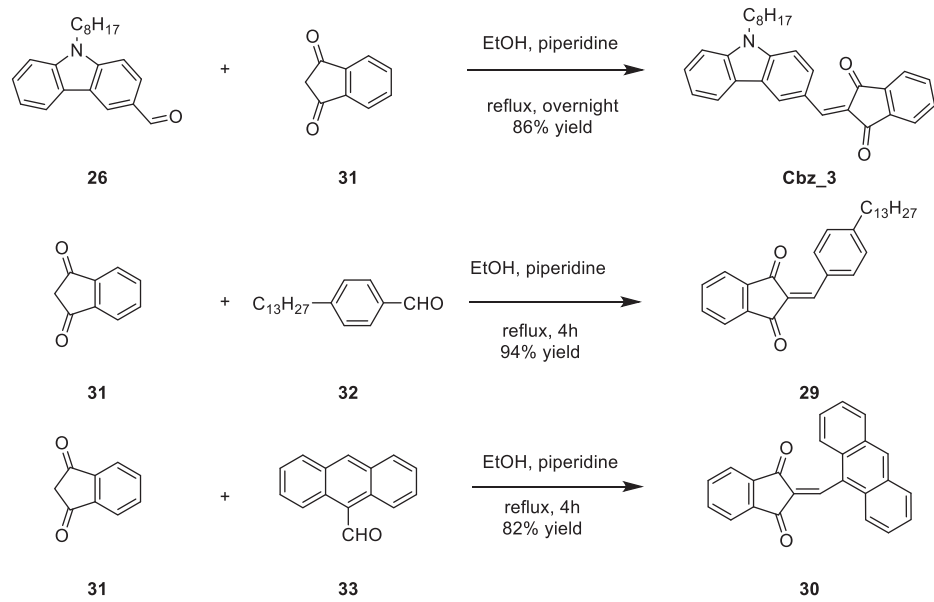
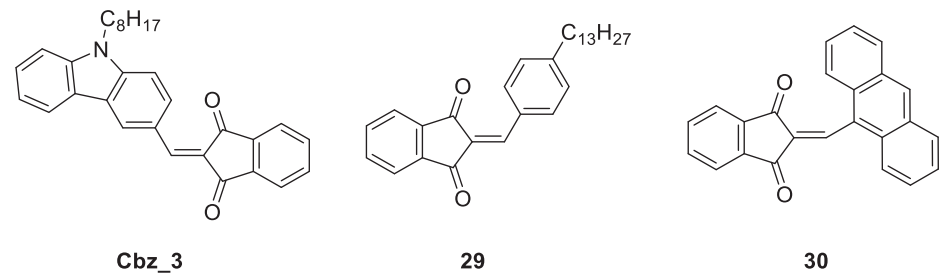
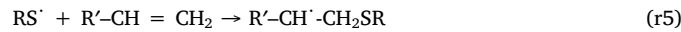
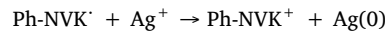
Scheme 4. Synthetic routes to **Cbz_3** and push-pull dyes **29** and **30**.

图10. Cbz_3 和参考化合物29和30的化学结构。



在更深入了解聚合机理时，对乙腈中**Cbz_3**/ Ph_2I^+ 混合物的光解研究发现，该染料的光漂白相对较快，而**29**/ Ph_2I^+ 和**30**/ Ph_2I^+ 混合物在10分钟后未观察到光漂白（见图13）。这些结果与r1中的反向电子转移以及这两种染料在光聚合中表现出的低效率一致。

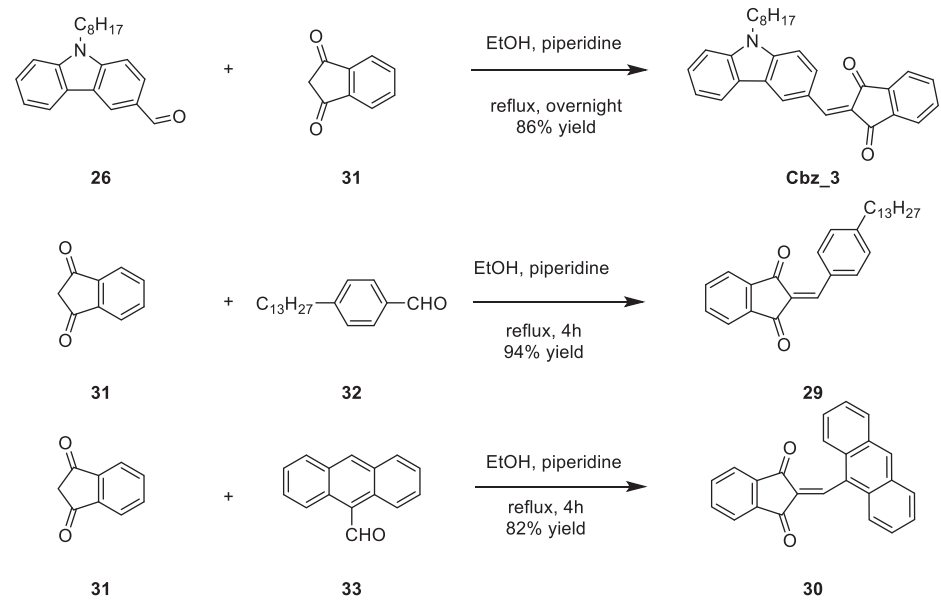
最后，研究了将Ag纳米粒子掺入EPOX和TMPTA基聚合物的可能性，因为这会深刻影响所得聚合物的机械性能。为生成这些纳米粒子，使用了银盐 AgSbF_6 。（参见图14）。Ag纳米粒子的引入主要通过 $\text{Ph-NVK}^{\cdot}$ 与这种银盐的氧化反应实现，将 Ag^+ 还原为 $\text{Ag}(0)$ ，反应式如下：



2017年，咪唑在一种新方法中被重新审视，其中咪唑不作为电子供体，而是作为共轭间隔基，连接引入其骨架两侧的给电子和吸电子基团。在这种创新方法中，硝基、二烷基氨基和醛基被作为外围基团引入**Cbz_4-Cbz_7**（参见图15）[109]。作为特点，四种咪唑衍生物**Cbz_4-Cbz_7**可用作光氧化还原催化剂，从而可以使用低光引发剂含量。为证明该方法的有效性，与两种基准光引发剂，即双(2,4,6-三甲基苯甲酰基)苯基氧化膦（BAPO）和9*H*-咪唑-9-乙醇（CARET）进行了比较。

首先值得注意的是，**Cbz_4**-

与最终单体转化率相比，



方案4. Cbz_3 和推拉型染料29和30的合成路线。

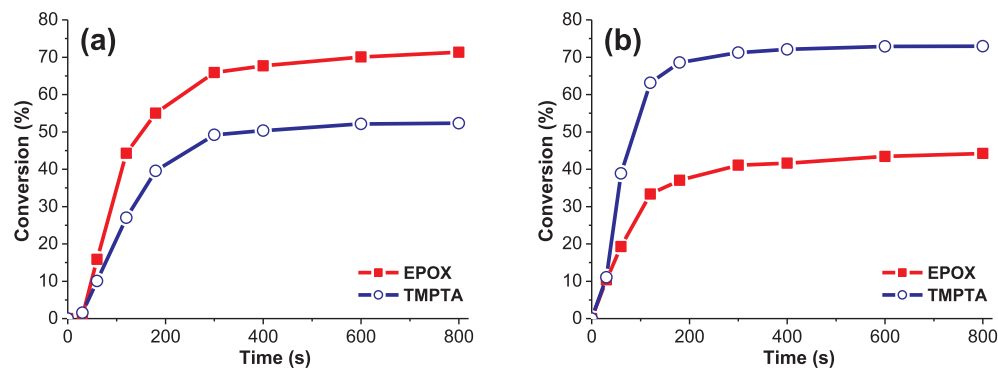


Fig. 11. Photopolymerization profiles of an EPOX/TMPTA blend (50%/50%, w/w) in the presence of **Cbz_3**/Ph₂I⁺/NVK (0.5%/2%/3%, w/w/w) under air (a) and in laminate (b) upon irradiation with a LED bulb at 462 nm. Reproduced with permission from Xiao et al. [69]. Copyright 2014 American Chemical Society.

carbazoles **Cbz_4-Cbz_7**, the final conversion obtained with BAPO was extremely low, only peaking at 15%. While decreasing the photo-initiator content from 1 to 0.5%, an improvement of ca 8% for the monomer conversion of EPOX could be obtained with **Cbz_6**. This improvement can be confidently assigned to internal filter effects at high photoinitiator concentrations, this compound being particularly colored [110,111]. Face to the high reactivity of the two-component systems, surface patterning with an epoxide resin was carried out. By using a LED projector emitting at 405 nm (100 mW/cm²) and by using the two-component system **Cbz_5**/Iod (0.5%/1% w/w), different structures of high resolution could be obtained, as shown in the Fig. 17.

Remarkable efficiencies were also obtained for the polymerization of TMPTA in laminates and similar monomer conversions were obtained with all dyes, peaking around 50%. While examining the polymerization of a methacrylic resin i.e. the dental resin composed of a mixture of bisphenol A-glycidyl methacrylate (BisGMA) and triethylene glycol dimethacrylate (TEGDMA) in a 70%/30% w/w ratio, a significant improvement of the monomer conversion was obtained upon addition of a sacrificial amine to the two-component system, namely ethyl 4-(dimethylamino)benzoate (EDB). While using the three-component system **Cbz_5**/(t-Bu)Ph₂I⁺/EDB (0.5%/1%/1%), the final monomer conversion could be increased up to 65% contrarily to 43% for the two-component system **Cbz_5**/(t-Bu)Ph₂I⁺ upon irradiation with a diode at 455 nm. Fluorescence quenching experiments carried out on the **Cbz_5**/EDB mixture revealed a fast quenching process, consistent with the formation of EDB[·] capable to initiate the FRP of acrylates. In this context, based on the different photolysis experiments and due to the major enhancement of the final monomer conversion

upon addition of EDB, authors concluded to the regeneration of the dye by oxidation of the sacrificial amine (See Scheme 5). Parallel to this, the excited dye can also react with EDB, producing EDB[·]. By mean of the coexisting oxidative and the reductive cycle, Ar[·] and EDB[·] are both generated, improving the initiation step and in turn the FRP of (meth) acrylates.

2.2. Carbazoles of extended aromaticity

Parallel to the design of push-pull dyes, another strategy developed with carbazoles consisted in improving the molar extinction coefficients by extending the aromaticity. If this strategy is efficient with regards to the molar extinction coefficient, it is also ineffective to red-shift the UV-centered absorption of carbazole towards the visible region. To illustrate this, substitution of carbazole with pyrene moieties enabled to produce **Cbz_8** exhibiting a maximum absorption at 346 nm with a molar extinction coefficient of 55 000 M⁻¹cm⁻¹, therefore, greatly higher than what could be obtained with pure carbazole. Examination of its absorption spectrum also revealed this compound to strongly absorb between 300 and 410 nm, the tail not extending beyond 410 nm (See Fig. 18).

On the opposite, **39** and **40** exhibited broader absorption spectra due to the simultaneous presence of three pyrene units, increasing the molar extinction coefficient. Interestingly, comparison of the photo-initiating abilities of **Cbz_8** with that of a series of eleven pyrene-based photoinitiators **34-44** revealed once again the carbazole derivative **Cbz_8** to outperform the others (See Fig. 19) [52].

While polymerizing EPOX with a Xe-Hg lamp and two-component

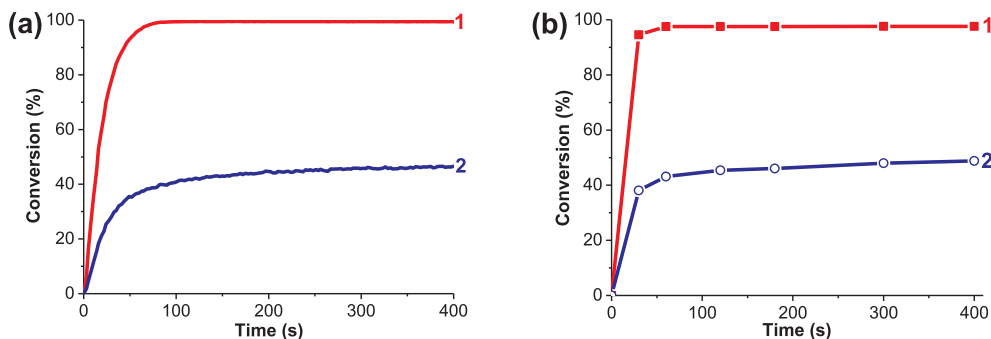
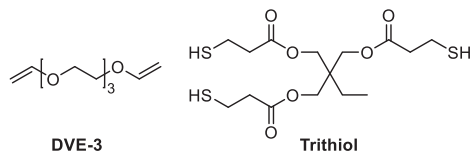


Fig. 12. Photopolymerization profiles of trithiol/DVE-3 blend (57%/43%, w/w) in laminate in the presence of **Cbz_3**/Iod (0.5%/2%, w/w) upon irradiation with a laser diode at 457 nm (a) and a LED bulb at 462 nm (b) curve 1: DVE-3 (vinyl double bond) conversion, curve 2: trithiol (S-H) conversion. Top: Chemical structures of the two monomers. Reproduced with permission from Xiao et al. [69]. Copyright 2014 American Chemical Society.

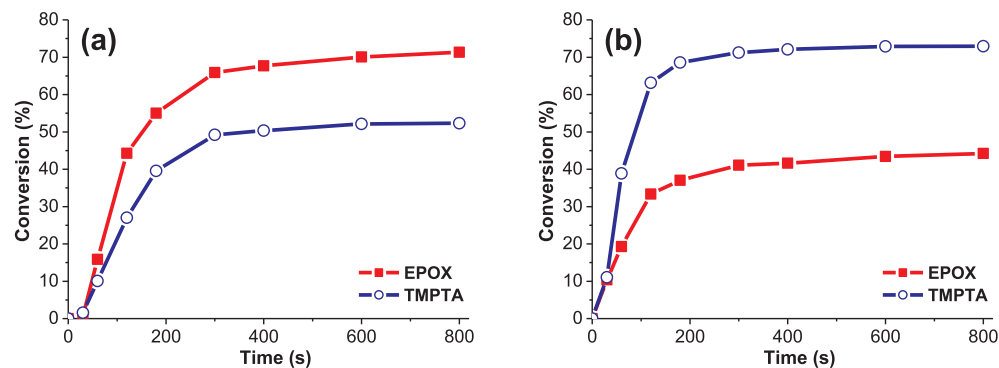


图 11. 在空气 (a) 和层压 (b) 条件下, 使用 462 nm LED 灯泡照射, EPOX/TMPTA 共混物 (50%/50%, w/w) 在 **Cbz_3**/Ph₂I⁺/NVK (0.5%/2%/3%, w/w/w) 存在下的光聚合曲线。经 Xiao 等人许可转载。 [69]。版权所有 2014 美国化学学会。

唑啉 **Cbz_4-Cbz_7**, 最终转化率使用BAPO时极低, 仅峰值达到15%。当光引发剂含量从1%降至0.5%时, 使用**Cbz6**可使EPOX单体转化率提高约8%。这一提</高可明确归因于高光引发剂浓度时的内部滤光效应, 该化合物颜色尤为显著 [110,111]。面对双组分体系的高反应活性, 使用环氧树脂进行了表面图案化。通过使用发射405 nm (100 mW/cm²) 的LED投影仪², 并采用双组分体系**Cbz_5**/Iod (0.5%/1% w/w), 可获得不同分辨率的高分辨率结构, 如图17所示。

对于TMPTA在层压板中的聚合也获得了显著的效率, 所有染料的单体转化率类似, 峰值约为50%。在检查甲基丙烯酸树脂 (即由双酚A-甲基丙烯酸缩水甘油酯 (BisGMA) 和双甲基丙烯酸三甘醇酯 (TEGDMA) 以70%/30%重量比组成的牙科树脂) 的聚合时, 向双组分体系中添加牺牲胺 (即4-(二甲氨基)苯甲酸乙酯 (EDB)) 显著提高了单体转化率。使用三组分体系**Cbz_5**/(t-Bu)Ph₂I⁺/EDB (0.5%/1%/1%) 时, 最终单体转化率可提高至65%, 而双组分体系**Cbz_5**/(t-Bu)Ph₂I⁺在455 nm二极管照射下的转化率仅为43%。对**Cbz_5**/EDB混合物进行的荧光猝灭实验揭示了快速猝灭过程, 与能够引发丙烯酸酯FRP的EDB[·] 的形成一致。在此背景下, 基于不同的光解实验以及由于最终单体转化率的显著提高

加入EDB后, 作者得出结论: 通过牺牲胺的氧化再生了染料 (参见 方案 5)。与此同时, 激发的染料也可以与EDB反应, 生成EDB[·]。通过共存的氧化循环和还原循环, Ar[·] 和EDB[·] 均被生成, 改善了引发步骤, 进而促进了 (甲基) 丙烯酸酯的FRP。

2.2. 扩展芳香性的唑啉

与推拉染料的设计平行, 另一种基于唑啉的策略是通过扩展芳香性来提高摩尔消光系数。虽然这一策略对于摩尔消光系数较为有效, 但它对于将唑啉的紫外中心吸收红移至可见光区域却无效。为了说明这一点, 用苊基团取代唑啉制备了 **Cbz_8**, 其最大吸收在346 nm, 摩尔消光系数为55 000 M⁻¹cm⁻¹, 因此远高于纯唑啉。对其吸收光谱的检查还显示, 该化合物在300至410 nm之间强烈吸收, 尾端不超过410 nm (参见图18)。

相反, **39** 和**40** 由于同时存在三个苊单元, 其吸收光谱更宽, 进一步提高了摩尔消光系数。有趣的是, 将**Cbz_8** 的光引发能力与一系列十一个基于苊的光引发剂**34-44** 进行比较, 再次显示唑啉衍生物**Cbz_8** 优于其他物质 (参见图19) [52]。

在使用Xe-Hg灯和双组分

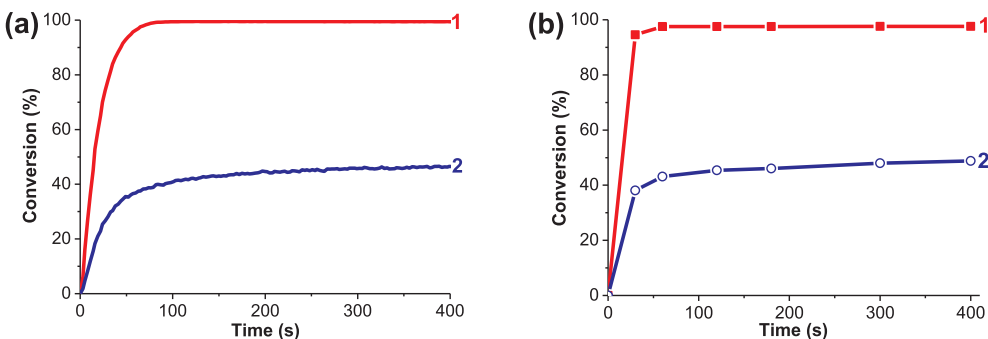
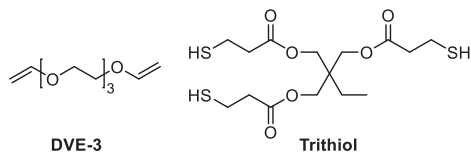


图 12.三硫醇/DVE-3 共混物 (57%/43%, w/w) 在 **Cbz_3**/Iod (0.5%/2%, w/w) 存在下, 分别在 457 nm 激光二极管 (a) 和 462 nm LED 灯泡 (b) 照射下的层压光聚合曲线: 曲线1为 DVE-3 (乙烯基双键) 转化率, 曲线2为三硫醇 (S-H) 转化率。上方: 两种单体的化学结构。经 Xiao 等人许可转载。 [69]。版权所有 2014 美国化学学会。

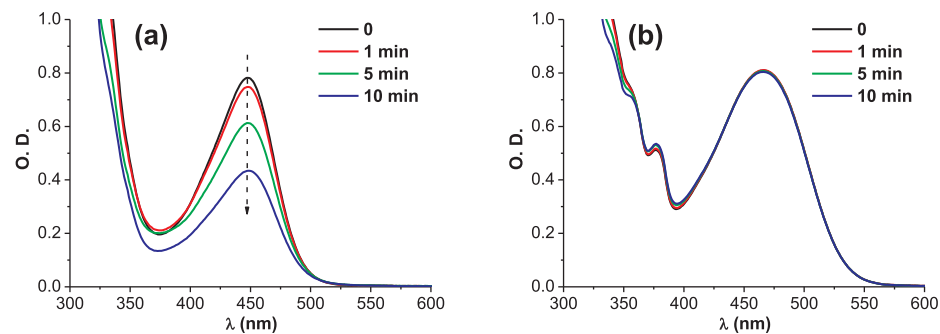


Fig. 13. Steady state photolysis of (a) **Cbz_3**/ Ph_2I^+ and (b) **30**/ Ph_2I^+ in acetonitrile upon the halogen lamp exposure. Reproduced with permission from Xiao et al. [69]. Copyright 2014 American Chemical Society.

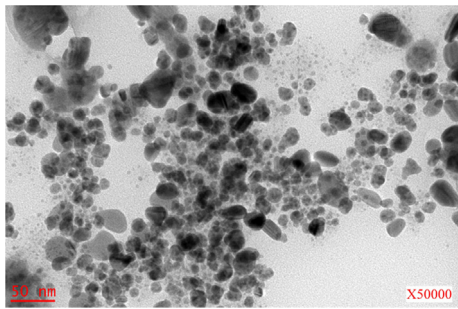


Fig. 14. TEM image for the polymerization of EPOX using the four-component system **Cbz_3**/ Ph_2I^+ /NVK/AgSbF₆ upon irradiation with a LED bulb at 462 nm. Reproduced with permission from Xiao et al. [69]. Copyright 2014 American Chemical Society.

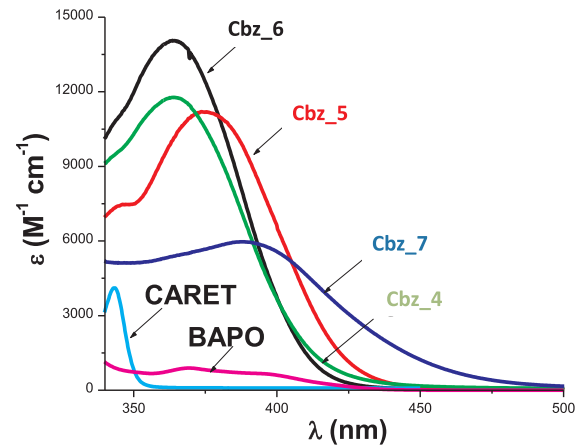


Fig. 16. Absorption spectra of the carbazole derivatives **Cbz_4**–**Cbz_7** in CH_3CN , and BAPO in CH_2Cl_2 . Reproduced with permission from Al Mousawi et al. [109]. Copyright 2017 American Chemical Society.

clear trend could be established concerning the structure-performance relationship. Notably, polymerization performances are the result of a subtle interplay between molar extinction coefficients, excited state lifetimes and rate constants of interaction between Iod and the dye (see equation below) and the ability of the dye^{•+} to initiate a ring-opening polymerization process.

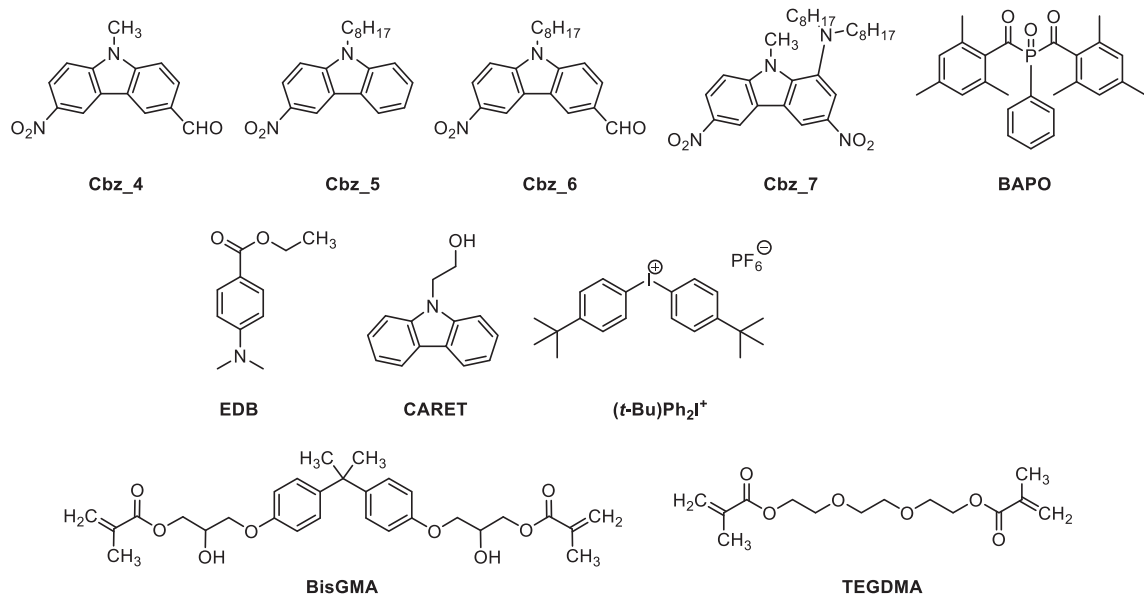


Fig. 15. Chemical structures of **Cbz_4**–**Cbz_7**.

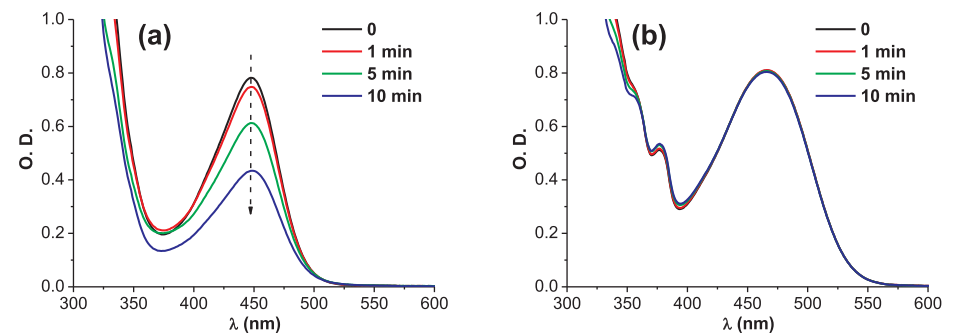


图13. 稳态光解 (a) **Cbz_3**/ Ph_2I^+ 和 (b) **30**/ Ph_2I^+ 在乙腈中经卤素灯照射。经Xiao等人许可转载。[69]。版权所有2014美国化学学会。

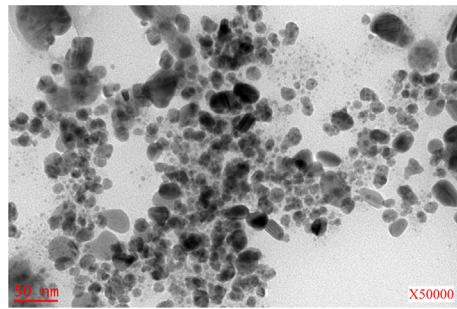


图14. 使用四组分体系**Cbz_3**/ Ph_2I^+ /NVK/AgSbF₆。在462 nm LED灯泡照射下聚合EPOX的TEM图像。经Xiao等人许可转载。[69]。版权所有2014美国化学学会。

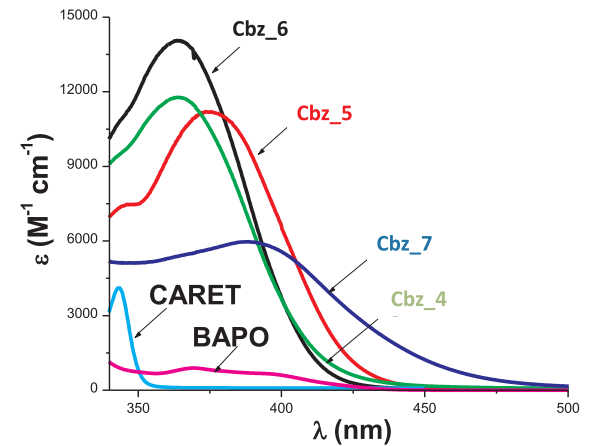


图16. 咔唑衍生物在 CH_3CN 中的吸收光谱**Cbz_4**–**Cbz_7**，以及BAPO在 CH_2Cl_2 中的吸收光谱。经Al Mousawi等人许可转载。[109]。版权所有2017美国化学学会。

光引发体系**Cbz_8** 或34–44的染料/Iod (0.2%/2% w/w) 聚合EPOX时，其中Iod代表 [甲基-4-苯基(甲基-1-乙基)-4-苯基]碘鎓四(五氟苯基)硼酸盐 (Iod)，经过200秒-光照后，Cbz 8的单体最终转化率达到86%，高于36 (78%) 并显著高于本研究中用作参考化合物的苝34 (50%)。鉴于这十一种染料是苝34的衍生物，Cbz_8、35–44与34相比光引发能力的显著差异证明了修饰这一基准光引发剂的价值。然而，在本工作中，没有

关于结构-性能关系，可以建立清晰的发展趋势。值得注意的是，聚合性能是摩尔消光系数、激发态寿命、碘鎓盐与染料相互作用速率常数（见下图）以及染料^{•+} 引发开环聚合能力之间微妙相互作用的结果。

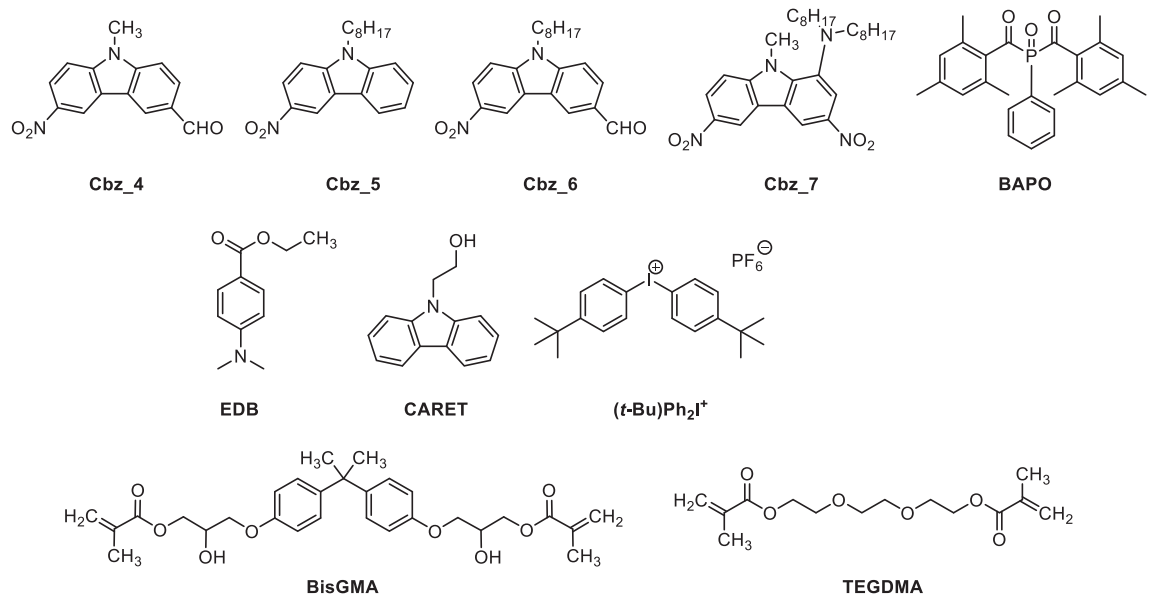


图15. **Cbz_4**–**Cbz_7**的化学结构。

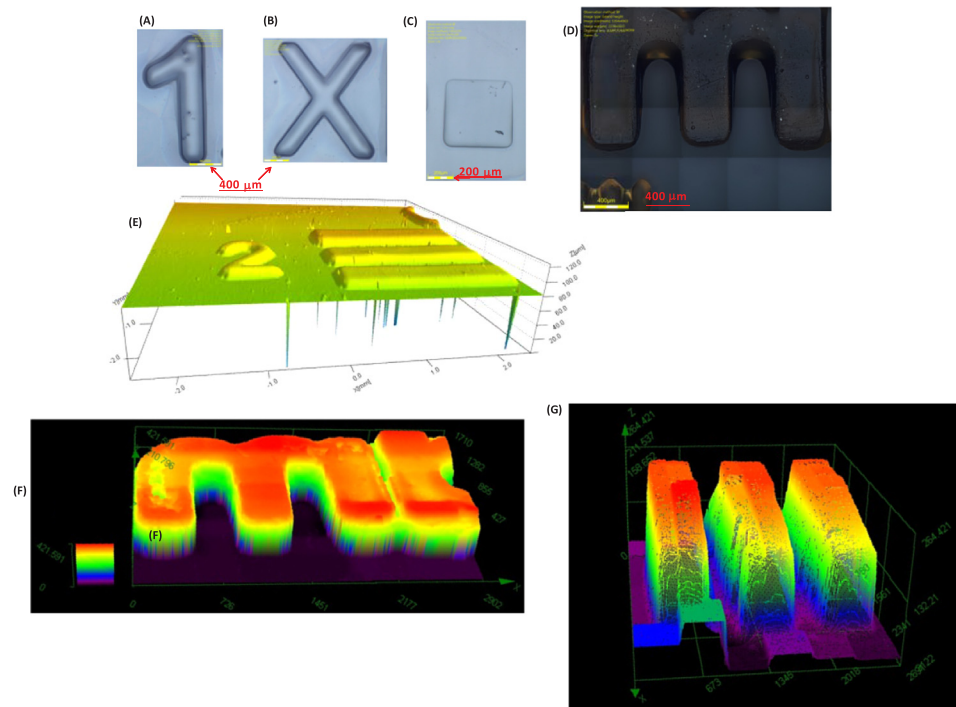
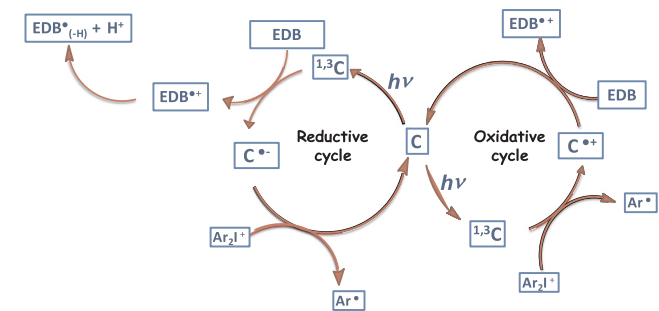


Fig. 17. Different structures obtained by cationic photopolymerization experiments of an epoxide resin using a LED projector emitting at 405 nm. Reproduced with permission from Al Mousawi et al. [109]. Copyright 2017 American Chemical Society.



Scheme 5. Catalytic cycles enabling the regeneration of the carbazole-based dye upon addition of EDB. Reproduced with permission from Al Mousawi et al. [109]. Copyright 2017 American Chemical Society.

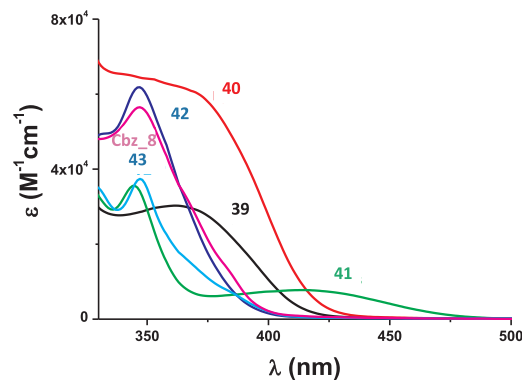
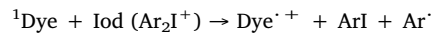


Fig. 18. UV–visible absorption spectra of **Cbz 8**, **39–43** in toluene. Reproduced with permission from Telitel et al. [52]. Copyright 2013 Beilstein J. Org. Chem.



While using a halogen lamp emitting beyond 380 nm, a final monomer conversion of 55% could be determined for EPOX with the

two-component system **Cbz 8**/Iod (0.2%/2% w/w), what could be increased up to 75% with the three-component system **Cbz 8**/Iod/NVK (0.2%/2%/2% w/w). This remarkable monomer conversion is notably attributable to the photocatalytic behavior of the three-component system with the formation of the highly reactive $\text{Ar-NVK}^{\cdot}$. Surprisingly, a low storage stability was found for the resins based on the two-component systems dye/Iod (1%/2% w/w) and incorporating **39**, **42**, **43** and **Cbz 8**. For these resins, a thermal polymerization occurring at room temperature could be evidenced and after 24 h of storage, a complete polymerization of the monomer was demonstrated. This unwanted polymerization at room temperature and in the dark constitutes a severe limitation of the future use of these pyrene derivatives as photoinitiators.

More stable formulations were obtained with the **Cbz 9**-**Cbz 15** series which differ from the previous one by the substitution pattern (See Fig. 20) [112]. In this work, an extension of conjugation by mean of a double bond was proposed to enhance the molar extinction coefficients while maintaining the absorption in the 300–400 nm region by the lack of intramolecular push-pull effects. Using this strategy, the molar extinction coefficient of **Cbz 13** could be increased up to $35\,000\text{ M}^{-1}\text{cm}^{-1}$ at 340 nm, far from that of 9H-carbazole-9-ethanol (CARET) ($4000\text{ M}^{-1}\text{cm}^{-1}$ at 340 nm). Benefiting from a significant absorption at 405 nm, **Cbz 9** ($1000\text{ M}^{-1}\text{cm}^{-1}$) and **Cbz 15** ($1000\text{ M}^{-1}\text{cm}^{-1}$) could outperform BAPO ($500\text{ M}^{-1}\text{cm}^{-1}$) for the polymerization of EPOX (44% of monomer conversion after 200 s of irradiation for **Cbz 9** and **Cbz 15** contrarily to 15% of conversion with BAPO) using a two-component system **Cbz x** ($x = 9, 15$)/(t-Bu) Ph_2I^+ (0.5%/1% w/w) (See Fig. 21).

The superiority of **Cbz 15** over the other carbazoles of this series was revealed during the FRP of acrylates. Upon irradiation with a LED at 405 nm, a final conversion of 57% was determined with the two component system **Cbz 15**/ (t-Bu) Ph_2I^+ (0.5%/1% w/w) after 100 s of irradiation. Conversely, a final conversion of only 46% was obtained in the same conditions with **Cbz 9**, attributable to a lower electron transfer quantum yield with (t-Bu) Ph_2I^+ . Besides, the order of reactivity determined during the CP of epoxides was confirmed with the FRP of acrylates. Interestingly, due to the lack of push-pull effects

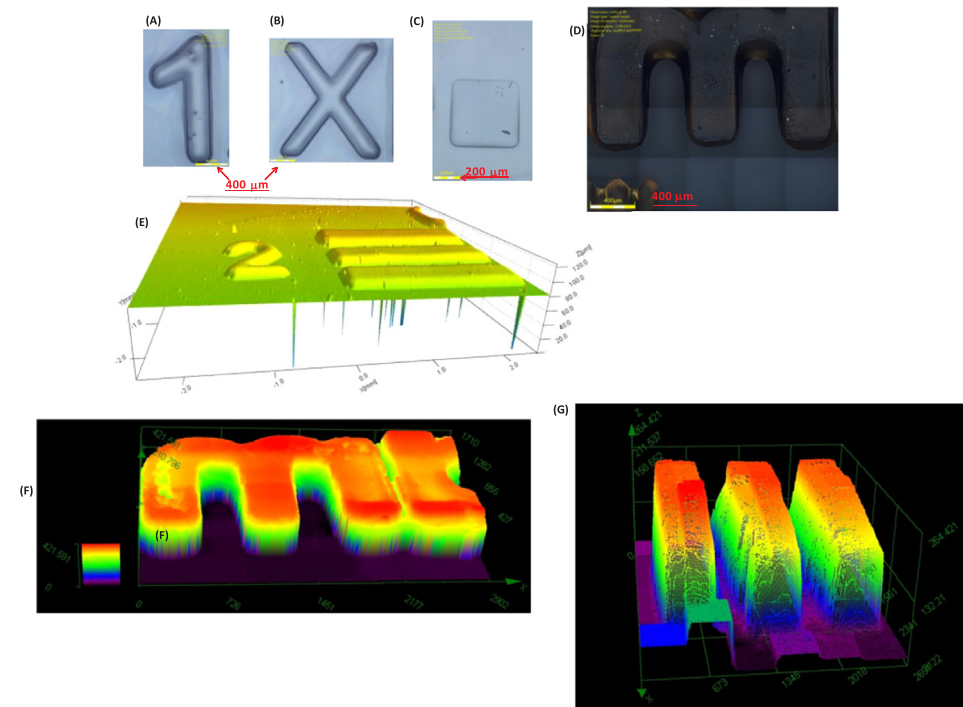
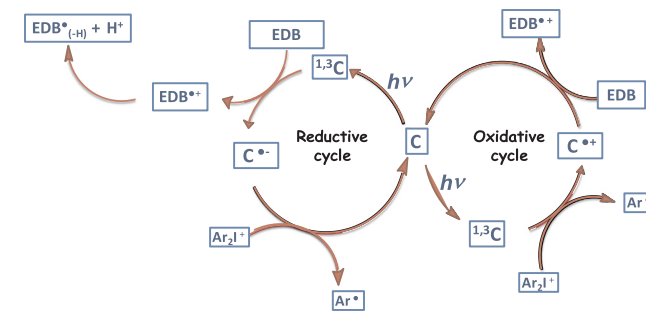


图 17. 通过使用发射波长为 405 nm 的 LED 投影仪对环氧树脂进行阳离子光聚合实验获得的不同结构。经 Al Mousawi 等人许可转载。[109]. 版权所有 2017 美国化学学会。



方案 5. 催化循环使得基于咔唑的染料在添加 EDB 后得以再生。经 Al Mousawi 等人许可转载。[109]. 版权所有 2017 美国化学学会。

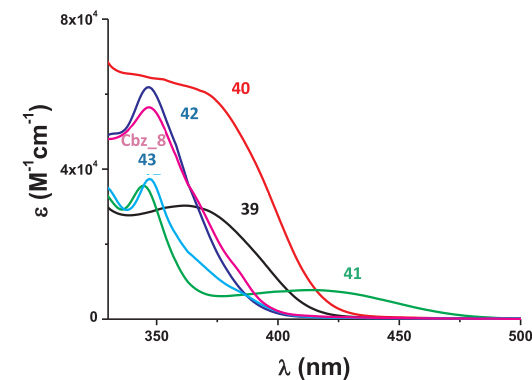
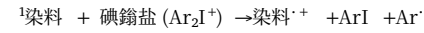


图 18. Cbz 的紫外-可见吸收光谱 **8**, **39–43** 在甲苯中。经 Telitel 等人许可转载。[52]. 版权所有 2013 Beilstein J. Org. Chem.



当使用发射波长超过 380 nm 的卤素灯时，EPOX 的最终单体转化率可达到 55%，使用

双组分体系 **Cbz 8**/碘鎓盐 (0.2%/2% w/w) 的单体转化率可提升至 75%——这是通过三组分体系 **Cbz 8**/碘鎓盐/NVK (0.2%/2%/2% w/w) 实现的。这种显著的单体转化率主要归因于三组分体系的光催化行为，形成了高反应活性的 $\text{Ar-NVK}^{\cdot}$ 。令人惊讶的是，基于双组分体系染料/碘鎓盐 (1%/2% w/w) 并加入 **39**、**42**、**43** 和 **Cbz 8** 的树脂，其储存稳定性较低。这些树脂在室温下会发生热聚合，且储存 24 小时后单体完全聚合。这种在室温及暗处的非预期聚合严重限制了这些衍生物未来作为光引发剂的应用。

通过改变取代模式 (见图 20)，**Cbz 9**-**Cbz 15** 系列获得了更稳定的配方 [112]。本研究提出通过双键延长共轭，旨在提高摩尔消光系数，同时由于缺乏分子内推拉效应，保持吸收在 300–400 nm 区域。采用这一策略，**Cbz 13** 在 340 nm 处的摩尔消光系数可提高至 $35,000\text{ M}^{-1}\text{cm}^{-1}$ ，远超 9H-咔唑-9-乙醇 (CARET) ($4000\text{ M}^{-1}\text{cm}^{-1}$, 340 nm)。得益于在 405 nm 处的显著吸收，**Cbz 9** ($1000\text{ M}^{-1}\text{cm}^{-1}$) 和 **Cbz 15** ($1000\text{ M}^{-1}\text{cm}^{-1}$) 在 EPOX 聚合中可超越 BAPO ($500\text{ M}^{-1}\text{cm}^{-1}$) (**Cbz 9** 和 **Cbz 15** 辐照 200 秒后单体转化率 44%，而 BAPO 仅为 15%)，使用双组分体系 **Cbz x** ($x = 9, 15$)/(t-Bu) Ph_2I^+ (0.5%/1% w/w) (见图 21)。

Cbz 15 在该系列咔唑类光引发剂中的优越性在丙烯酸酯的 FRP 中得以体现。在 405 nm LED 照射下，使用双组分体系 **Cbz 15**/(t-Bu) Ph_2I^+ (0.5%/1% w/w) 照射 100 秒后，最终转化率为 57%。相比之下，在相同条件下使用 **Cbz 9** 仅获得 46% 的最终转化率，这归因于与 (t-Bu) Ph_2I^+ 的电子转移量子产率较低。此外，在环氧化合物的 CP 中确定的反应活性顺序与丙烯酸酯的 FRP 结果一致。有趣的是，由于缺乏推拉效应

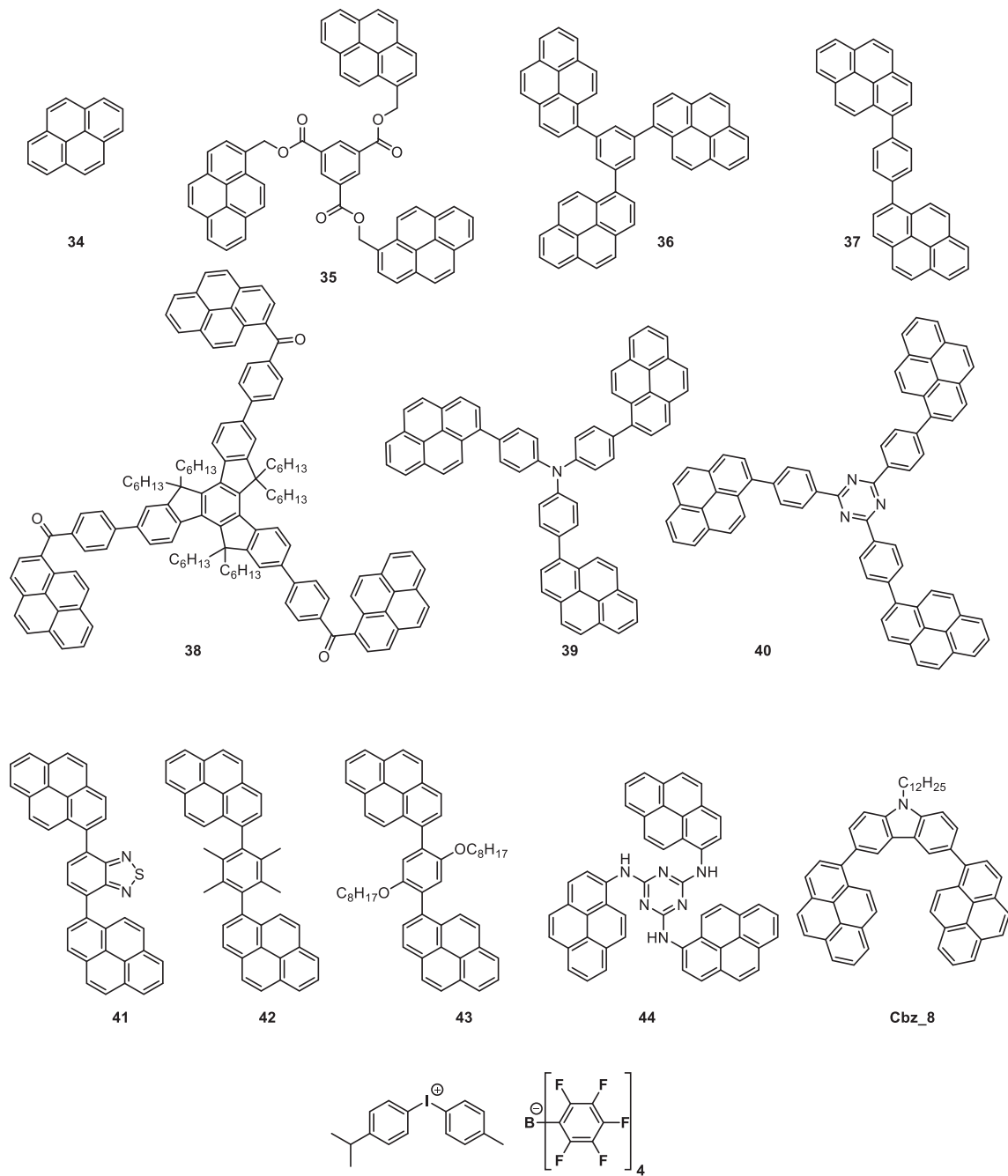


Fig. 19. Chemical structure of Cbz_8 and 34–44.

within **Cbz_9-Cbz_15**, colorless polymer films could be obtained with these different photoinitiators (See Fig. 21).

Finally, during the polymerization of the TMPTA/EPOX blend with the two component system **Cbz_15**/(*t*-Bu)Ph₂I⁺ (0.5%/1% w/w), finally monomer conversions of 76 and 23% were respectively determined for the conversion of TMPTA and EPOX. The three-fold enhancement of the TMPTA conversion relative to that of EPOX was assigned to the higher reactivity of the radicals compared to the cationic species [113]. Similar conclusions concerning CP and FRP could be established with another series of carbazole derivatives **Cbz_16-Cbz_21** (See Fig. 22) [114]. Here again, the higher final monomer conversions obtained with **Cbz_17** and **Cbz_20** during the FRP of acrylate or the FRPCP of epoxides was directly related their higher molar extinction coefficients. Interestingly, the best performances were

obtained with the carbazoles substituted with an aldehyde function. In this case, a small electronic delocalization exists between the nitrogen and the aldehyde function, inducing a small push-pull effect and thus enhancing the absorption.

To extend the π -conjugation, another strategy consisted in introducing electron donating groups such as triphenylamine as peripheral groups of carbazole [115]. In this case, the adjunction of electron-donating groups to an electron donor cannot induce a push-pull effect within the molecules, no electronic interaction being possible between electron donors. As a consequence of this, the absorption positions of **Cbz_22** and **Cbz_23** were not significantly modified compared to that of an unsubstituted carbazole whereas an enhancement of the absorption coefficients could be obtained for the two dyes. By the presence of the numerous aromatic rings in **Cbz_22** and **Cbz_23**, increase of the molar

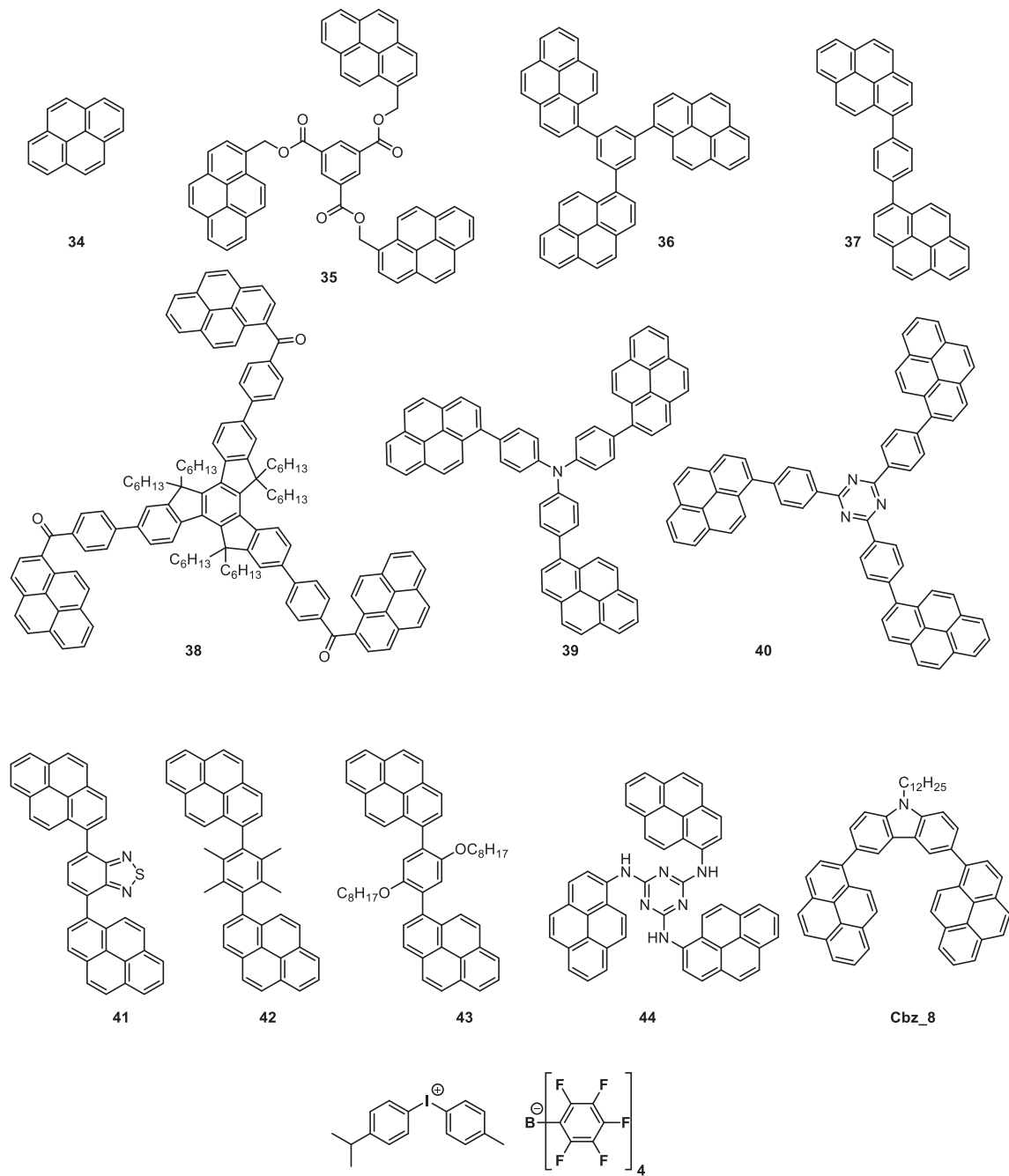


图 19. Cbz_8 和34–44的化学结构。

在**Cbz_9-Cbz_15**中，可以获得无色聚合物膜薄膜，使用这些不同的光引发剂（参见不同的光引发剂（参见图21）。

最终，在TMPTA/EPOX混合物与双组分体系聚合时**Cbz_15**/(*t*-Bu)Ph₂I⁺ (0.5%/1% w/w)，最终测定的TMPTA和EPOX单体转化率分别为76%和23%。TMPTA转化率相对于EPOX的三倍增强被归因于自由基比阳离子物种具有更高的反应活性。 [113]关于CP和FRP的类似结论可以通过另一系列咪唑衍生物得出。**Cbz16-Cbz_21** （参见图22） [114]。同样地，在丙烯酸酯的FRP或环氧化物的FRPCP中，使用**Cbz_17** 和**Cbz_20** 获得的更高最终单体转化率直接与其更高的摩尔消光系数相关。有趣的是，最佳性能是在

带有醛基取代的咪唑上获得的。在这种情况下，氮原子与醛基之间存在微弱的电子离域，产生微小的推拉效应，从而增强了吸收。

为了扩展 π -共轭，另一种策略是引入电子给体基团，例如三苯胺作为咪唑 [115]的外围基团。在这种情况下，给电子基团连接到电子给体上无法在分子内产生推拉效应，因为电子给体之间不可能存在电子相互作用。因此，与未取代的咪唑相比，Cbz_22和Cbz_23的吸收位置没有显著变化，但两种染料的吸收系数有所增强。由于Cbz_22和Cbz_23中存在大量芳香环，摩尔

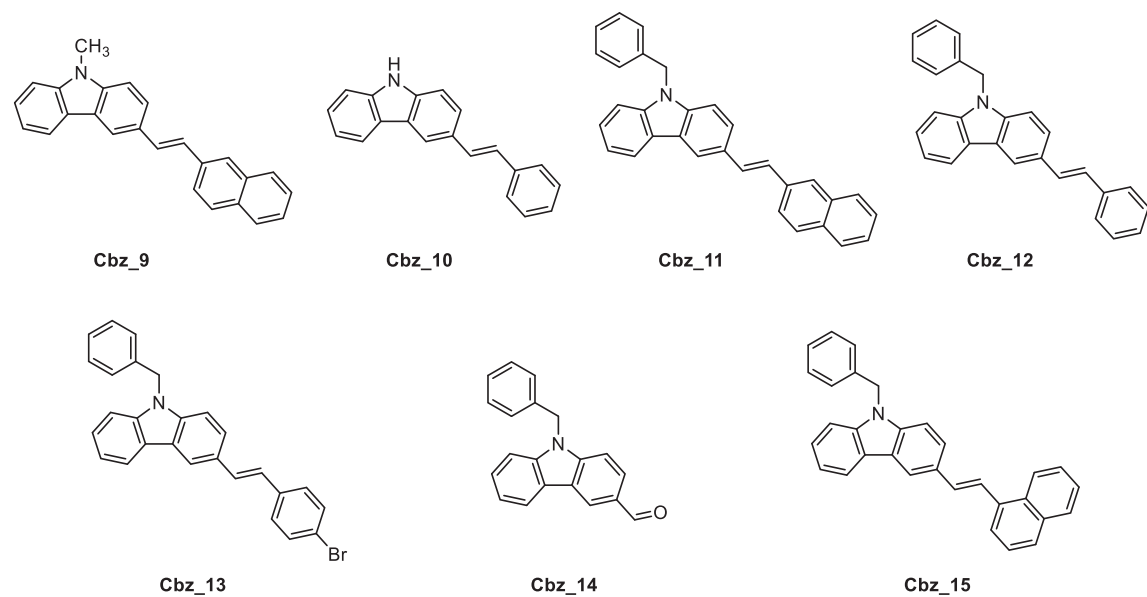
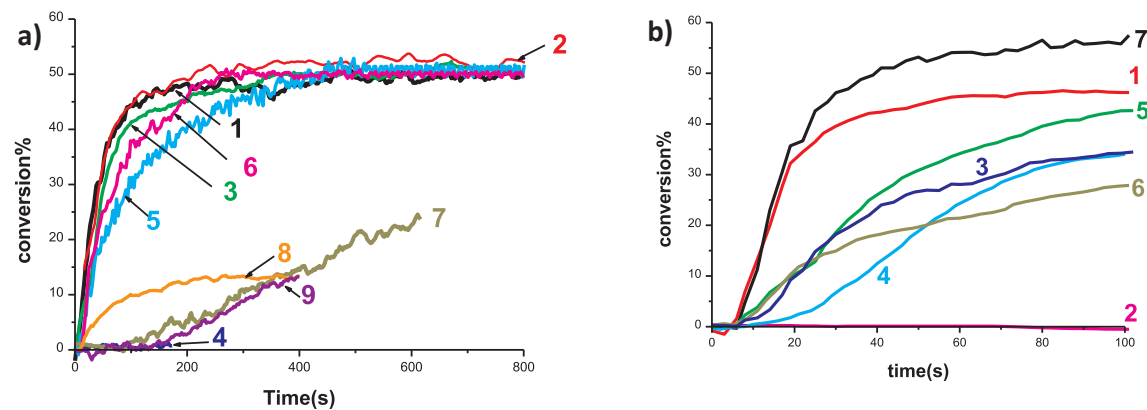


Fig. 20. Chemical structures of the Cbz_9-Cbz_15 series.

extinction coefficients resulted an absorption that became not negligible in the 400–450 nm region for **Cbz_22** and **Cbz_23** so that a polymerization process could be initiated for irradiations occurring beyond 400 nm. As other specificity, **Cbz_22** and **Cbz_23** were designed as non-extractable photoinitiators by introducing polymerizable groups such as styrene or vinylic groups in **Cbz_22** and **Cbz_23** respectively

(See Fig. 22). Introduction of polymerizable functions on carbazole is not new and carbazole specifically designed for the FRPCP of epoxides was previously reported in 2009, possessing in this case oxiranes groups (**Cbz_24** and **Cbz_25**) [116]. By carefully selecting the two component photoinitiating systems and by replacing $(t\text{-Bu})\text{Ph}_2\text{I}^+$ by *N*-phenylglycine (NPG), colorless polymers could be obtained by polymerizing a



c) Before polymerization

After polymerization

Fig. 21. (A) Polymerization profiles of EPOX (monomer conversion vs. irradiation time) upon exposure to a LED@405 nm under air using different photoinitiating systems: (1) **Cbz_15**/ $(t\text{-Bu})\text{Ph}_2\text{I}^+$ (0.5%/1%w/w); (2) **Cbz_9**/ $(t\text{-Bu})\text{Ph}_2\text{I}^+$ (0.5%/1%w/w); (3) **Cbz_11**/ $(t\text{-Bu})\text{Ph}_2\text{I}^+$ (0.5%/1%w/w); (4) **Cbz_10**/ $(t\text{-Bu})\text{Ph}_2\text{I}^+$ (0.5%/1%w/w); (5) **Cbz_12**/ $(t\text{-Bu})\text{Ph}_2\text{I}^+$ (0.5%/1%w/w); (6) **Cbz_13**/ $(t\text{-Bu})\text{Ph}_2\text{I}^+$ (0.5%/1%w/w); (7) **Cbz_14**/ $(t\text{-Bu})\text{Ph}_2\text{I}^+$ (0.5%/1%w/w); (8) BAPO/ $(t\text{-Bu})\text{Ph}_2\text{I}^+$ (0.5%/1%w/w); (9) CARET/ $(t\text{-Bu})\text{Ph}_2\text{I}^+$ (1%/1%w/w); (B) Polymerization profiles of TMPTA (acrylate function conversion vs. irradiation time) in laminate upon exposure to LED@405 nm in the presence of: (1) **Cbz_9**/ $(t\text{-Bu})\text{Ph}_2\text{I}^+$ (0.5%/1%w/w); (2) **Cbz_10** (0.5%/1%w/w); (3) **Cbz_11**/ $(t\text{-Bu})\text{Ph}_2\text{I}^+$ (0.5%/1%w/w); (4) **Cbz_12**/ $(t\text{-Bu})\text{Ph}_2\text{I}^+$ (0.5%/1%w/w); (5) **Cbz_13**/ $(t\text{-Bu})\text{Ph}_2\text{I}^+$ (0.5%/1%w/w); (6) **Cbz_14**/ $(t\text{-Bu})\text{Ph}_2\text{I}^+$ (0.5%/1%w/w); (7) **Cbz_15**/ $(t\text{-Bu})\text{Ph}_2\text{I}^+$ (0.5%/1%w/w); (C) colorless films obtained during the FRP of acrylates. Reproduced with permission from Al Mousawi et al. [112]. Copyright 2017 MDPI.

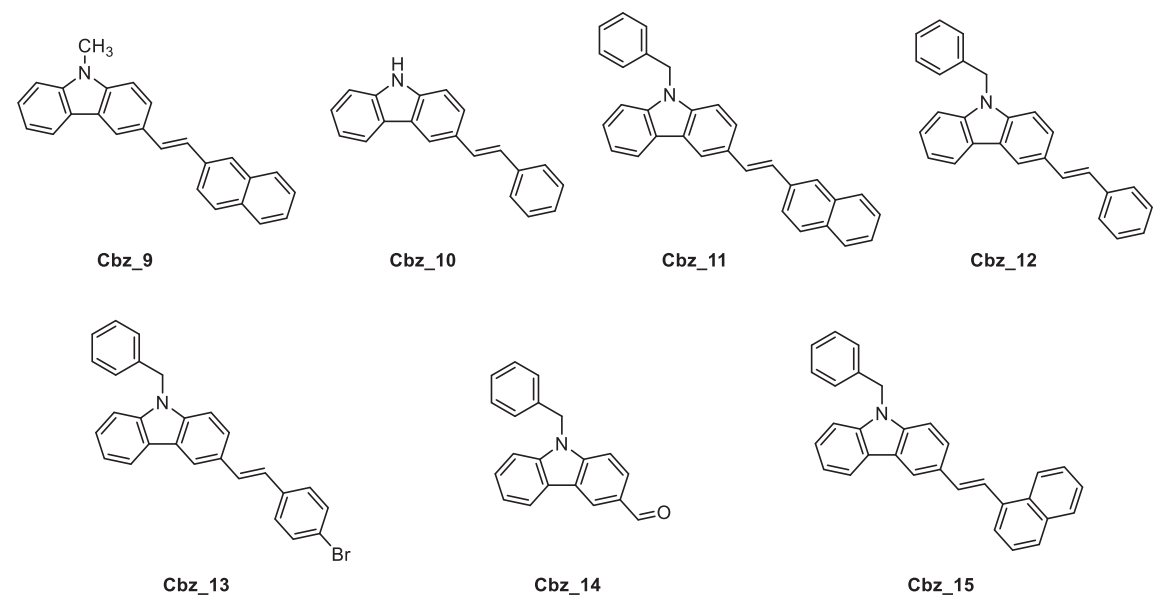
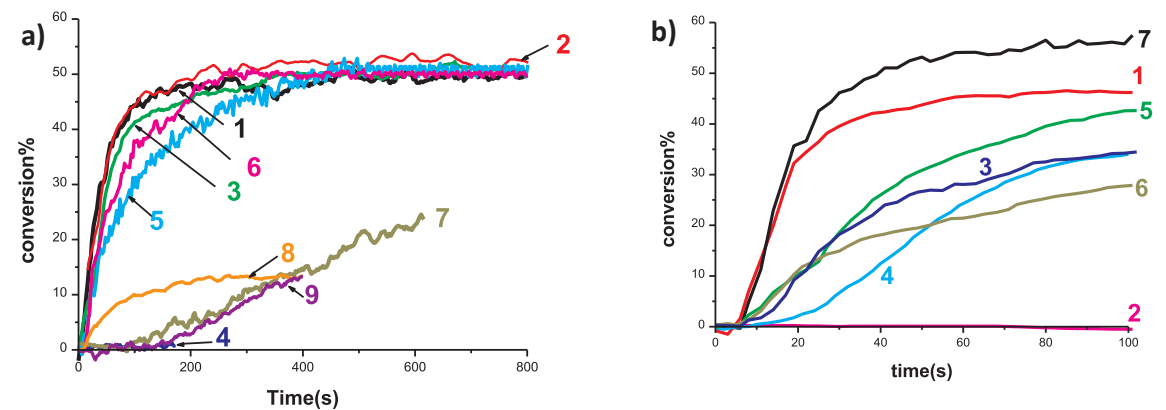


图 20. Cbz_9-Cbz_15 系列的化学结构

消光系数的增加导致**Cbz_22** 和**Cbz_23** 在400–450 nm区域的吸收不可忽略，因此可以在超过400 nm的照射下引发聚合过程。另一个特点是，**Cbz_22** 和**Cbz_23** 被设计为不可提取的光引发剂，通过在**Cbz_22** 和**Cbz_23** 中分别引入苯乙烯或乙烯基等可聚合基团来实现。

(参见图22)。在咔唑上引入可聚合官能团并非新概念，专门为环氧树脂的FRPCP设计的咔唑此前于2009年已有报道，其中含有环氧基团 (**Cbz_24** 和**Cbz_25**) [116]。通过精心选择两种组分的光引发体系，并将 $(t\text{-Bu})\text{Ph}_2\text{I}^+$ 替换为*N*-苯基甘氨酸 (NPG)，可以在空气氛围下于 405 nm聚合



c) Before polymerization

After polymerization

图 21. (A) 在空气环境下，使用LED@405 nm照射，不同光引发体系下EPOX的聚合曲线（单体转化率 vs. 照射时间）：(1) **Cbz_15**/ $(t\text{-Bu})\text{Ph}_2\text{I}^+$ (0.5%/1% w/w); (2) **Cbz_9**/ $(t\text{-Bu})\text{Ph}_2\text{I}^+$ (0.5%/1% w/w); (3) **Cbz_11**/ $(t\text{-Bu})\text{Ph}_2\text{I}^+$ (0.5%/1% w/w); (4) **Cbz_10**/ $(t\text{-Bu})\text{Ph}_2\text{I}^+$ (0.5%/1% w/w); (5) **Cbz_12**/ $(t\text{-Bu})\text{Ph}_2\text{I}^+$ (0.5%/1% w/w); (6) **Cbz_13**/ $(t\text{-Bu})\text{Ph}_2\text{I}^+$ (0.5%/1% w/w); (7) **Cbz_14**/ $(t\text{-Bu})\text{Ph}_2\text{I}^+$ (0.5%/1% w/w); (8) BAPO/ $(t\text{-Bu})\text{Ph}_2\text{I}^+$ (0.5%/1% w/w); (9) CARET/ $(t\text{-Bu})\text{Ph}_2\text{I}^+$ (1%/1% w/w); (B) 在层压条件下，使用LED@405 nm照射，不同体系下TMPTA的聚合曲线（丙烯酸酯官能团转化率 vs. 照射时间）：(1) **Cbz_9**/ $(t\text{-Bu})\text{Ph}_2\text{I}^+$ (0.5%/1% w/w); (2) **Cbz_10** (0.5%/1% w/w); (3) **Cbz_11**/ $(t\text{-Bu})\text{Ph}_2\text{I}^+$ (0.5%/1% w/w); (4) **Cbz_12**/ $(t\text{-Bu})\text{Ph}_2\text{I}^+$ (0.5%/1% w/w); (5) **Cbz_13**/ $(t\text{-Bu})\text{Ph}_2\text{I}^+$ (0.5%/1% w/w); (6) **Cbz_14**/ $(t\text{-Bu})\text{Ph}_2\text{I}^+$ (0.5%/1% w/w); (7) **Cbz_15**/ $(t\text{-Bu})\text{Ph}_2\text{I}^+$ (0.5%/1% w/w); (C) 在丙烯酸酯自由基聚合过程中得到的无色薄膜。经 Al Mousawi 等人 [112] 许可转载。版权 2017 MDPI。

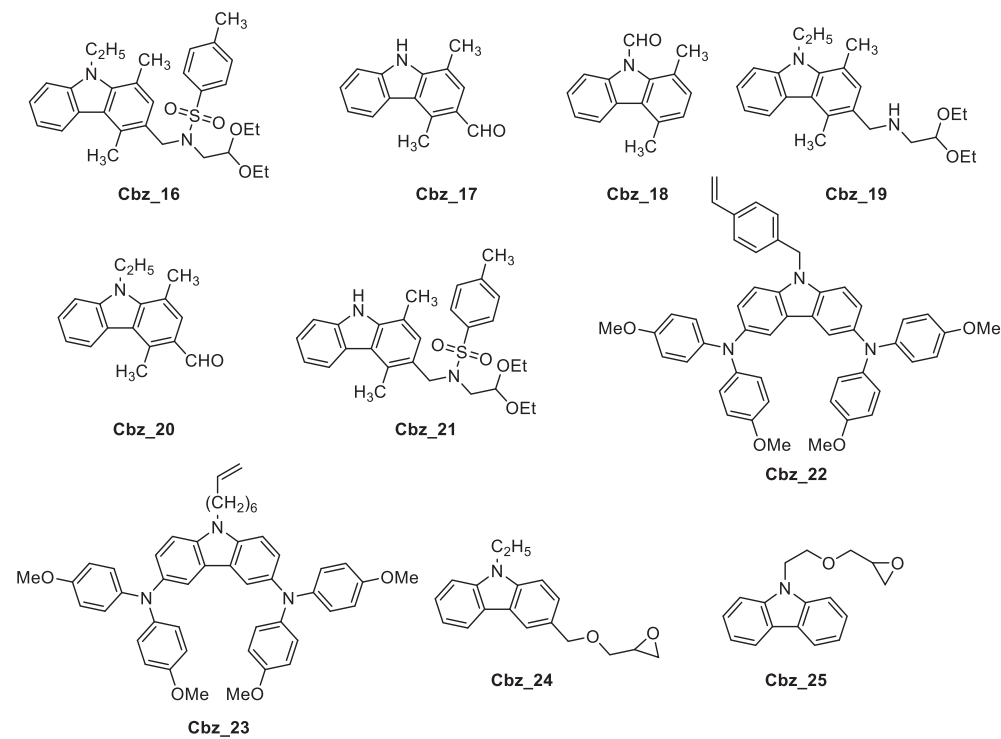


Fig. 22. Chemical structures of the Cbz_16-Cbz_25 series.

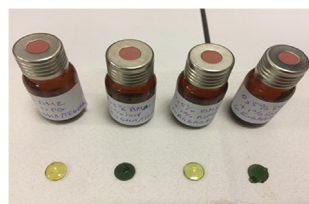
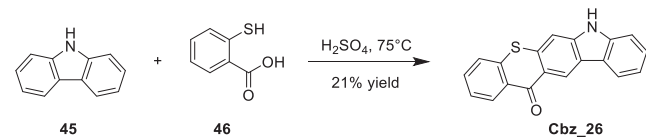


Fig. 23. Photographs of a BisGMA/TEGDMA thick film (1.4 mm) upon irradiation with the LED@405 nm for 100 s in the presence of **Cbz_23**/NPG (0.5%/1% w/w), **Cbz_23**/(*t*-Bu)Ph₂I⁺ (0.5%/1% w/w), **Cbz_22**/NPG (0.5%/1% w/w) and **Cbz_22**/(*t*-Bu)Ph₂I⁺ (0.5%/1% w/w), respectively (from left to right), under air after polymerization. Reproduced with permission from Al Mousawi et al. [112]. Copyright 2017 MDPI.

BisGMA/TEGDMA blend under air at 405 nm using **Cbz_22** and **Cbz_23**. Especially, an enhancement by about 20% of the monomer conversion was obtained by replacing (*t*-Bu)Ph₂I⁺ by *N*-phenylglycine in the two-component systems (See Fig. 23).

Finally, the access to photocomposites was also examined. Indeed, by introducing glass fibers into the polymer network (50% glass fibers/50% resin w/w), the mechanical properties of the final polymers can be greatly modified. If this specific point was not examined in this study, a proof of concept was done while using **Cbz_23** in the two-component system **Cbz_23** /NPG (0.5%/1% w/w).

In 2010, Yagci and coworkers proposed to fuse within a unique structure two efficient photoinitiators, namely the carbazole core to the thioxanthone moiety [117]. Thioxanthone is a standard type II photoinitiator extensively studied in the literature [118–121] and used in combination with numerous hydrogen donors such as alcohols, [122] thiols [123], ethers [124] or amines [125,126]. When thioxanthones are substituted with thiols [127–129] or carboxylic groups [130–132], photopolymerization without coinitiators became possible, these photoinitiators already containing hydrogen donors. This is in this context that **Cbz_26** was developed, possessing a NH group onto carbazole. From a synthetic point of view, **Cbz_26** could be prepared in 21% yield, by reacting carbazole **45** with thiosalicylic acid **46** in concentrated

Scheme 6. Synthetic route to **Cbz_26**.

sulfuric acid (See Scheme 6).

Compared to carbazole and thioxanthone considered separately, an improvement of the molar extinction coefficient associated with a redshift of the maximum absorption was demonstrated in solution. Notably, a shift of 100 nm was observed between thioxanthone and **Cbz_26**. Photolysis experiments with and without hydrogen donors proved the carbazole moiety to act as a hydrogen donor so that the following mechanism described in Scheme 7 could be proposed to support the FRP of methacrylates. In the absence of external hydrogen donors, **Cbz_26**[•] is the initiating species, what could be demonstrated by examining the photoluminescence of the poly(methyl methacrylate) (Poly(MMA)). Indeed, in this last case, **Cbz_26** is connected to the polymer as the end group. A low inhibition to oxygen was demonstrated during the FRP of MMA, resulting from the hydrogen donor directly connected to the initiating radical. Benefiting from this, the FRP could be initiated even with photoinitiator concentration as low as 1×10^{-3} M. Besides, if the concept is appealing, comparison of the photoinitiating ability of **Cbz_26** with that of camphorquinone in the same conditions revealed camphorquinone to outperform **Cbz_26**, whatever the experimental conditions were.

2.3. Carbazole-based TADF photoinitiators

As previously mentioned, the photoinitiating ability of type II photoinitiators is directly related to their excited state lifetimes. Indeed, long-living excited states lifetimes favor intermolecular interactions of the photoinitiator with the different additives, optimizing the generation of initiating species. In 2012, Chihaya Adachi developed an innovative approach to improve the photoluminescence quantum yields

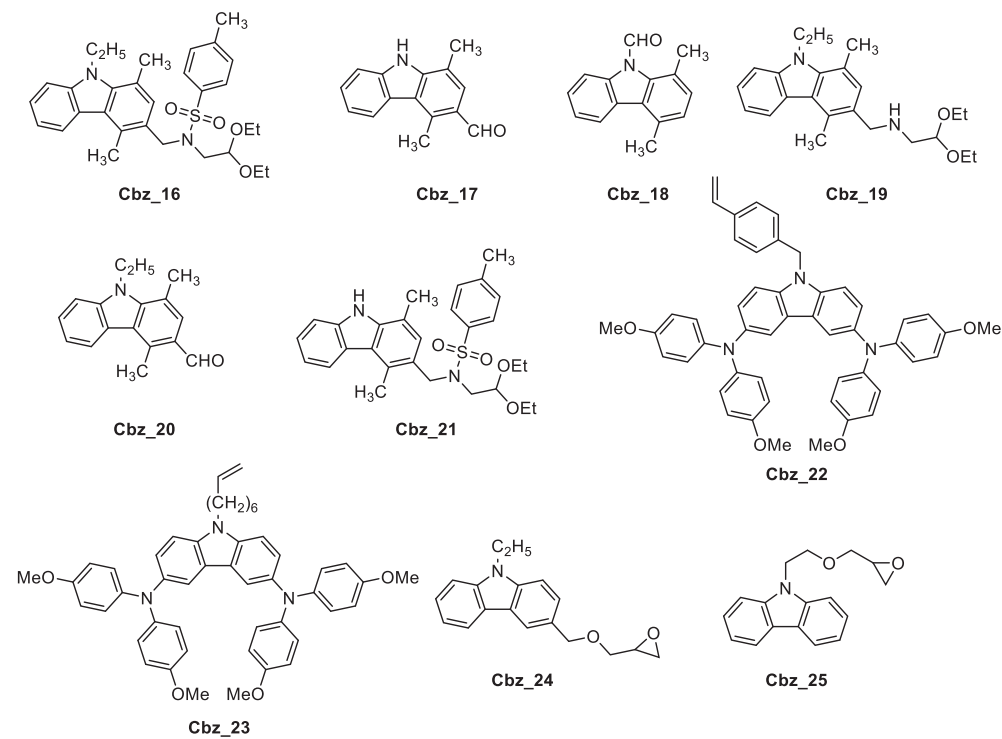


图22. Cbz_16-Cbz_25 系列的化学结构。

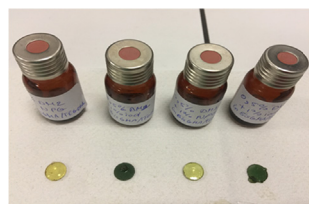
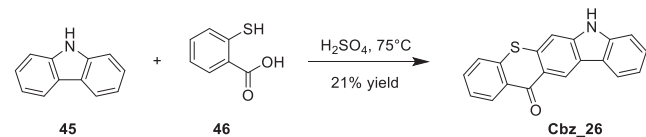


图23. BisGMA/TEGDMA厚膜 (1.4 mm) 在LED@405 nm照射100秒, 分别在 **Cbz_23**/NPG (0.5%/1% w/w)、**Cbz_23**/(*t*-Bu)Ph₂I⁺ (0.5%/1% w/w)、**Cbz_22**/NPG (0.5%/1% w/w) 和**Cbz_22**/(*t*-Bu)Ph₂I⁺ (0.5%/1% w/w) 存在下 (从左到右), 在空气中聚合后的照片。经Al Mousawi等人许可转载。[112]. 版权2017 MDPI。

BisGMA/TEGDMA混合物, 使用**Cbz_22** 和**Cbz_23**。特别地, 通过将 (*t*-Bu)Ph₂I⁺ 替换为*N*-苯基甘氨酸作为双组分体系, 单体转化率提高了约20% (参见图23)。

最后, 还研究了光复合材料的制备。实际上, 通过将玻璃纤维引入聚合物网络 (50%玻璃纤维/50%树脂 w/w), 最终聚合物的力学性能可以得到大幅改善。尽管本研究未对此进行详细探讨, 但在使用**Cbz_23** 与**Cbz_23** /NPG (0.5%/1% w/w) 双组分体系时进行了概念验证。

2010年, Yagci及其同事提出在独特结构中融合两种高效光引发剂, 即咪唑核与噻吨酮部分 [117]。噻吨酮是标准II型光引发剂, 在文献中广泛研究 [118–121], 并与多种氢供体 (如醇、[122]硫醇 [123], 醚 [124] 或胺 [125,126]) 联合使用。当噻吨酮被硫醇取代时 [127–129], 或带有羧基时 [130–132], 无需共引发剂即可实现光聚合, 因为这些光引发剂已含有氢供体。正是在此背景下, **Cbz_26** 被开发出来, 其咪唑上带有NH基团。从合成角度看, **Cbz_26** 可通过咪唑**45** 与硫代水杨酸**46** 在浓中反应制备, 产率21%。

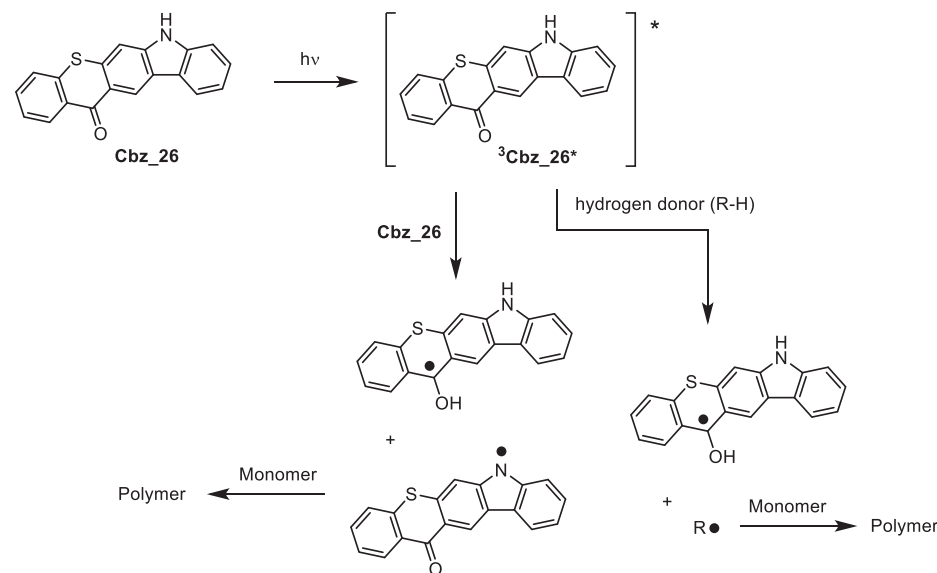
方案 6. **Cbz_26** 的合成路线。

硫酸 (参见方案6)。

与分别考虑的咪唑和噻吨酮相比, 在溶液中观察到摩尔消光系数提高且最大吸收红移。值得注意的是, 噻吨酮与**Cbz_26**之间观察到100 nm的位移。有无氢供体的光解实验证明咪唑部分充当氢供体, 因此可提出以下方案7中描述的机制来支持甲基丙烯酸酯的FRP。在无外部氢供体的情况下, **Cbz_26**[•] 是引发物种, 这可通过检查聚甲基丙烯酸甲酯 (Poly(MMA))的光致发光来证明。确实, 在后一种情况下, **Cbz_26** 作为端基连接至聚合物。在MMA的FRP过程中, 由于氢供体直接连接至引发自由基, 表现出低的氧抑制效果。利用这一点, 即使光引发剂浓度低至 1×10^{-3} M, 也能引发FRP。此外, 尽管这一概念有吸引力, 但在相同条件下将**Cbz_26** 的光引发能力与樟脑醌相比, 发现无论实验条件如何, 樟脑醌的性能均优于**Cbz_26**。

2.3. 基于咪唑的TADF光引发剂

如前所述, II型光引发剂的光引发能力与其激发态寿命直接相关。确实, 长寿命的激发态有利于光引发剂与不同添加剂之间的分子间相互作用, 从而优化引发物种的生成。2012年, Chihaya Adachi提出了一种创新方法, 以提高用于有机发光二极管 (OLED) 的有机和有机金属化合物的光致发光量子产率



Scheme 7. The two possible photoinitiating systems with and without hydrogen donors using **Cbz_26** as the photosensitizer.

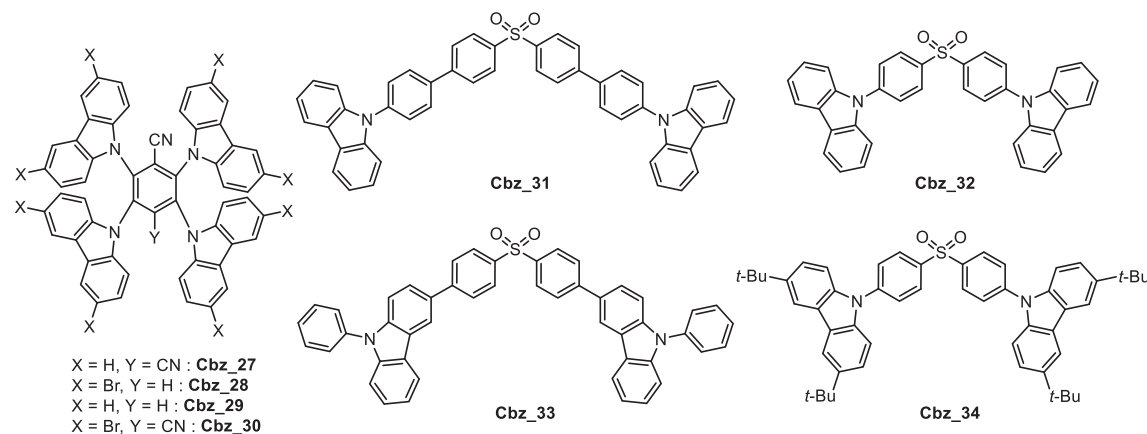


Fig. 24. Chemical structures of **Cbz_27-Cbz_34**.

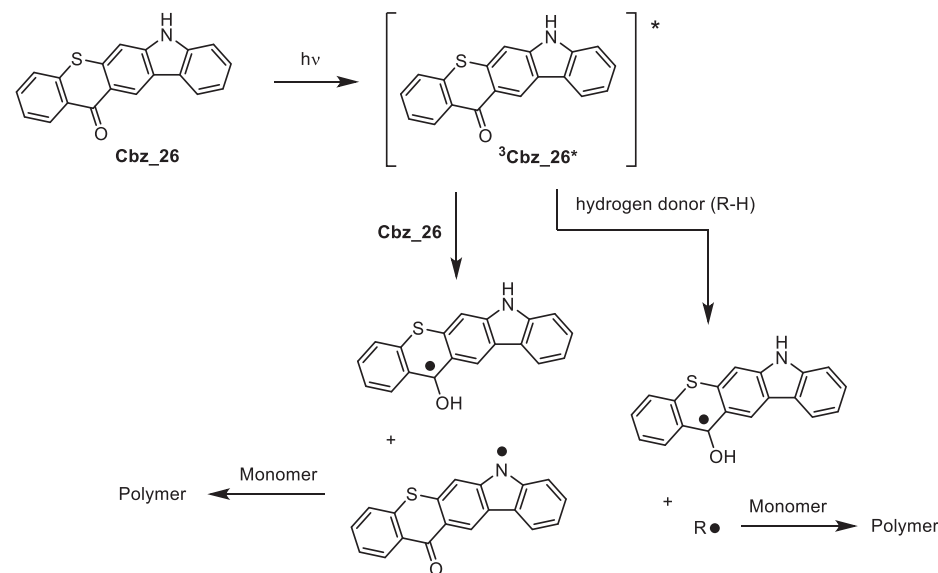
(PLQY) of organic and organometallic compounds used in organic light-emitting diodes (OLEDs) [133,134]. By introducing sterically demanding groups around a central core, the Highest Occupied Molecular Orbital (HOMO) and the Lowest Unoccupied Molecular Orbital (LUMO) could be isolated from each other by interrupting the π -conjugation. By carefully selecting the electron donating and electron-accepting groups, compounds exhibiting a small energy splitting between the singlet and the triplet excited state could be obtained. As a result of this small energy S_1 - T_1 splitting existing within these compounds, electrons from the triplet state can be up-converted to the singlet one, inducing a thermally activated delayed fluorescence process. Beyond the simple enhancement of the PLQY which is valorized in OLEDs, another parameter evidencing the presence of a Thermally Activated Delayed Fluorescence (TADF) property is a significant elongation of the excited state lifetime. Considering that the increase of the S_1 lifetime remains a challenge in photochemistry, in 2017, this TADF property was exploited for the design of photoinitiators. In this work, a fine tuning of the excited state lifetimes could be obtained upon bromination of the peripheral carbazoles of the **Cbz_27-Cbz_30** series (See Fig. 24) [135].

Thus, introduction of bromine onto carbazoles enabled to drastic elongate the excited state lifetimes under nitrogen, from 1.9 μs for **Cbz_27** to 4.8 μs for **Cbz_30**, from 7.2 μs for **Cbz_29** to 7.8 μs for **Cbz_28** respectively. Parallel to this, by introducing cyano groups on **Cbz_27** and **Cbz_30**, energy level of the LUMO orbital could be lowered, thereby redshifting the absorption of these two compounds relative to

Cbz_29 and **Cbz_28**. FRPCP of EPOX using a two-component system **Cbz_x** ($x = 27-30$)/($t\text{-Bu}$) Ph_2I^+ (0.5%/1% w/w) under air was characterized by high polymerization rates and high final monomer conversions (47–55%) both upon irradiation at 405 and 455 nm. In the same conditions, CARET and BAPO which are two benchmark photoinitiators could only furnish a final monomer conversion of 12 and 15% after 400 s of irradiation at 405 nm. At 455 nm, no polymerization could even be initiated with these two references as a result of a lack of absorption for these two compounds. Considering that CARET is also a carbazole derivative, all the benefits of the chemical engineering in **Cbz_27-Cbz_30** was demonstrated in terms of polymerization performances.

Concerning the FRP of TMPTA, a photoredox catalyst behavior could be demonstrated with all TADF photoinitiators while using a three-component system **Cbz_x** ($x = 27-30$)/($t\text{-Bu}$) Ph_2I^+ /EDB (0.5%/1%/1% w/w). Finally, examination of the polymerization of a BISGMA/TEGDMA blend (70/30) with the best photoinitiator i.e. **Cbz_28** could provide tack-free and colorless polymers within 100 s with a monomer conversion of 60% upon irradiation with a LED at 405 nm under air. Conversely, at 477 nm, the monomer conversion remained low (around 5%) due to the lack of absorption for **Cbz_28** (see Fig. 25).

While getting a deeper insight into the mechanism, the ability of **Cbz_27-Cbz_30** to act as electron acceptors in the **Cbz_x** ($x = 27-30$)/EDB two-component photoinitiating system or as electron donors in the two-component system **Cbz_x** ($x = 27-30$)/($t\text{-Bu}$) Ph_2I^+ could be easily



Sche图7. 使用**Cbz_26** 作为光敏剂时，有/无氢供体的两种可能光引发体系

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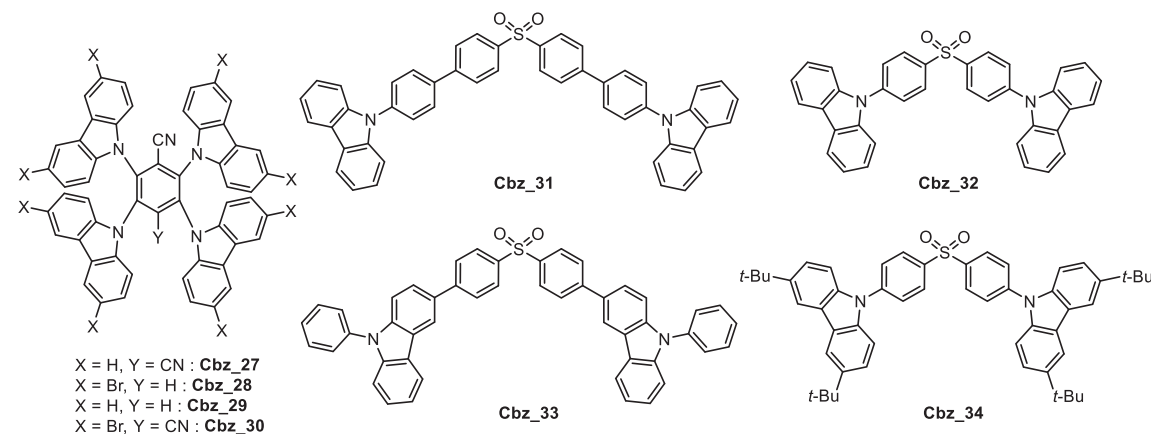


图 24. **Cbz_27-Cbz_34**的化学结构。

(PLQY) [133,134]. 通过在中心核心周围引入空间位阻基团，最高占据分子轨道 (HOMO) 和最低未占分子轨道 (LUMO) 可以通过中断 π -共轭而相互隔离。通过仔细选择给电子和吸电子基团，可以获得在单重态和三重激发态之间具有小能量分裂的化合物。由于这些化合物中存在这种小的 S_1 - T_1 分裂，三重态的电子可以上转换到单重态，从而诱导热激活延迟荧光过程。除了在 OLED 中应用的 PLQY 的简单提升之外，另一个证明热激活延迟荧光 (TADF) 特性的参数是激发态寿命的显著延长。考虑到增加 S_1 寿命在光化学中仍然是一个挑战，2017 年，这种 TADF 特性被用于光引发剂的设计。在这项工作中，通过对外围咔唑进行溴化，可以精细调节激发态寿命，从而获得 **Cbz_27-Cbz_30** 系列 (参见图 24) [135]。

因此，在咔唑上引入溴使得氮气下的激发态寿命大幅延长，**Cbz_27** 从 1.9 μs 延长至 **Cbz_30** 的 4.8 μs ，**Cbz_29** 从 7.2 μs 延长至 **Cbz_28** 的 7.8 μs 。与此同时，通过在 **Cbz_27** 和 **Cbz_30** 上引入氨基，LUMO 轨道的能级得以降低，从而使这两种化合物相对于

Cbz_29 和 **Cbz_28** 的吸收发生红移。使用双组分体系 **Cbz_x** ($x = 27-30$)/($t\text{-Bu}$) Ph_2I^+ (0.5%/1% w/w) 在空气中对 EPOX 进行 FRPCP，在 405 nm 和 455 nm 辐照下均表现出高聚合速率和高最终单体转化率 (47–55%)。在相同条件下，两种基准光引发剂 CARET 和 BAPO 在 405 nm 辐照 400 s 后仅提供 12% 和 15% 的最终单体转化率。在 455 nm 下，这两种参考化合物甚至无法引发聚合，原因是它们缺乏吸收。考虑到 CARET 也是咔唑衍生物，**Cbz_27-Cbz_30** 在化学工程方面的所有优势已在聚合性能上得到证明。

关于 TMPTA 的 FRP，使用三组分体系时，所有 TADF 光引发剂均可表现出光氧化还原催化行为 **Cbz_x** ($x = 27-30$)/($t\text{-Bu}$) Ph_2I^+ /EDB (0.5%/1%/1% w/w)。最后，使用最佳光引发剂即 **Cbz_28** 对 aBISGMA/TEGDMA 共混物 (70/30) 进行聚合测试，在 405 nm LED 照射下，空气中 100 秒内即可获得无粘性、无色的聚合物，单体转化率达到 60%。相反，在 477 nm 处，由于 **Cbz_28** 缺乏吸收 (见图 25)，单体转化率仍然很低 (约 5%)。

为了更深入地了解其机制，通过荧光淬灭实验可以轻松证明 **Cbz_27-Cbz_30** 在 **Cbz_x** ($x = 27-30$)/EDB 双组分光引发体系中作为电子受体，或在 **Cbz_x** ($x = 27-30$)/($t\text{-Bu}$) Ph_2I^+ 双组分体系中作为电子供体的能力，导致光引发剂被氧化或还原，形成非发光化合物。

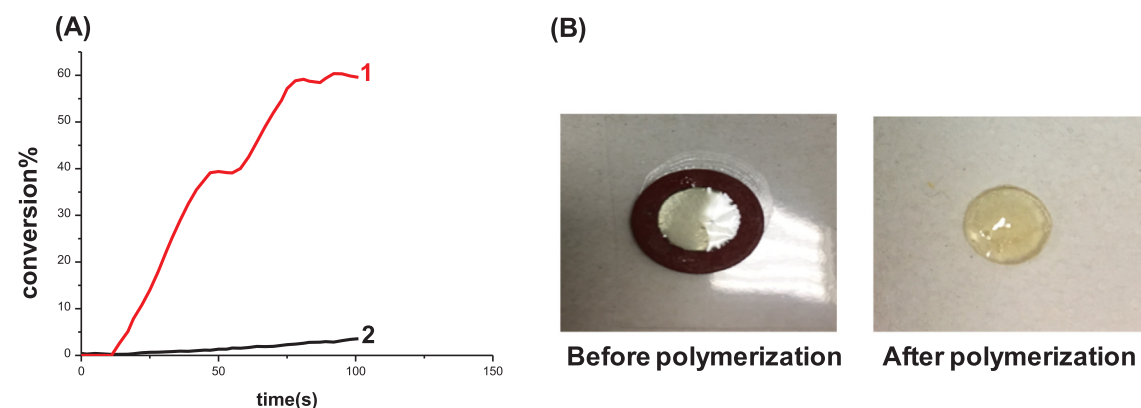
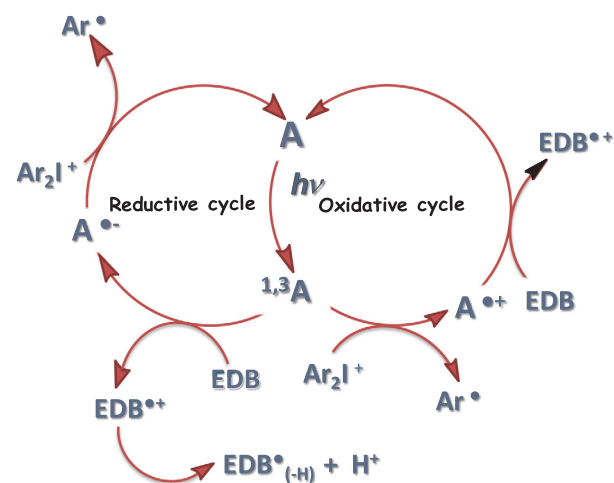


Fig. 25. Polymerization profiles of BisGMA/TEGDMA (methacrylate function conversion vs. irradiation time) under air (1.4 mm thick sample) in the presence of two-component photoinitiating system **Cbz_28**/(*t*-Bu)Ph₂I⁺ (0.5%/1% w/w) upon exposure to (1) LED@405 nm; (2) LED@477 nm; The irradiation starts at *t* = 10 s. (B) Photos of a BisGMA/TEGDMA thick film (1.4 mm) upon irradiation with the LED@405 nm for 100 s in the presence of **Cbz_28**/(*t*-Bu)Ph₂I⁺ (0.5%/1%w/w) under air before and after polymerization. The irradiation starts for *t* = 10 s Adapted with permission from Al Mousawi et al. [135]. Copyright 2017 American Chemical Society.



Scheme 8. The two concomitant photoredox catalytic cycles existing in the three component photoinitiating systems **Cbz_x** (*x* = 27–30)/(*t*-Bu)Ph₂I⁺/EDB. Adapted with permission from Al Mousawi et al. [135]. Copyright 2017 American Chemical Society.

demonstrated by the fluorescence quenching experiments, the oxidation or the reduction of the photoinitiators resulting in the formation of none-luminescent compounds. Consequently, in the **Cbz_x** (*x* = 27–30)/(*t*-Bu)Ph₂I⁺/EDB three-component systems, a competition between the two routes exists and the two processes also occur simultaneously, the electron transfer quantum yields between **Cbz_x** (*x* = 27–30)/(*t*-Bu)Ph₂I⁺ or **Cbz_x** (*x* = 27–30)/EDB being similar. Due to the concomitant regeneration of **Cbz_27**–**Cbz_30** in both the

oxidative and the reductive cycle, the improved efficiency of the three component systems over the two components systems **Cbz_x** (*x* = 27–30)/(*t*-Bu)Ph₂I⁺ or **Cbz_x** (*x* = 27–30)/EDB can be easily explained and a full picture of the mechanism is presented in the Scheme 8.

Besides, in a next study, the same authors evidenced the limitation of the TADF approach with another series of carbazoles **Cbz_31**–**Cbz_34** (See Fig. 24) [136]. If these four dyes have previously been reported as TADF emitters for OLEDs [137], the TADF process doesn't constitute the main deexcitation pathway (10–15% of the overall deexcitation process) for these compounds so that no modification of the polymerization rates and the monomer conversions could be evidenced between the TADF and the none-TADF photoinitiators. Therefore, it can be concluded that beyond the elongation of the excited state lifetime, the desexcitation from the singlet excited state should constitute the main desexcitation pathway for the molecule in order the photo-initiating ability to be improved.

2.4. Carbazole-based BOPIDYs

BODIPYs have been extensively studied as fluorescent probes for bioimaging applications [138–141] but only scarcely for the design of photoinitiators [142–144]. The first carbazole-based BODIPY used as photoinitiator was reported in 2019 (**Cbz_35**) in the context of a polymerization carried out in the near infrared region (See Fig. 26) [50]. As anticipated, **Cbz_35** exhibited a strong absorption centered in the visible region, extending from 400 nm to 550 nm with a maximum absorption located at 492 nm (68 080 L·mol⁻¹·cm⁻¹). Even if the absorption coefficients were extremely low at 785 nm (170 L·mol⁻¹·cm⁻¹) and 940 nm (100 L·mol⁻¹·cm⁻¹), the possibility to initiate the FRP of methacrylates in the NIR region was brilliantly

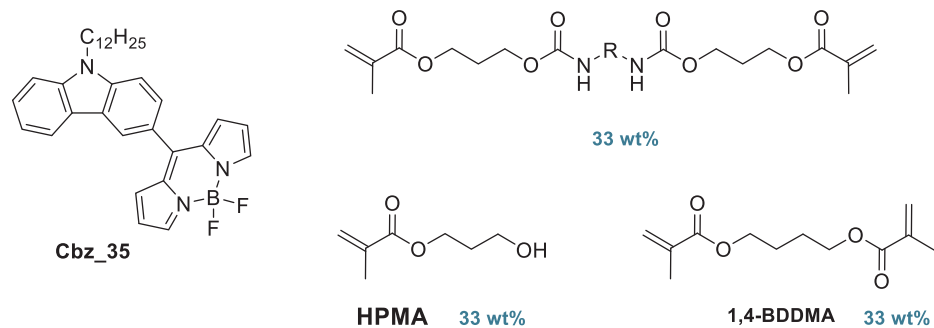


Fig. 26. Chemical structures of **Cbz_35** and the different monomers used in the mixture.

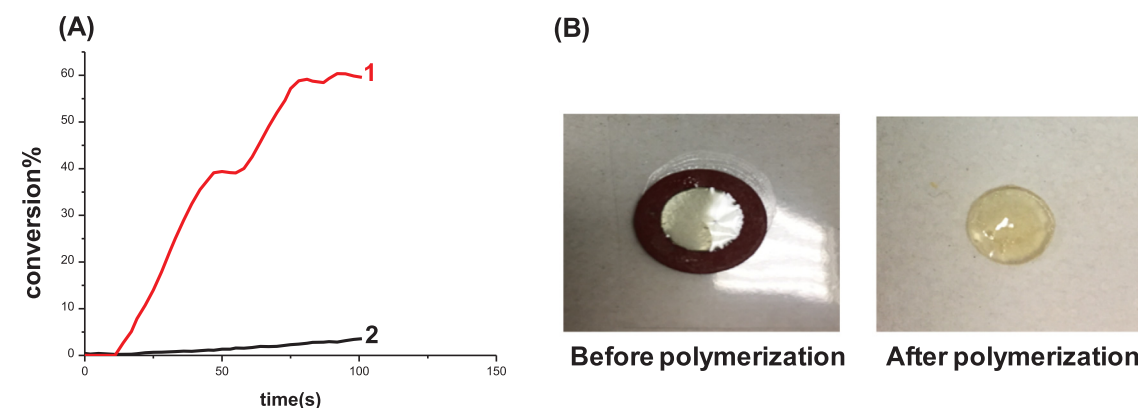
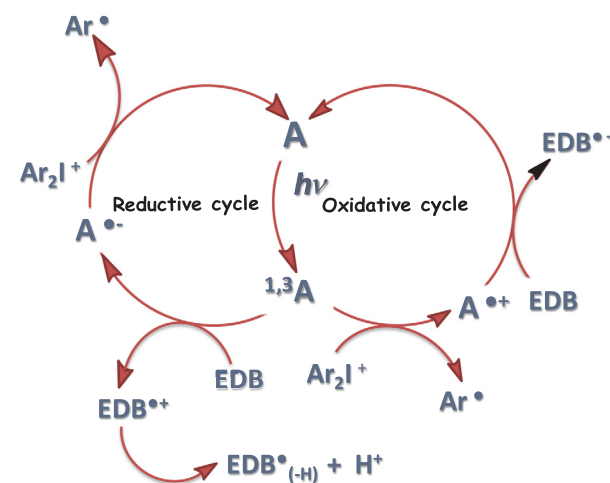


图25. 聚合过程曲线: BisGMA/TEGDMA (甲基丙烯酸酯官能团转化率 vs. 照射时间) 在空气下 (1.4 mm厚样品), 存在双组分光引发体系**Cbz_28**/(叔丁基)Ph₂I⁺ (0.5%/1% w/w), 暴露于(1) LED@405 nm; (2) LED@477 nm; 照射从*t* = 10 秒开始。(B) BisGMA/TEGDMA厚膜(1.4 mm)在LED@405 nm下照射100 s, 存在**Cbz_28**/(叔丁基)Ph₂I⁺ (0.5%/1% w/w)在空气下聚合前后的照片。照射从*t* = 10 秒开始。经Al Mousawi等人许可可改编。 [135]。版权2017美国化学学会。



方案8.三组分光引发体系中存在的同时发生的光氧化还原催化循环**Cbz_x**(*x* = 27–30)/(*t*-Bu)Ph₂I⁺/EDB. 改编自Al Mousawi等人的许可。 [135]。版权2017美国化学学会。

因此, 在**Cbz_x** (*x* = 27–30) / (*t*-Bu) Ph₂I⁺/EDB三组分体系中, 两条路径之间存在竞争, 两个过程也同时发生, **Cbz_x** (*x* = 27–30) / (*t*-Bu) Ph₂I⁺ 或**Cbz_x** (*x* = 27–30) /EDB之间的电子转移量子产率相似。由于**Cbz_27**–**Cbz_30**的伴随再生,

氧化还原循环中, 三组分体系相对于两组分体系的效率提升可通过**Cbz_x**(*x* = 27–30)/(*t*-Bu)Ph₂I⁺或**Cbz_x** (*x* = 27–30)/EDB 轻松解释, 完整机理如图Scheme 8所示。

此外, 在后续研究中, 同一作者通过另一系列咔唑 **Cbz_31**–**Cbz_34** (参见Fig. 24) [136]证明了 TADF 方法的局限性。尽管这四种染料此前已被报道为用于 OLED 的 TADF 发射体, [137], 但 TADF 过程并非这些化合物的主要去激发途径 (占总去激发过程的10–15%), 因此在 TADF 和非 TADF 光引发剂之间未能观察到聚合速率和单体转化率的差异。因此, 可以得出结论: 除了延长激发态寿命外, 从单重激发态的去激发应构成分子的主要去激发途径, 才能提高光引发能力。

2.4. 咔唑基BOPIDYs

BODIPY 作为荧光探针在生物成像应用中已被广泛研究, 但在光引发剂的设计中却很少被使用 [138–141]。 [142–144]。第一种用作光引发剂的咔唑基 BODIPY 于 2019 年被报道 (**Cbz_35**), 用于近红外区域的聚合反应 (见图 26)。[50]正如预期, **Cbz_35** 在可见光区域表现出强吸收, 范围从 400 nm 到 550 nm, 最大吸收位于 492 nm (68 080 L·mol⁻¹·cm⁻¹)。即使在 785 nm (170 L·mol⁻¹·cm⁻¹)–940 nm (100 L·mol⁻¹·cm⁻¹) 处吸收系数极低, 但利用该化合物在近红外区域引发甲基丙烯酸酯的自由基聚合仍被精彩地

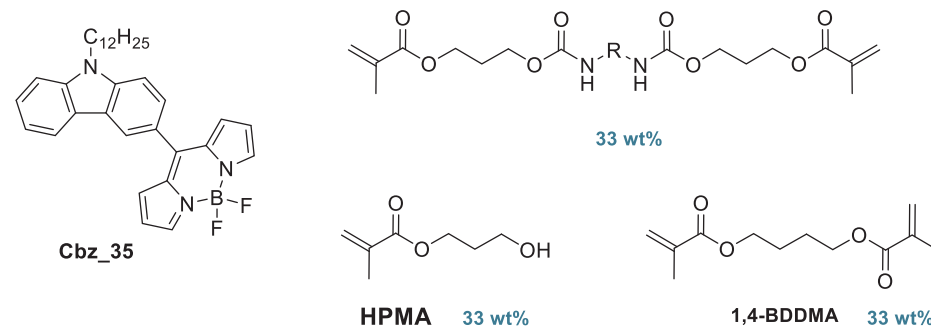


图26. **Cbz_35** 和混合物中使用的不同单体的化学结构。

demonstrated. Indeed, photopolymerization at these wavelengths is dominated by cyanines [145–150] but these compounds are expensive so that alternatives are actively researched. While polymerizing a resin composed of a mixture of three monomers (33.3 wt% of (hydroxypropyl)methacrylate (HPMA), 33.3 wt% of 1,4-butanediol dimethacrylate (1,4-BDMA) and 33.3 wt% of a urethane dimethacrylate monomer (UDMA), final monomer conversions of 92% upon irradiation at 785 nm and 60% upon irradiation at 940 nm could be obtained with the four component system **Cbz_35**/(t-Bu)Ph₂I⁺/4-dppba/BlocBuilder®MA (0.1 wt%/3 wt%/2 wt%/2 wt%) where 4-dppba and Block-Builder®MA stand for 4-(diphenylphosphino)benzoic acid and the acronym for a commercial alkoxyamine respectively. Remarkably, in this study, authors combined two polymerization modes, one being based on a pure photochemical system generating radicals through light activation and a second one being based on a photothermal system based on the conversion of light to heat. Indeed, infrared dyes are well-known to generate heat upon photoexcitation. By combining the two polymerization modes, the heat released by photoexcitation but also by the polymerization of methacrylates was sufficient to decompose Block-Builder®MA which is a thermal initiator of polymerization, improving the final monomer conversion. Interestingly, when the two polymerization modes were considered separately, no or low monomer conversions were obtained, demonstrating that the combination of these two systems is required to get good monomer conversions.

2.5. Carbazole-based helicoidal photoinitiators

The solubility of photoinitiators in the photocurable resins is a major issue as it drastically affects the final monomer conversion. This point is of crucial importance, especially when polyaromatic structures (perylene, pyrenes, pentacene, etc...) are used as photoinitiators. To address this issue, helicenes were proposed as possible structures. Indeed, helicenes are constituted of *ortho*-fused aromatic rings, enforcing the structure to adopt an helicoidal shape when several aromatic rings are connected together. As a result of this non-planar geometry, the π - π staking between molecules is efficiently avoided, addressing the solubility issue. In this work, a series of azahelicenes **Cbz_36**–**Cbz_38** differing by the number of angularly annulated aromatic rings and substituents was proposed (See Fig. 27) [151]. Interestingly, presence of an additional aromatic ring in **Cbz_36** enabled to redshift the absorption of 50 nm compared to that of **Cbz_38** while increasing the molar extinction coefficient (See Fig. 28). Upon addition of a bromine in **Cbz_37**, only a slightly redshifted absorption was found compared to that of **Cbz_38**.

The two components systems **Cbz_x** (x = 36–38)/(t-Bu)Ph₂I⁺ (0.5%/1% w/w) could efficiently initiate the FRPCP of EPOX or the FRP of TMPTA. Especially, higher polymerization rates were found for the polymerization of EPOX and TMPTA, with polymerization processes ended after a few seconds. However, the most interesting finding concerned the photoluminescence of the epoxy films. Indeed, upon photoexcitation of the polymer films for a while and switching the light off, a luminescence persisting over six seconds was observed, tentatively assigned to the presence of photoproducts generated during the

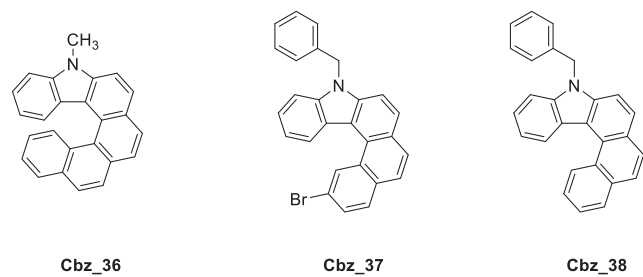


Fig. 27. Chemical structures of **Cbz_36**–**Cbz_38**.

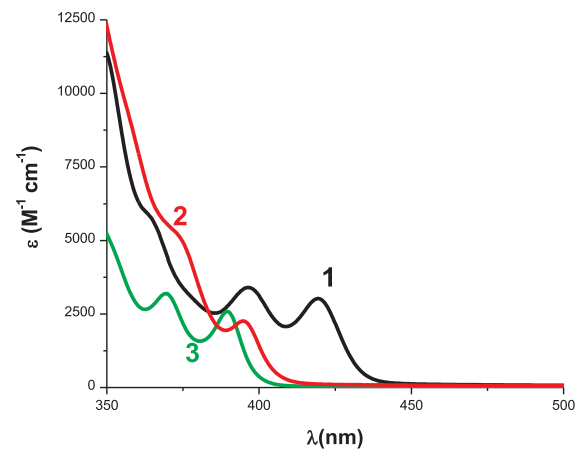


Fig. 28. UV–visible absorption spectra of **Cbz_36** (1), **Cbz_37** (2) and **Cbz_38** (3) in acetonitrile. Reprinted with permission from Al Mousawi et al. [151] Copyright © 2017 John Wiley & Sons, Inc.

polymerization process and/or the presence of unreacted azahelicenes **Cbz_36**–**Cbz_38**, despite the non-catalytic character of the two-component system.

2.6. Carbazole-based ferrocenium

Ferrocenium salts are attractive structures for the cationic polymerization of epoxides as these compounds are readily available from inexpensive starting materials [152–155]. Notably, these compounds can be prepared in one step by ligand exchange on ferrocene while using a high boiling point apolar solvents and Al/AlCl₃ as the metal catalysts (See Scheme 9). Finally, metathesis with an hexafluorophosphate salt enables to convert the water-soluble ferrocenium salt as a metal salt now soluble in organic solvents. This simple and efficient synthetic method was reported in the early 60's by Nesmeyanov and co-workers and numerous ferrocenium salts have been prepared using this procedure [156,157]. Among ferroceniums, Irgacure 261 i.e. (η₆-(1-methylethyl)phenyl)(η₅-cyclopentadienyl)iron hexafluorophosphate is a popular photoinitiator of cationic polymerization as this compound is commercially available [158–164]. However, the maximum absorption is centered at 241 nm and the absorption of Irgacure 261 extends from 200 to 300 nm so that this ferrocenium salt is a UV-centered photoinitiator. This photoinitiator is thus not for photopolymerization processes carried out in the near UV/visible range.

Numerous works were devoted to redshift the absorption of ferrocenium salts towards the visible range and relevant examples of this hard work were reported in 2006. In fact, the first carbazole-based photoinitiators **Cbz_39** and **Cbz_40** were published in 2006 and prepared according to the abovementioned procedure (See Fig. 29) [165]. Compared to Irgacure 261 which exhibits a maximum absorption at 241 nm, a redshift as high as 100 nm was obtained upon replacing the isopropylbenzene ligand of ferrocenium by a carbazole. At 347 nm, introduction of the carbazole moiety resulted in a 30-fold enhancement of the molar extinction coefficients of **Cbz_39** and **Cbz_40** compared to that of Irgacure 261.

Comparison of the photoinitiating ability of **Cbz_39** and **Cbz_40** with that of Irgacure 261 for wavelength higher than 320 nm revealed the two carbazole derivatives to outperform Irgacure 261 while simply introducing 2% of photoinitiators in EPOX. Fast polymerization processes were reported in this work, some polymerization processes being ended after 20 s. However, it has to be noticed that high intensity lamps were used, the intensity ranging between 700 W for the high-pressure Hg lamp to 1000 W for the D-lamp. But the remarkable efficiency of the cationic polymerization is also supported by the proposed mechanism. Notably, a ligand exchange with EPOX resulting from the

展示。事实上，在这些波长下的光聚合主要由花菁类化合物主导 [145–150]，但这些化合物价格昂贵，因此研究者们正积极寻找替代品。在使用由三种单体（33.3 wt% 的羟丙基甲基丙烯酸酯 (HPMA)、33.3 wt% 的 1,4-丁二醇二甲基丙烯酸酯 (1,4-BDMA) 和 33.3 wt% 的聚氨酯二甲基丙烯酸酯单体 (UDMA)) 混合而成的树脂进行聚合时，使用四组分体系 Cbz/(t-Bu)PhI/4-dppba/BlocBuil-der®MA (0.1 wt%/3 wt%/2 wt%/2 wt%)，其中 4-dppba 和 Block-Builder®MA 分别代表 4-(二苯基膦基)苯甲酸和一种商业烷氧胺的缩写，在 785 nm 辐照下最终单体转化率可达 92%，在 940 nm 辐照下可达 60%。值得注意的是，在这项研究中，作者结合了两种聚合模式：一种基于纯光化学体系，通过光激活产生自由基；另一种基于光热体系，将光转化为热。实际上，红外染料在光激发下会产生热量。通过结合这两种聚合模式，光激发以及甲基丙烯酸酯聚合释放的热量足以分解 Block-Builder®MA（一种热引发剂），从而提高了最终单体转化率。有趣的是，当单独考虑这两种聚合模式时，得到的单体转化率很低甚至为零，这表明需要将这两个体系结合才能获得良好的单体转化率。

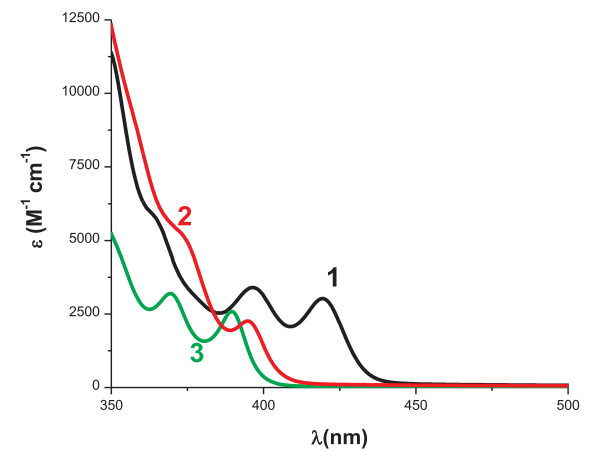


图28.紫外–可见吸收光谱：**Cbz_36**（1）、**Cbz_37**（2）和**Cbz_38**（3）在乙二醇中。经Al Mousawi等人许可转载。 [151]版权所有©2017 John Wiley & Sons, Inc.

聚合过程中产生的光产物和/或未反应的氮杂螺烯 Cbz_36–Cbz_38 的存在，尽管该两组分体系不具有催化特性。

2.6. 咔唑基二茂铁

二茂铁盐是环氧树脂阳离子聚合的有吸引力的结构，因为这些化合物易于从廉价的起始原料中获得 [152–155]。值得注意的是，这些化合物可以通过在二茂铁上进行配体交换一步制备，同时使用高沸点非极性溶剂和 Al/AlCl₃ 作为金属催化剂（参见方案9）。最后，与六氟磷酸盐进行复分解反应，可以将水溶性的二茂铁盐转化为可溶于有机溶剂的金属盐。这种简单而高效的合成方法由 Nesmeyanov 及其同事在 60 年代初报道，并且已经使用该程序制备了许多二茂铁盐 [156,157]。在二茂铁中，Irgacure 261 即 (η₆-(1-甲基乙基)苯基) (η₅-环戊二烯基) 铁六氟磷酸盐是一种流行的阳离子聚合光引发剂，因为该化合物是市售的 [158–164]。然而，最大吸收中心位于 241 nm，Irgacure 261 的吸收范围从 200 到 300 nm，因此这种二茂铁盐是紫外中心的光引发剂。因此，该光引发剂不适用于近紫外/可见光范围内的光聚合过程。

大量工作致力于将二茂铁盐的吸收红移至可见光范围，2006 年报道了这项艰巨工作的相关实例。事实上，首个咔唑基光引发剂 **Cbz_39** 和 **Cbz_40** 于 2006 年发表，并根据上述程序制备（见图 29） [165]。与最大吸收在 241 nm 的 Irgacure 261 相比，将二茂铁盐的异丙苯配体替换为咔唑后，获得了高达 100 nm 的红移。在 347 nm 处，引入咔唑基团使摩尔消光系数 **Cbz_39** 和 **Cbz_40** 比 Irgacure 261 提高了 30 倍。

比较 **Cbz_39** 和 **Cbz_40** 与 Irgacure 261 在波长大于 320 nm 时的光引发能力，发现两种咔唑衍生物的性能优于 Irgacure 261，而只需在 EPOX 中引入 2% 的光引发剂。本研究报告了快速聚合过程，一些聚合过程在 20 秒后结束。然而，需要注意的是，实验中使用了高强度灯，强度范围从高压汞灯的 700 W 到 D 灯的 1000 W。但阳离子聚合的显著效率也由所提出的机理支持。值得注意的是，由光激发后咔唑解配位导致的与 EPOX 的配体交换被认为是合理的机理。

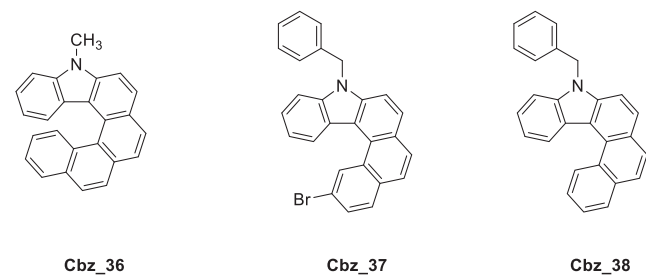


图27. Cbz_36–Cbz_38 的化学结构。

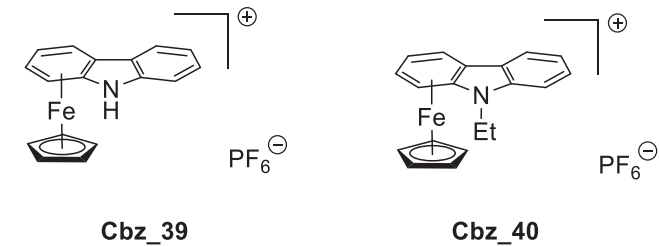
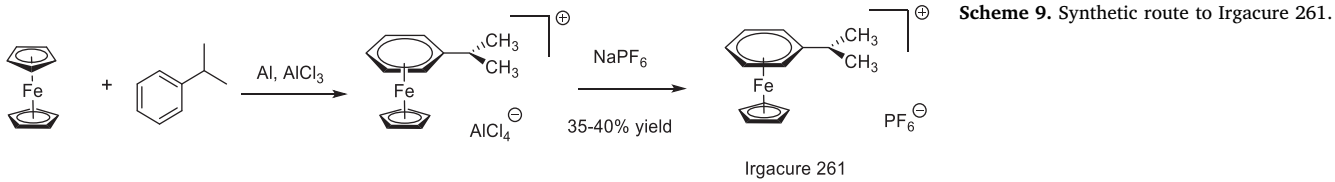


Fig. 29. Chemical structures for Cbz_39 and Cbz_40.

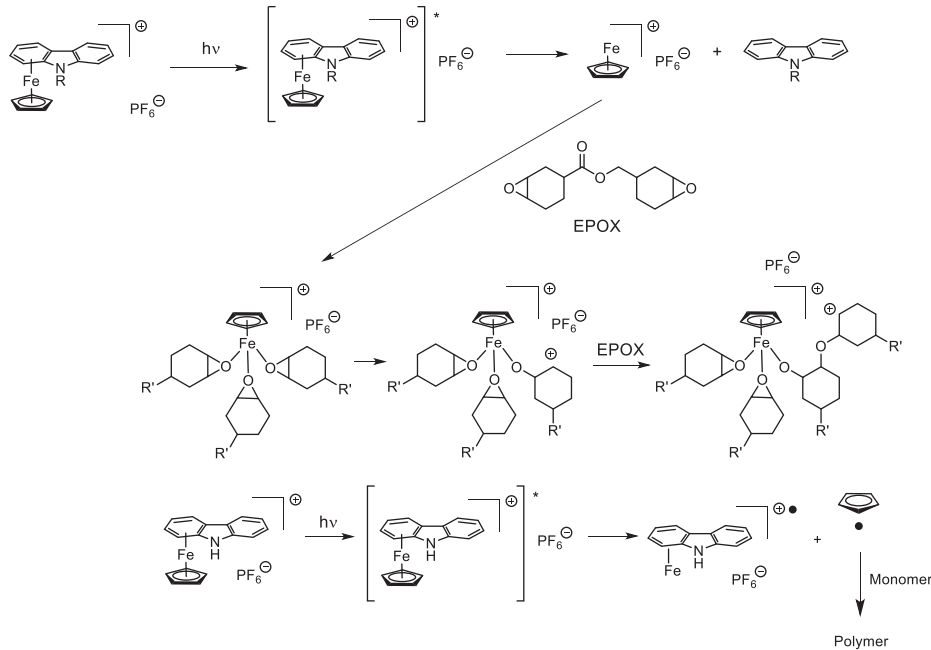
decoordination of carbazole upon photoexcitation was suggested as a plausible mechanism (See [Scheme 10](#)). Such a mechanism has been proposed as soon as 1986 by Riediker and coworkers [\[166\]](#). Indeed, upon decoordination of the carbazole ligands, a strong Lewis acid is formed, enabling three EPOX units to enter in the coordination sphere of iron and to activate the ring opening of epoxide and the formation of a cation, responsible of the initiation process.

Ten years later, **Cbz_39** was revisited in totally different experimental conditions for the CP of EPOX on the basis of new formulations and the new light sources at disposal [\[167\]](#). The great achievement of this work compared to the previous one was first the possibility to initiate a polymerization while using low intensity light sources, and second, to develop photoinitiating systems activable under visible light. Notably, authors evidenced that **Cbz_39** was efficient to initiate the CP of EPOX, alone, even upon irradiation of the resins with low light intensity. Thus, final monomer conversions of 57 and 53% were determined upon irradiation with LEDs at 385 and 405 nm and by using **Cbz_39** without additives. While using the two component **Cbz_39**/Ph₂I⁺ (0.2%/2%, w/w) photoinitiating system, insignificant modifications of the final monomer conversions were found compared to the

previous conditions (55 and 57% respectively determined upon irradiation of the resins with LEDs emitting at 385 nm and 405 nm). However, higher polymerization rates could be determined for the three-component system. While using the **Cbz_39**/Ph₂I⁺/NVK (0.2%/2%/3%, w/w) system, the monomer conversion could be increased up to 66% for an irradiation with a LED@405 nm, resulting from the formation of the highly reactive Ph-NVK⁺ species. Finally, ferrocenium are well-known cationic photoinitiators but the possibility to initiate the FRP of acrylates was also demonstrated. Indeed, based on the previous reports mentioning the photodissociation of the Fe-cyclopentadiene bond, a cyclopentadienyl radical can be released, initiating the FRP (See [Scheme 10](#)). Besides, the different experiments revealed the ability of the cyclopentadienyl radicals to remain limited, the FRP of TMPTA being limited to 8% for the two-component system **Cbz_39**/2,4,6-*tris*(trichloromethyl)-1,3,5-triazine. However, upon addition of EDB, the monomer conversion could be increased up to 44%, demonstrating the validity of the approach.

2.7. Carbazolium derivatives

The previous works have clearly demonstrated the carbazole derivatives to efficiently interact with onium salts but also to both accelerate and improve the cationic photopolymerization of epoxides. In this field, NVK is an unavoidable representant of carbazoles. Besides, by quaternization of the amine of carbazole, photoinitiators for FRP can also be obtained as exemplified with **Cbz_41** (See [Scheme 11](#)) [\[168\]](#). Chemical structure of **Cbz_41** is directly inspired from the structures of phenacyl onium salts that are highly photosensitive compounds [\[169\]](#). Notably, phenacyl onium salts are well-known to initiate a homolytic and/or a heterolytic bond cleavage upon irradiation so that radicals and cations can be both generated upon photolysis [\[170–178\]](#). In the case of **Cbz_41**, cyclic voltammetric and photobleaching experiments revealed



Scheme 10. Mechanism suggested to support the photoinitiating ability of carbazole-based ferrocenium salt.

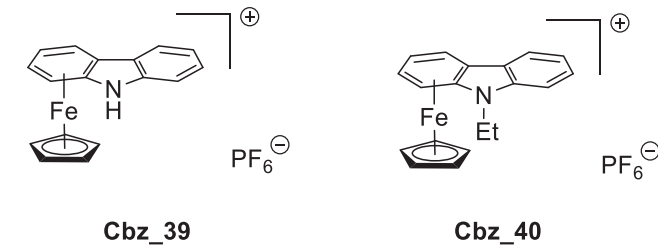
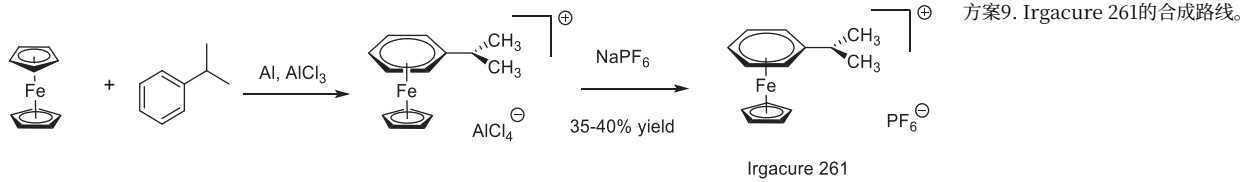


图29. Cbz_39 和 Cbz_40 的化学结构。

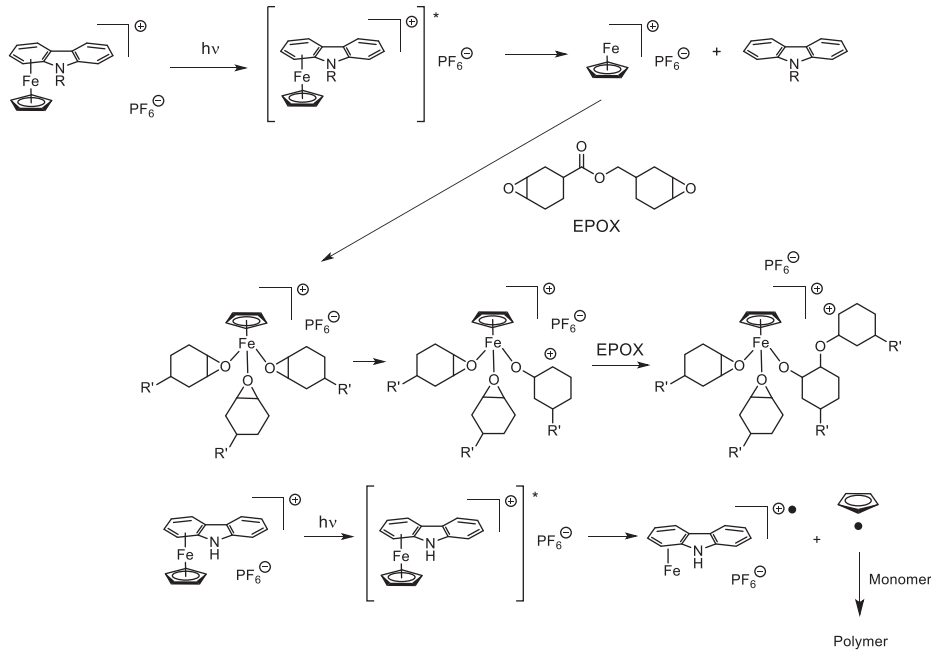
光激发后咪唑解配位导致的配体交换被认为是合理的机理（参见[方案10](#)）。这一机理早在1986年由Riediker及其同事提出 [\[166\]](#)。实际上，咪唑配体解配位后，形成强路易斯酸，使得三个EPOX单元进入铁的配位球，从而激活环氧化物的开环和阳离子的形成，阳离子负责引发过程。

十年后，基于新的配方和可用的新型光源，Cbz_39 在完全不同的实验条件下被重新用于环氧化物的阳离子聚合。与之前的工作相比，这项工作的主要成就是：首先，能够使用低强度光源引发聚合；其次，开发出在可见光下可激活的光引发体系。值得注意的是，作者证明Cbz_39 单独使用时就能有效引发环氧化物的阳离子聚合，即使在低光强度下照射树脂也是如此。因此，在使用385 nm和405 nm LED照射并单独使用Cbz_39（无添加剂）时，最终单体转化率分别达到57%和53%。而使用双组分Cbz_39/Ph₂I⁺（0.2%/2%，w/w）光引发体系时，与之前条件相比（在385 nm和405 nm LED照射下分别得到55%和57%的转化率），最终单体转化率没有显著变化。

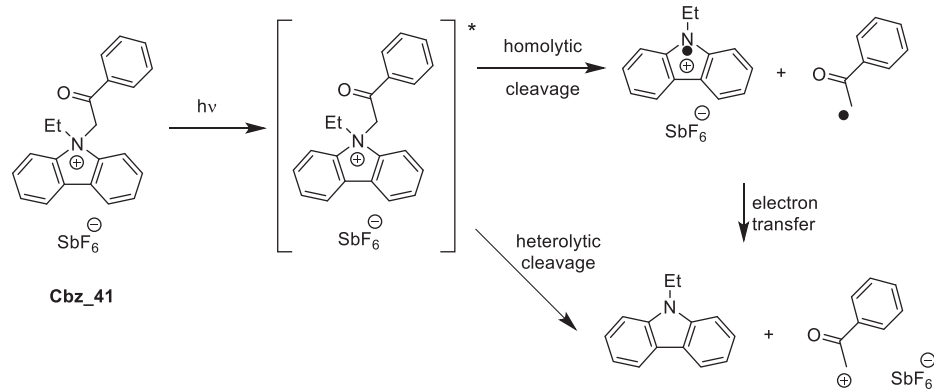
然而，对于三组分体系，可以测定出更高的聚合速率。使用**Cbz_39**/Ph₂I⁺/NVK（0.2%/2%/3%，w/w）体系时，在LED@405 nm照射下，单体转化率可提高至66%，这是由于形成了高反应活性的Ph-NVK⁺物种。最后，二茂铁盐是众所周知的阳离子光引发剂，但同时也证明了它们能够引发丙烯酸酯的自由基聚合。实际上，基于先前报道中Fe-环戊二烯键的光解离，可以释放出环戊二烯基自由基，从而引发自由基聚合（见[方案10](#)）。此外，不同实验揭示了环戊二烯基自由基的活性有限，对于双组分体系**Cbz_39**/2,4,6-三(三氯甲基)-1,3,5-三嗪，TMPTA的自由基聚合转化率仅为8%。然而，在添加EDB后，单体转化率可提高至44%，证明了该方法的有效性。

2.7. 咪唑鎓衍生物

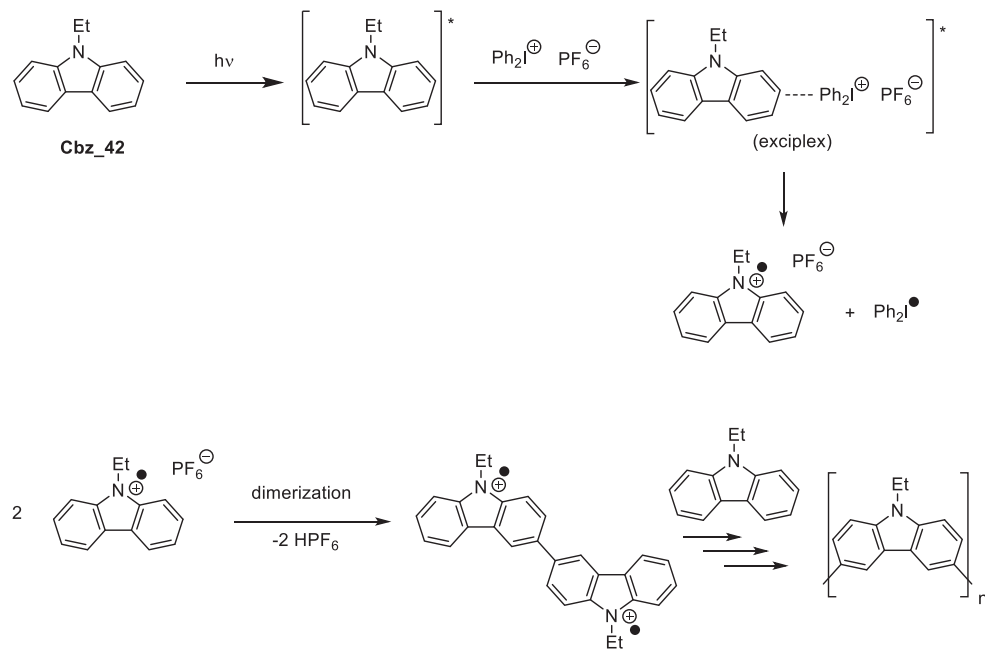
先前的工作已清楚表明，咪唑衍生物能有效与鎓盐相互作用，同时加速并改善环氧化物的阳离子光聚合。在这一领域，NVK是咪唑中不可避免的代表。此外，通过咪唑胺的季铵化，也可以获得FRP的光引发剂，如所示**Cbz_41**（参见[方案11](#)）[\[168\]](#)。的化学结构**Cbz_41**直接受到高光敏性化合物苯酰鎓盐结构的启发 [\[169\]](#)。值得注意的是，苯酰鎓盐在光照下会发生均裂和/或异裂键断裂，从而同时产生自由基和阳离子 [\[170–178\]](#)。在**Cbz_41**的情况下，循环伏安和光漂白实验表明



方案10. 提出的支持咪唑基二茂铁盐光引发能力的机理。



Scheme 11. The different photoproducts obtained during photolysis of phenacyl onium salts.



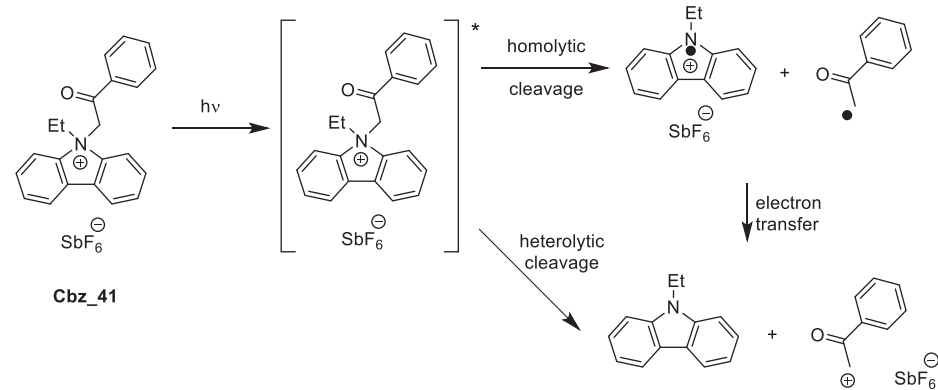
Scheme 12. Step-growth photopolymerization mechanism of poly(*N*-ethylcarbazole).

the carbazole moieties to give rise to an extension of conjugation, attributed to the dimerization of carbazole radical cations. Indeed, the radical cations formed during the homolytic bond cleavage can further interact, as shown in the [Scheme 10](#). To investigate the applicability of **Cbz_41**, the photopolymerization of various monomers such as methyl methacrylate, methacrylic acid, styrene and NVK was investigated. Polymerization of all monomers was successfully achieved, providing polymers of high monomer conversions. Consequently, photolysis of the **Cbz_41** salt by homolytic cleavage was clearly demonstrated, the polymerization of the previous monomers being only possible by the photogeneration of free radicals. Conversely, all attempts of cationic polymerization failed due to the improbability of an electron transfer to yield cationic species and the tendency of photofragments to dimerize under irradiation.

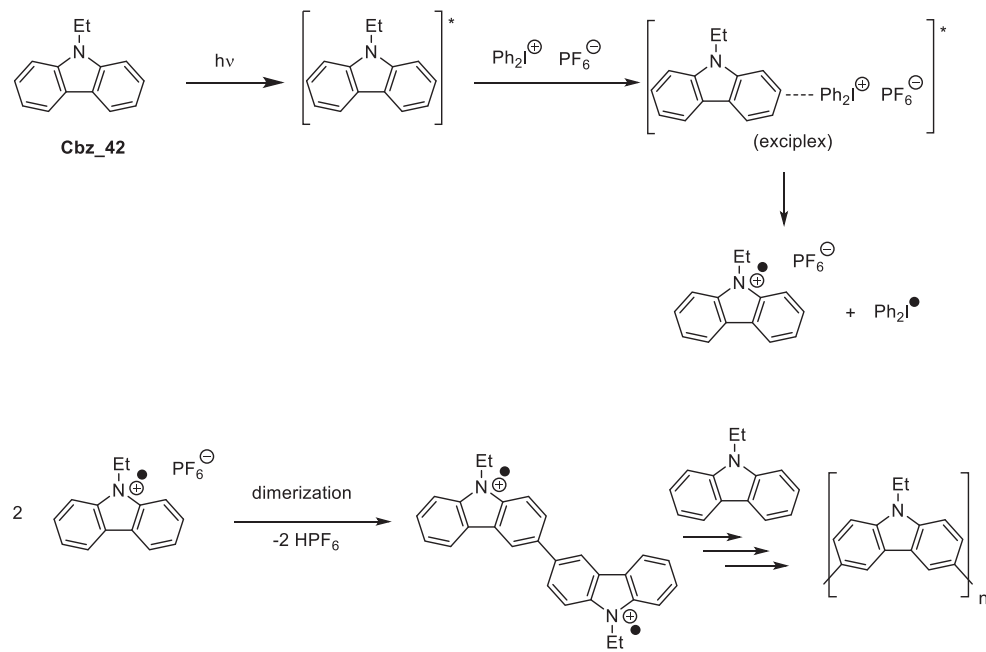
2.8. Carbazoles used as monomers and photoinitiators

At present, all the carbazole derivatives reported in this review have been used as photoinitiators and not as monomers for their own photopolymerizations. This original strategy was reported in 2018 with *N*-ethylcarbazole **Cbz_42** [179]. As specificity, light-induced step-growth polymerization doesn't require any photoinitiator to proceed and the initiation, propagation and termination steps cannot be distinguished

[180]. Contrarily to the previous works where non-conjugated polymers are obtained, photopolymerization of **Cbz_42** could furnish a poly(carbazole). Poly(carbazoles) are extensively studied in Organic Electronics as these polymers are semi-conductors exhibiting an excellent film-forming ability which is of crucial importance to avoid shortcuts in devices [75,181]. In this field, poly(*N*-vinylcarbazole) (PVK) is one of the most representative polymers. PVK is notably commonly used as hole-transport materials for the design of solution-processed organic light-emitting diodes (OLEDs) due to its remarkable morphological stability and its good solubility in most of the common organic solvents. Therefore, devices can be prepared by wet-process while using PVK [76,182–184]. Parallel to PVK derivatives in which carbazole units are connected by mean of a none-conjugated main chain, carbazoles can also be connected by forming carbon–carbon bonds between aromatic rings, generating π -conjugated polymers. The most popular strategies are notably oxidative polymerization [185], electropolymerization [186] and metal-catalyzed cross-coupling reactions [187,188]. In 2018, Yagci and coworkers proposed an original synthesis based on an electron-transfer between carbazole and an iodonium salt. When mixed in solution, appearance of a broad band between 500 and 750 nm in the UV–visible absorption spectrum of **Cbz_42** confirmed the formation of a doped conjugated polymer upon irradiation. Additionally, the photopolymerized polymer which is obtained as a doped polymer could be



方案11. 苯甲酰甲基鏽盐光解过程中获得的不同光产物。



方案12. 聚（N-乙基咔唑）的逐步增长光聚合机理。

咔唑部分导致共轭延伸，归因于咔唑自由基阳离子的二聚化。实际上，均裂键断裂期间形成的自由基阳离子可以进一步相互作用，如方案10所示。为了研究**Cbz_41**的适用性，研究了甲基丙烯酸甲酯、甲基丙烯酸、苯乙烯和NVK等各种单体的光聚合。所有单体均成功聚合，得到高单体转化率的聚合物。因此，通过均裂键断裂对**Cbz_41**盐进行光解已得到明确证明，先前单体的聚合只能通过光生自由基实现。相反，所有阳离子聚合的尝试均失败，因为电子转移生成阳离子物质的可能性极低，且光碎片在辐照下倾向于二聚化。

2.8. 咔唑作为单体和光引发剂

目前，本综述报道的所有咔唑衍生物均用作光引发剂，而非用于其自身光聚合的单体。这种原创策略于2018年以N-乙基咔唑**Cbz_42** [179]报道。其特异性在于，光诱导逐步增长聚合无需任何光引发剂即可进行，且引发、增长和终止步骤无法区分。

[180]。与以往获得非共轭聚合物的研究相反，**Cbz_42**的光聚合可能提供聚咔唑。聚咔唑在有机电子学中被广泛研究，因为这些聚合物是半导体，具有优异的成膜能力，这对避免器件短路至关重要 [75,181]。在此领域，聚(*N*-乙基咔唑) (PVK) 是最具代表性的聚合物之一。PVK因其出色的形态稳定性和在大多数常见有机溶剂中的良好溶解性，常被用作溶液处理有机发光二极管 (OLED) 设计中的空穴传输材料。因此，器件可以通过湿法工艺使用PVK制备[76,182–184]。与PVK衍生物（其中咔唑单元通过非共轭主链连接）并行，咔唑还可以通过形成芳香环之间的碳–碳键连接，生成 π 共轭聚合物。最常用的策略包括氧化聚合 [185]，电聚合[186]和金属催化的交叉偶联反应 [187,188]。2018年，Yagci及其同事提出了一种基于咔唑与碘鎓盐之间电子转移的原始合成方法。当在溶液中混合时，在–可见吸收光谱中于500至750 nm之间出现宽谱带，**Cbz_42**证实了辐照下掺杂共轭聚合物的形成。此外，光聚合得到的掺杂聚合物可以通过

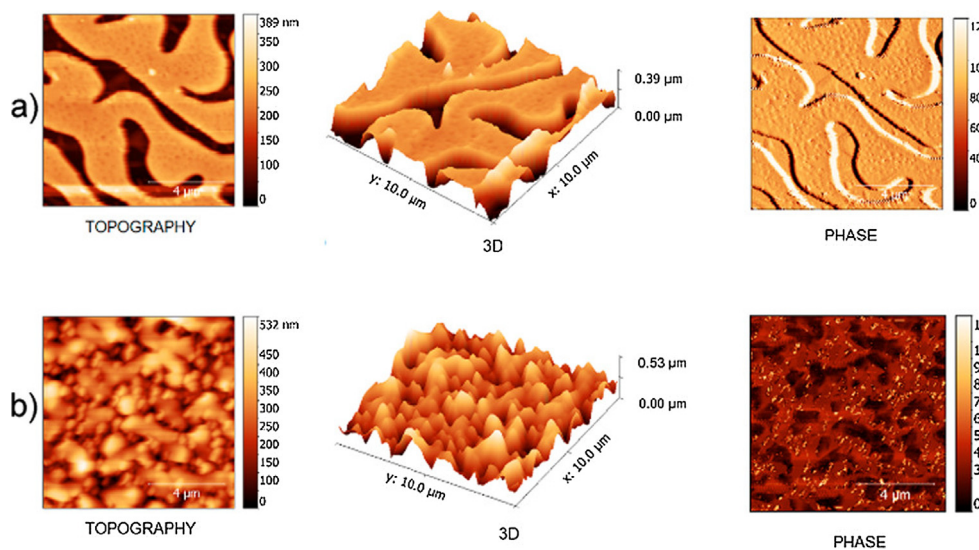


Fig. 30. AFM images of polymer films of doped (a) and dedoped (b) poly(*N*-ethylcarbazole). Reproduced with permission from Sari et al. [179]. Copyright 2018 American Chemical Society.

dedoped by treatment with hydrazinium hydroxide in methanol. Authors also evidenced the step-growth polymerization of poly(*N*-ethylcarbazole) to result from successive cycles of photoinduced electron transfer and coupling of radical cations, as shown in the Scheme 12. In fact, this mechanism is similar to that previously reported for the photopolymerization of polythiophenes [189–191]. Examination of the polymer films by atomic Force Spectroscopy (AFM) revealed the morphology of the doped and dedoped polymer layers to be drastically different. Notably, an increased roughness for the dedoped polymer layers was observed, assigned to a strong π - π stacking between carbazole as a result of PF_6^- anions removal (See Fig. 30).

3. Conclusions -perspectives

During the last decades, a wide range of carbazole derivatives have been proposed as photoinitiators of polymerization. The advantages of carbazole-based photoinitiators compared to other newly developed photoinitiators are numerous. Among the most interesting features, their low cost, low toxicity, high reactivity, good thermal and photochemical stability, good solubility in monomers, ease to handle (powders), high absorptions in the near visible/visible range making these photoinitiators ideal candidates for 3D-printing applications, possibility to prepare organometallic structures by using carbazole as a ligand can be cited as appealing characteristics. Contrarily to numerous photoinitiators that have fallen by the wayside due to a difficult synthetic access, chemistry of carbazole is relatively simple and most of the carbazole derivatives reported in this review have been obtained in high yields by simple purifications processes (precipitation, recrystallisation). From this point of view, carbazoles are promising candidates for future industrial transfer and materials applications. But the different families of carbazole-based photoinitiators listed in this review are not equivalent as far as their photoreactivities is concerned.

Among the series of forty carbazole derivatives reported in the literature, the most important family is undoubtedly based on nitrogen-containing polyaromatic hydrocarbons. Using this strategy, compounds with absorption in the near-UV/visible region have been developed and the different setups previously used for an initiation of the polymerization process in the UV range can be still used with these photoinitiators. Besides, with regards to the development of the new, compact, low-consuming and cheap irradiation setups, a new generation of photoinitiators totally disconnected from the UV photoinitiators have been proposed, as exemplified with the push-pull dyes, Bodipys or

TADF photoinitiators. With these different structures, absorption spectra strongly centered in the visible range are obtained. The last family of photoinitiators i.e. the TADF photoinitiators whose TADF property have only been discovered by Chihaya Adachi in 2012 clearly deserves to be further investigated due to the significant enhancement of the excited state lifetime that could be obtained by chemical engineering. Parallel to the neutral photoinitiators, two families of cationic photoinitiators, namely ferrocenium and carbazolium have respectively been designed for the CP of epoxides or the FRP of acrylates. Concerning carbazoliums, the performances reported at present are far behind that of benchmark photoinitiators so that an extensive work needs to be carried out to improve the photoinitiating ability of these derivatives. Considering that carbazole is a versatile building block, further developments of photoinitiators will consist in elaborating near-infrared dyes, the light penetration into the photocurable resins being the main parameter governing the reactivity of photoinitiators and thus the polymerization speed.

As other promising future application for carbazole-based photoinitiators is the preparation of materials. Indeed, carbazoles, by their strong absorption in the near-visible/visible range, are perfect candidates for photopolymerization at 405 nm, and thus constitutes ideal candidates for 3D-printing applications. The recent works done by Yagci and coworkers on the step-growth polymerization of poly(*N*-ethylcarbazole) also paves the way towards applications in Organic Electronics. Future developments of carbazole-based photoinitiators in this vast research field is directly related to the ability to design π -conjugated polymers, opening the way towards new and unexpected applications.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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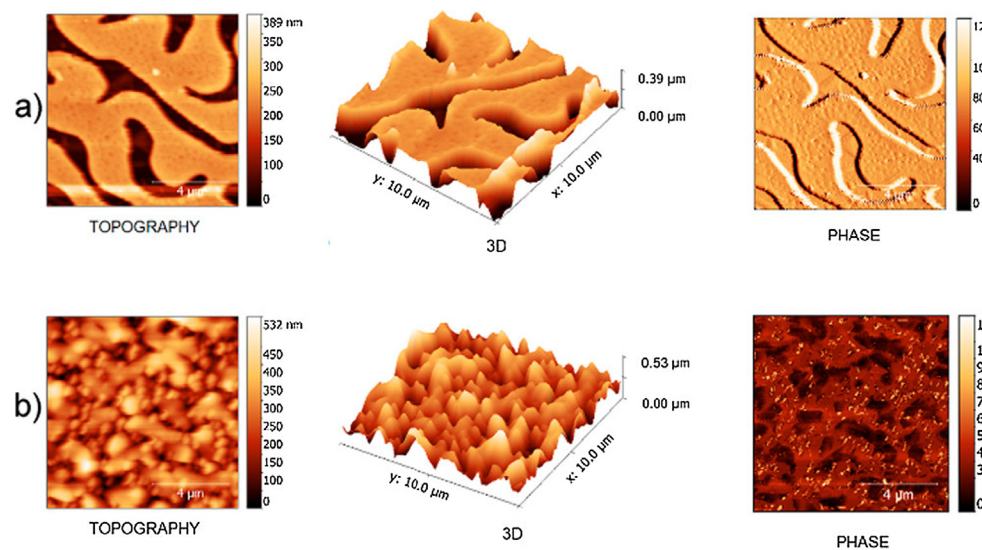


图30. d的聚合物薄膜的AFM图像 掺杂 (a) 和脱掺杂 (b) 聚 (N-乙基咔唑)。经Sari等人许可转载 [179]。版权2018 美国化学学会。

用甲醇中的氢氧化胼处理进行脱掺杂。作者还证明了聚(*N*-乙基咔唑)的逐步增长聚合源于连续的光诱导电子转移和自由基阳离子偶联循环, 如方案12所示。事实上, 该机制与先前报道的聚噻吩光聚合机制相似 [189–191]。通过原子力显微镜 (AFM) 检查聚合物薄膜发现, 掺杂和脱掺杂聚合物层的形态截然不同。特别是, 观察到脱掺杂聚合物层的粗糙度增加, 这归因于 PF_6^- 阴离子移除后咔唑之间强烈的 π - π 堆积 (见图30)。

3. 结论与展望

在过去的几十年里, 人们提出了多种咔唑衍生物作为聚合光引发剂。与其他新开发的光引发剂相比, 咔唑类光引发剂具有众多优势。其中最吸引人的特性包括: 低成本、低毒性、高反应活性、良好的热稳定性和光化学稳定性、在单体中良好的溶解度、易于操作 (粉末状)、在近可见/可见光范围内的高吸收性使其成为3D打印应用的理想候选物, 以及利用咔唑作为配体制备有机金属结构的可能性。与许多因合成困难而被淘汰的光引发剂不同, 咔唑的化学性质相对简单, 本综述中报道的大多数咔唑衍生物均通过简单的纯化过程 (沉淀、重结晶) 以高产率获得。从这一角度来看, 咔唑是未来工业转移和材料应用的有前景的候选物。但本综述中列出的不同咔唑类光引发剂家族在光反应活性方面并不同等。

TADF光引发剂。通过这些不同的结构, 获得了强烈集中在可见光范围的吸收光谱。最后一类光引发剂, 即TADF光引发剂 (其TADF特性仅在2012年由Chihaya Adachi发现), 显然值得进一步研究, 因为通过化学工程可以显著增强激发态寿命。与中性光引发剂并行, 两类阳离子光引发剂, 即二茂铁鎓和咔唑鎓, 分别被设计用于环氧化物的阳离子聚合或丙烯酸酯的自由基聚合。关于咔唑鎓, 目前报道的性能远低于基准光引发剂, 因此需要进行大量工作以提高这些衍生物的光引发能力。考虑到咔唑是一种多功能构建块, 光引发剂的进一步开发将包括制备近红外染料, 光穿透光固化树脂是控制光引发剂反应性以及聚合速度的主要参数。

咔唑基光引发剂另一个有前途的未来应用是材料的制备。实际上, 咔唑因其在近可见/可见光区域的强吸收, 是405 nm光聚合的完美候选物, 因此成为3D打印应用的理想选择。Yagci及其同事最近在聚(*N*-乙基咔唑)逐步增长聚合方面的工作也为有机电子学的应用铺平了道路。未来在这个广阔研究领域中咔唑基光引发剂的发展与设计 π -共轭聚合物的能力直接相关, 为新的和意想不到的应用开辟了道路。

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